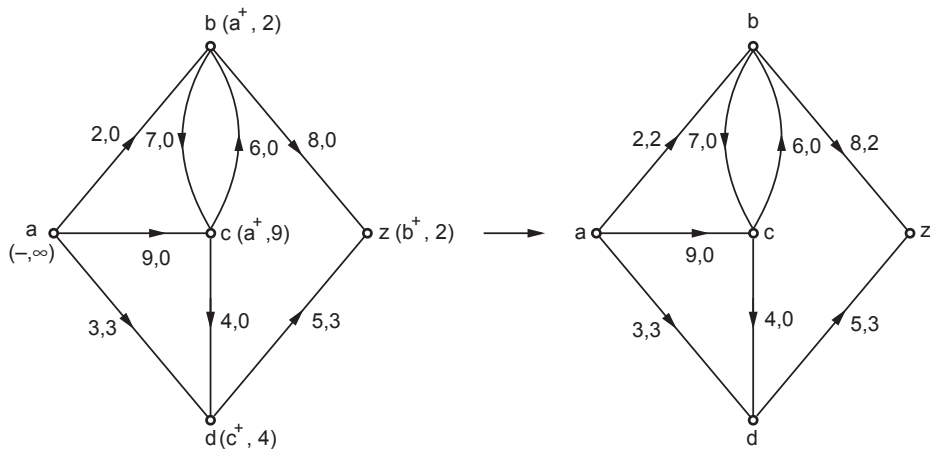
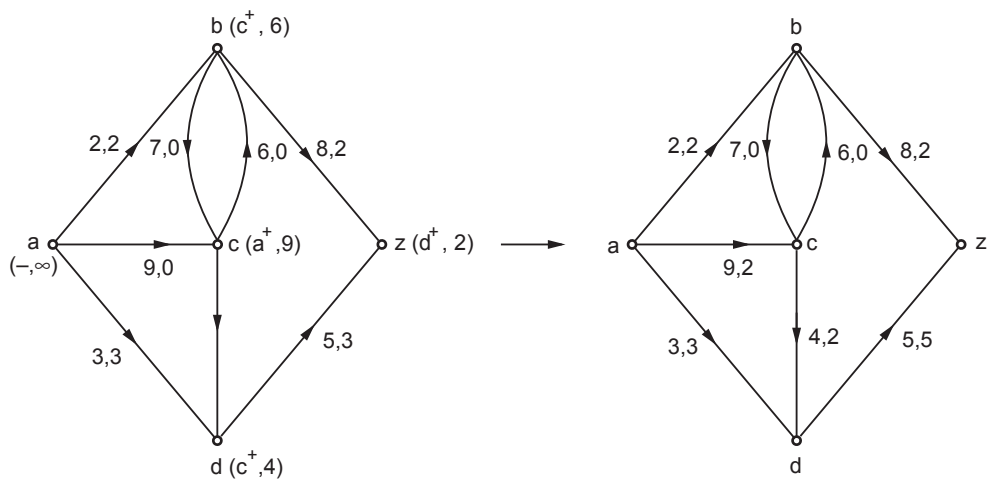
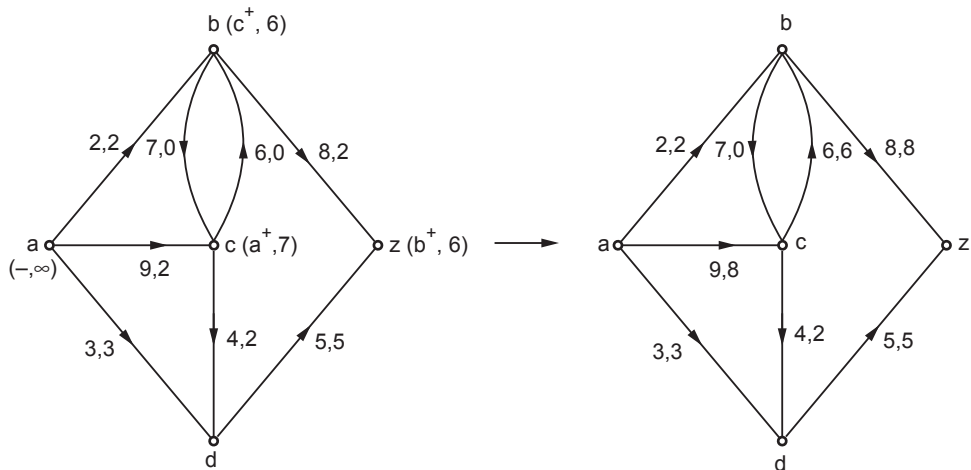


Step 3 :**Step 4 :****Step 5 :**

Example 8.9.3 Use labelling procedure to find a maximum flow in the transport network shown in the following Fig. 8.9.2. Determine the corresponding minimum cut.

SPPU : May-07, 17, Dec.-14, 16, 18

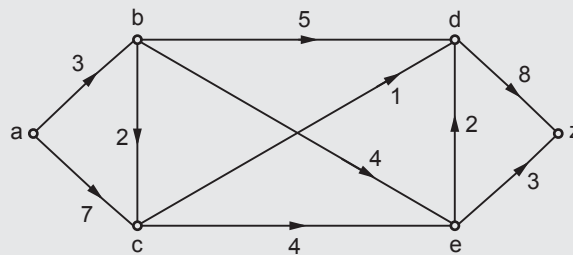
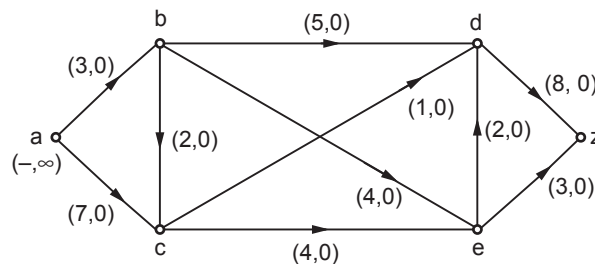


Fig. 8.9.2

Solution : To find the maximum flow in the given transport network, assign the flow zero to each edge and label $(-, \infty)$ to the source a .

Step 1 :



Step 2 : Since the vertices b and c are adjacent to the source a , therefore label the vertices b and c as $(a^+, \Delta b)$ and $(a^+, \Delta c)$ respectively.

Where $\Delta b = ((a, b) - f(a, b) = 3$

$\Delta c = ((a, c) - f(a, c) = 7$

$\therefore$ Label of b is $(a^+, 3)$ and label of c is $(a^+, 7)$

Now the vertices d and e are adjacent to labeled vertices b $(a^+, 3)$ and c $(a^+, 7)$ therefore we label the vertices d and e as $(b^+, \cdot d)$ and $(c^+, \cdot e)$ where

$$\begin{aligned}\Delta d &= \min(\Delta b, c(b, d) - f(b, d)) \\ &= \min(3, 5, -0) \\ &= 3\end{aligned}$$

$$\begin{aligned}\text{Similarly } \Delta e &= \min(\Delta c, c(c, e) - f(c, e)) \\ &= \min(7, 4, -0) \\ &= 4\end{aligned}$$

$\therefore$ Label of d is $(b^+, 3)$ and label of e is $(c^+, 4)$

Now the sink z is adjacent to both the vertices d and e , we can choose any vertex d or e to label the sink z .

Let us choose the vertex e , which is labeled as $(c^+, 4)$

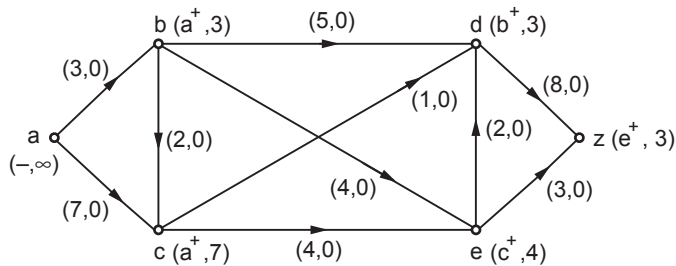
The label of z will be $(e^+, \Delta z)$ where

$$\Delta z = \min(4, 3, -0)$$

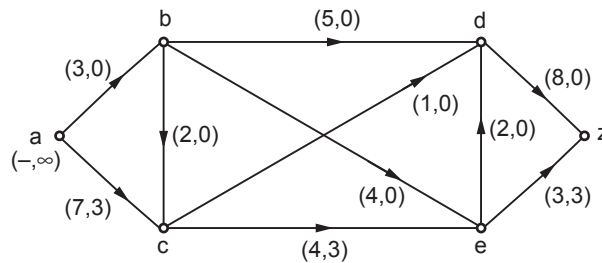
$$= 3$$

Hence label of z is $(e^+, 3)$

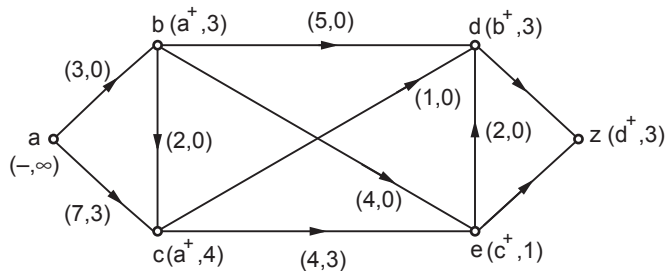
The labels of vertices can be shown as follows :



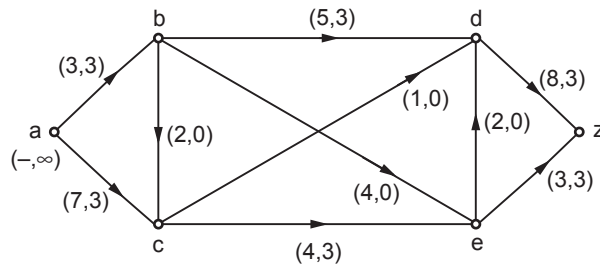
According to the label of the sink z , adjust the flow in the edges (e, z) , (c, e) and (a, c)



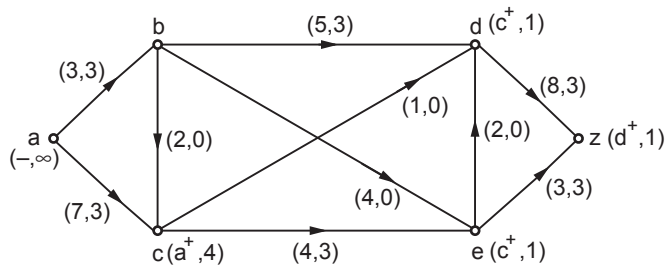
Repeat the step 2 in each pass, the new value of the flow is obtained.



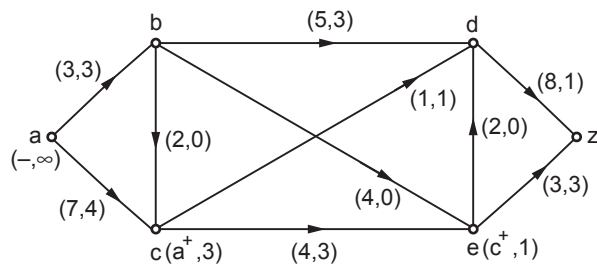
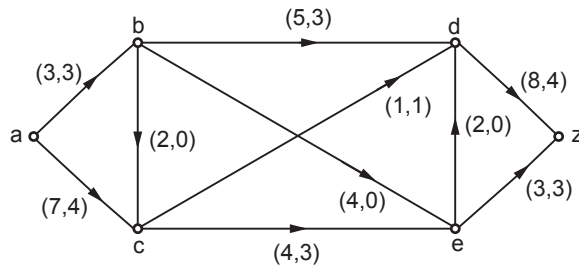
After adjusting the flow we have



Step 4 :



Adjust the flow according to sink z label.



Since the vertex z (sink) cannot be labelled, we have to stop. The vertices b, d and z cannot be labelled because either the edges approaching to vertices are saturated or the edges in the opposite direction have flow zero. At this stage, we have minimum cut $(P, \bar{P})$.

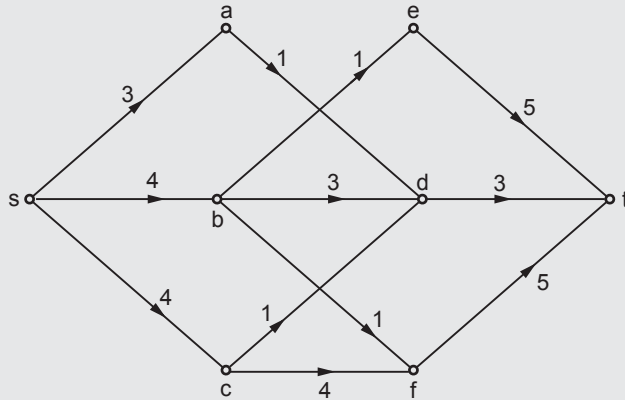
P is the set of labelled vertices $P = \{a, c, e\}$. $\bar{P}$ is the set of unlabeled vertices, $\bar{P} = \{b, d, z\}$

The edges in cut $P \bar{P} \{(a, b), (c, d), (e, z)\}$ capacity of the cut is $c(a, b) + c(c, d) + c(e, z) = 3 + 1 + 3 = 7$.

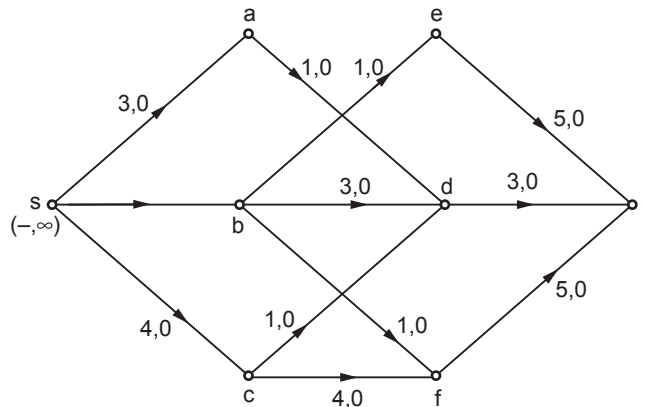
Hence the maximum flow in the given transport network 7.

Example 8.9.4 Find the maximum flow in the following network using labeling algorithm.

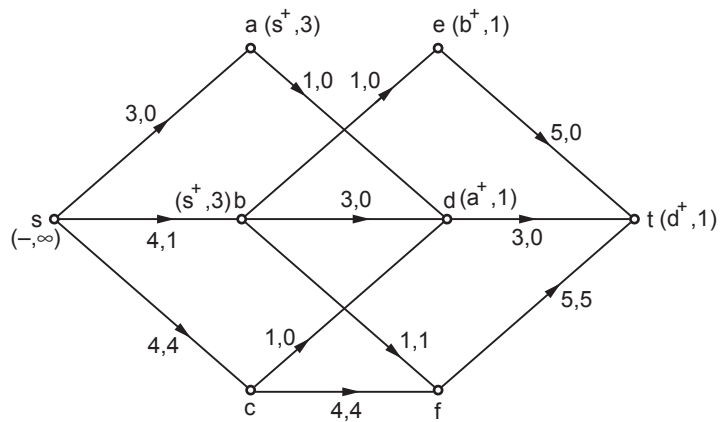
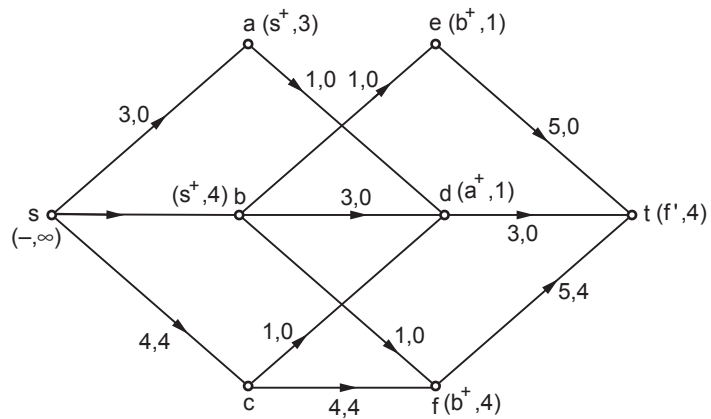
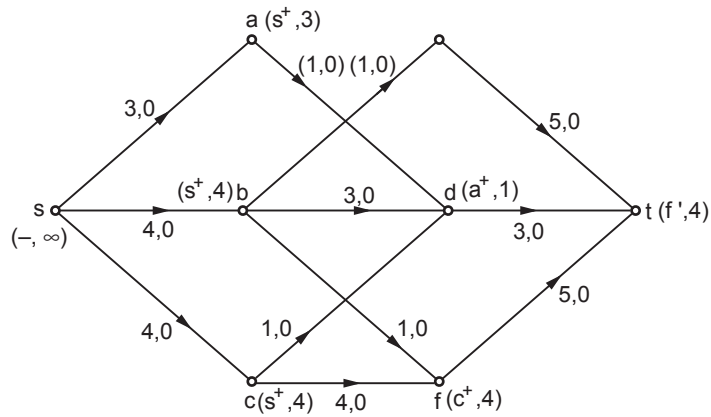
SPPU : April-11

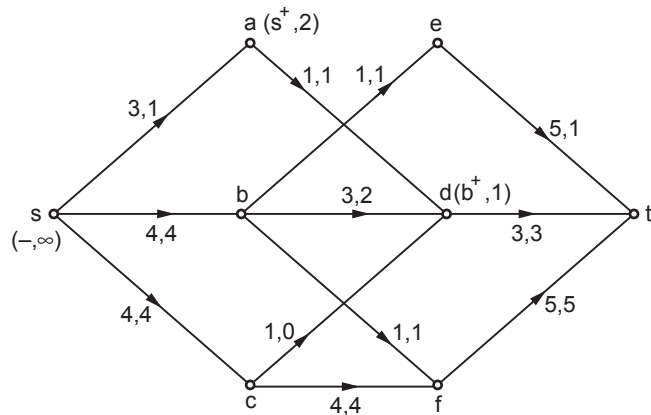
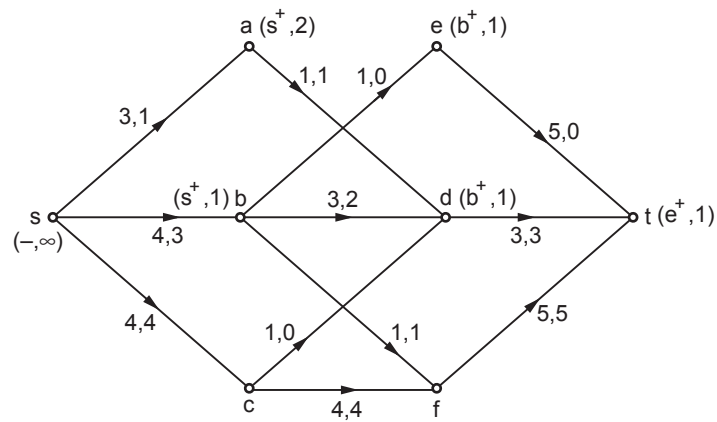
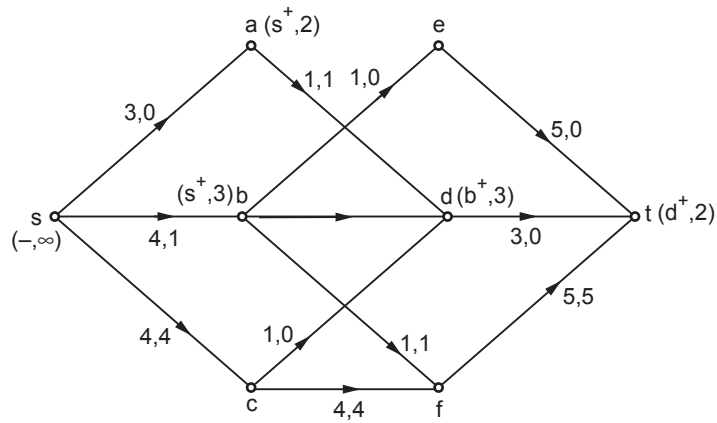


Solution : To find the maximum flow in the given transport network, assign the flow zero to each edge and label $(-, \infty)$ to the sources.



Using labelling procedure the maximum flow can be obtained using following steps.





Since t cannot be labelled we will stop the procedure minimum cut from the set.

$$P = \{s, a\} \text{ to } \bar{P} = \{b, c, d, e, f, t\} \text{ is } \{(s, c), (s, b), (c, d)\}$$

The capacity of the cut is $c\{s, c\} + c\{s, b\} + c\{c, d\}$

$$= 4 + 4 + 1 = 9$$

$\Rightarrow$ Maximum flow in the network is 9.

8.10 Game Tree**SPPU : Dec.-17, May18**

A directed graph whose nodes are positions in a game and edges are moves, is called a game tree.

The complete game tree for a game is the game tree starting at the initial position and containing all possible moves from each position. The complete tree is the same tree as that obtained from the extensive form game representation.

We look at using trees for game planning, in particular the problems of searching game trees. We classify all games into following three types.

1) Single player path finding problems :**For examples :**

- a) Travelling salesman problem
- b) Sliding puzzle
- c) Hamiltonian paths
- d) Rubik's cube

2) Two player games :**For examples :**

- a) Chess
- b) Badminton
- c) Tennis
- d) Checkers

3) Constraint satisfaction problems :**For examples :**

- a) Sudoku
- b) Eight queen problem
- c) Mathematical puzzles
- d) Four queen problem

Each game consists of a problem space, an initial state and a set of goal states.

A problem space is a mathematical abstraction in a form of a tree :

- The root represents current state
- Nodes represent states of the game
- Edges represents moves

- Leaves represent final states (Win, loss or draw)

For example, in the 8-puzzle game

- Nodes : The different permutations of the tiles.
- Edges : Moving the black file up, down, right or left.

Minimax :

We consider game with two players in which one players gains are the result of another players losses so called zero sum games.

The minimax algorithm is a specialized search algorithm which returns the optimal sequence of moves for a player in an zero sum game.

In the game tree that result from the algorithm, each level represents a move by either of two players, say A and B player. The minimax algorithm explores the entire game tree using a depth-first search.

At each node in the tree where A-player has to move, A-player would like to play the move that maximizes the payoff. Thus, A-player will assign the maximum score amongst the children to the node where max makes a move. Similarly, B-player will minimize the pay off to A-player. The maximum and minimum scores are taken at alternating levels of the tree, Since A and B alternate turns. The minimax algorithm computes the minimx decision for the leaves of the game tree and than backs up through the tree to give the final value to the current state.

8.11 Case Studies

- 1) Algebraic expression tree : Refer section 8.5.2.
- 2) Tic - Tac - Toe game tree : A game free is a directed graph whose nodes are positions in a game and edges are move Tic - Tac - Toe is a paper - pencil game for two players, X and O, who take turns making the spaces in a 3×3 grid. The player who succeeds in placing three of their marks in a horizontal, vertical or diagonal row wins the game.

O	O	X
X	X	X
X	O	O

Minimax algorithm : Min-Max is a decision making algorithm which uses decision theory, game theory, statistics and philosophy to calculate the optimal move. It is a two-player game. The mechanism evaluates minimum loss and maximum profit.

In the Tic - Tac - Toe game, a player tries to ensure two cases :

- i) Maximize a player's own chances of win
- ii) Minimize opponent's chances of win.

Maximize profit : The profit can be maximized by either fork or win.

Fork : Initially player will create opportunity where he or she can win in two ways.

Win : If there are two same 'X' or 'O' in a row, then play the third to get three in a row.

Minimize loss : The loss can be minimized by a block.

Block : If two 'X' or 'O' of the opponent are in a row then block it or else block opponents fork.

8.12 University Questions with Answers

Dec. 2016

- Q.1** Find maximum flow in the transport network using labeling procedure. Determine the corresponding min cut. (Refer similar example 8.9.3) [7]

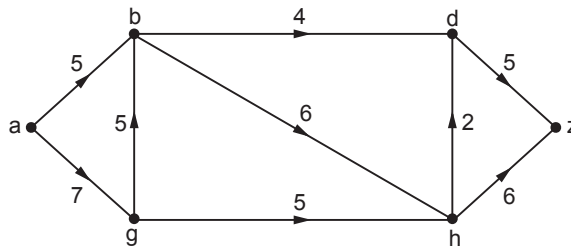


Fig. 8.12.1

- Q.2** Define :
 i) Forest (Refer section 8.2) ii) Height of a tree (Refer section 8.4.2)
 iii) Ordered tree (Refer section 8.4)
 iv) Properties of tree (Refer theorem 1 to 4 statements only of section 8.2) [4]
- Q.3** State whether the given code is prefix code. Justify :
 {000, 001, 01, 10, 11} (Refer example 8.6.6) [2]

- Q.4** Give the stepwise construction of minimum spanning tree using Prim's algorithm for the following graph. Obtain the total cost of minimum spanning tree.
 (Refer example 8.7.2) [6]

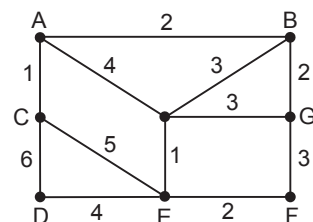


Fig. 8.12.2

Q.5 Explain fundamental system of cutsets with suitable examples.
(Refer section 8.8)

[4]

Q.6 Explain binary tree, binary search tree and ordered tree with suitable examples.
(Refer section 8.6)

[3]

May 2017

Q.7 Use labeling procedure of find a maximum flow in the transport network given in the following figure. determine the corresponding minimum cut.
(Refer example 8.9.3)

[7]

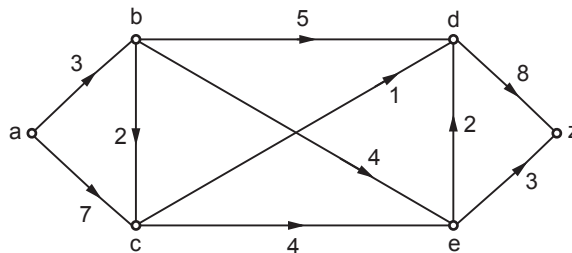


Fig. 8.12.3

Q.8 Explain the following :

- 1) Difference between binary tree and binary search tree.
- 2) Rooted free 3) Cut-sets. (Refer sections 8.5 and 8.6.4)

[6]

Q.9 Find minimum spanning tree for given graph using Kruskal's algorithm.

[6]

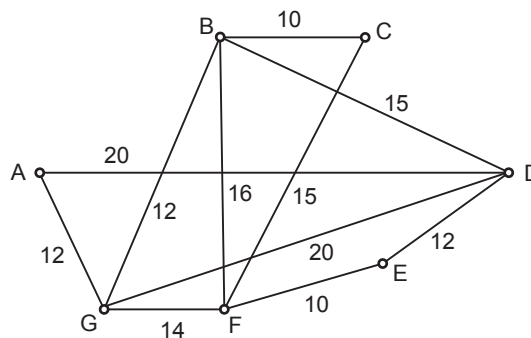
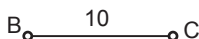


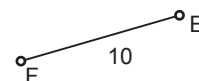
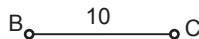
Fig. 8.12.4

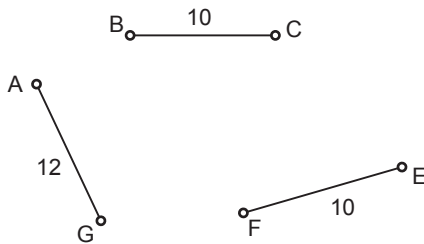
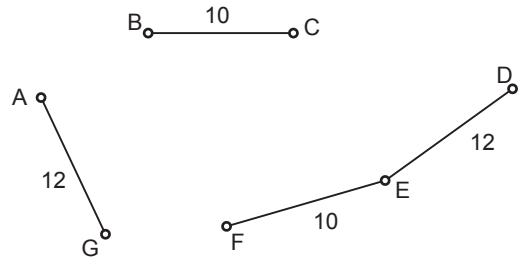
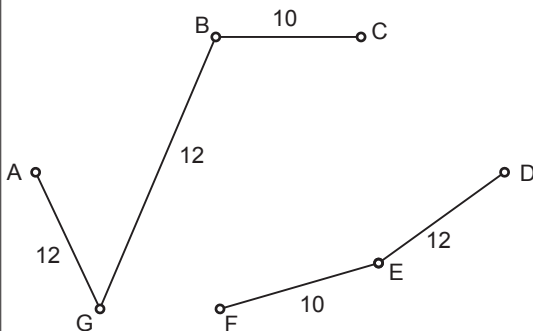
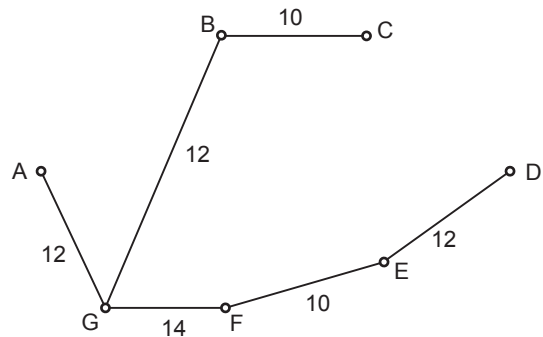
Ans. :

Step 1 :



Step 2 :



Step 3 :**Step 4 :****Step 5 :****Step 6 :**

Weight of MS = $10+10+10+12+12+14 = 68$

Q.10 Explain the following terms :

- Application of cutset in computer engineering domain.
- Prefix code construction using Huffman coding. (Refer section 8.6.2)
- Properties of trees. (Refer section 8.2)

[7]

Ans. : i) Applications of cutset

- transport network
- Map coloring
- Ad-hoc network
- Network security
- Fault tolerant computing system.

Dec. 2017

Q.11 Use Prim's algorithm to find the minimum spanning tree for the connected weighted graph G as shown in Fig. 8.12.5.

[7]

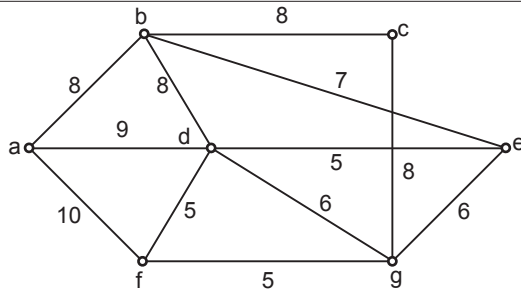


Fig. 8.12.5 Connected weighted graph

Ans. : Consider the following steps for the construction of the minimum spanning tree by prim's algorithm.

Step 1 : Vertex a is starting vertex

$$T = \{\{a\}, \phi\}$$

Step 2 : $T = \{\{a, b\}, e_1\}$

a ○

Fig. 8.12.6



Fig. 8.12.7

Step 3 : $T = \{\{a, b, e\}, \{e_1, e_2\}\}$

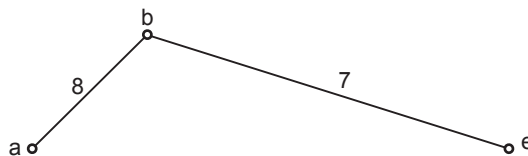


Fig. 8.12.8

Step 4 : $T = \{\{a, b, e, d\}, \{e_1, e_2, e_3\}\}$

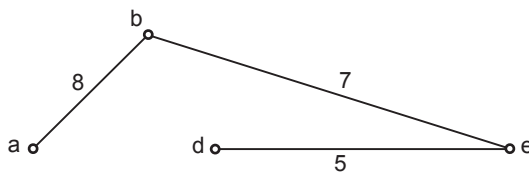


Fig. 8.12.9

Step 5 : $T = \{\{a, b, e, d, f\}, \{e_1, e_2, e_3, e_4\}\}$

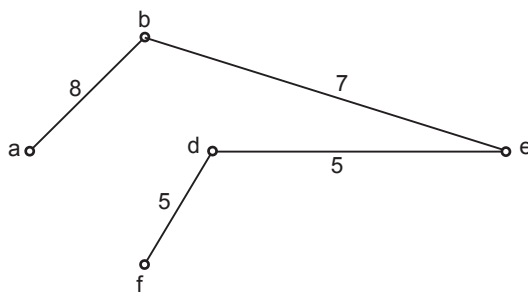


Fig. 8.12.10

Step 6 : $T = \{\{a, b, e, d, f, g\}, \{e_1, e_2, e_3, e_4, e_5\}\}$

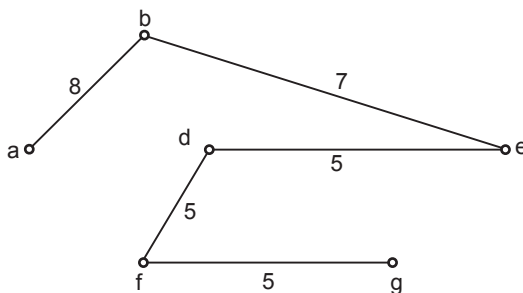


Fig. 8.12.11

Step 7 : $T = \{\{a, b, e, d, f, g, c\}, \{e_1, e_2, e_3, e_4, e_5, e_6\}\}$

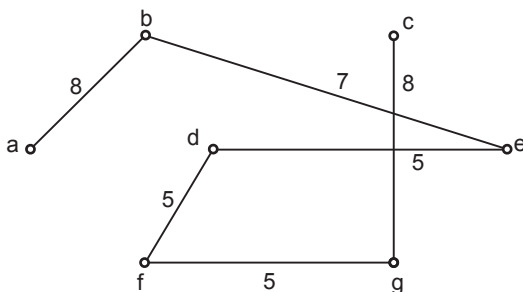


Fig. 8.12.12

The above graph is the minimum spanning tree and its total weight is $8 + 7 + 5 + 5 + 5 + 8 = 38$.

Q.12 Explain any two of the following :

- i) Min-max tree case study ii) Transport network iii) Decision free.

[6]

Ans. : i) Refer section 8.10

ii) Refer section 8.9.

iii) **Decision tree** : A decision tree is a decision support tool that uses a tree of decisions and their possible consequences, including chance event, outcomes, resource cost and utility. It is one way to display an algorithm that only contains conditional control statements.

Decision trees are used in decision analysis to help identify a strategy most likely to reach goal, but are also popular tool in machine learning.

Q.13 Explain the Kruskal's algorithm in detail. (Refer section 8.7.3)

[7]

Q.14 Find the pre, post and inorder traversal of a tree shown in Fig. 8.12.13.

[6]

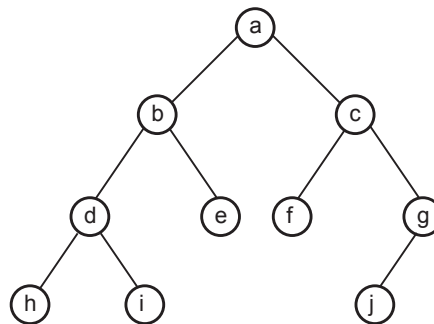


Fig. 8.12.13 Tree

Ans. : Preorder : a, b d h i e c f g j

Inorder : h d i b e a f c j g

Postorder : h i d e b f j g c a

May 2018

Q.15 Determine the maximum flow in the transport network shown in Fig. 8.12.14 using labelling procedure. Determine the corresponding min. cut. [7]

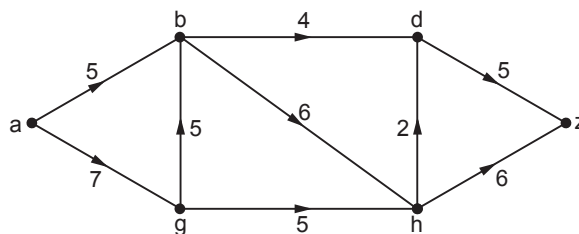


Fig. 8.12.14

Ans. : The maximum flow in the transport network using labelling procedure can be obtained using following steps.

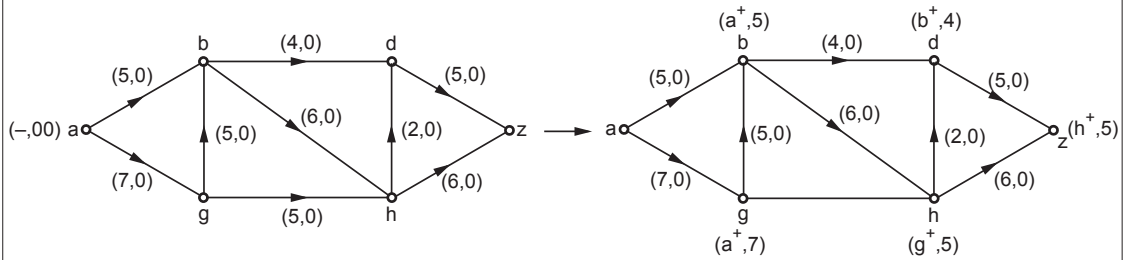


Fig. 8.12.15

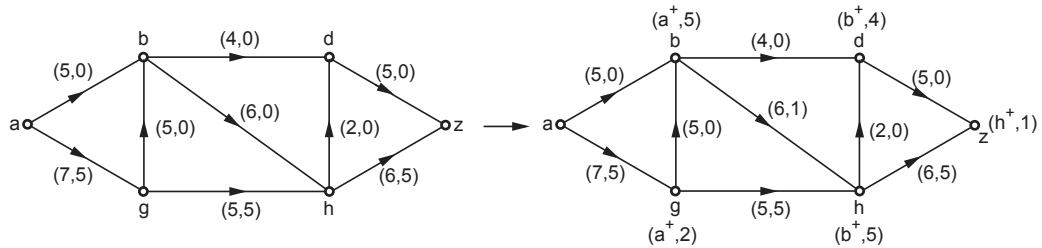


Fig. 8.12.16

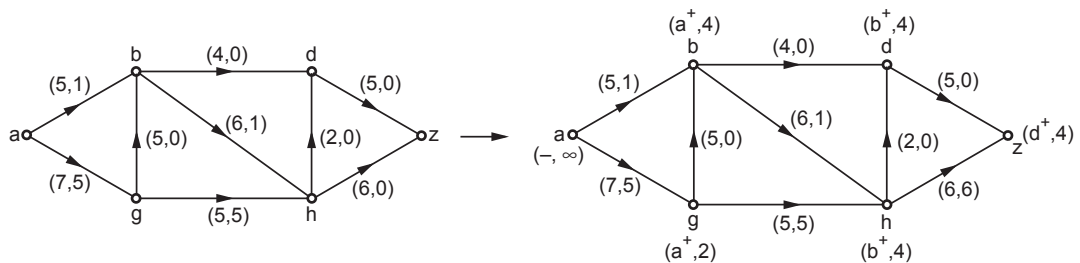


Fig. 8.12.17

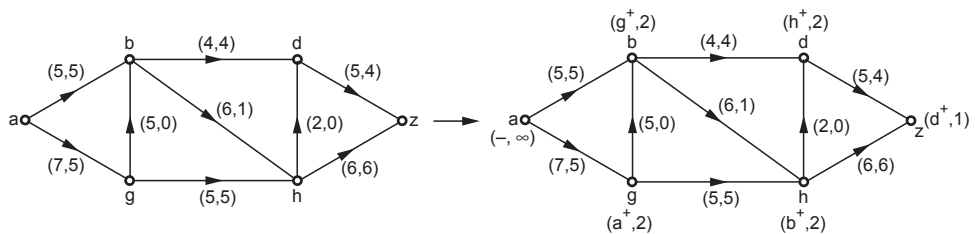


Fig. 8.12.18

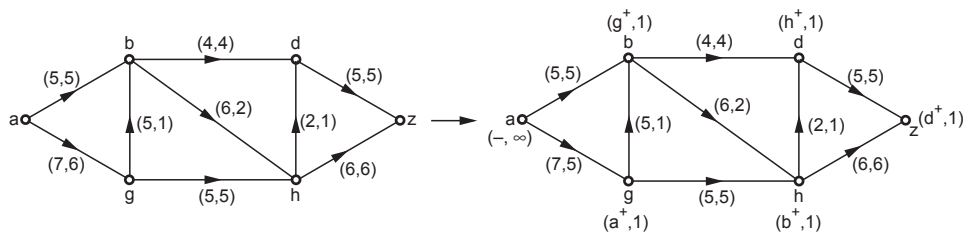


Fig. 8.12.19

Since z can not be labeled so we have to stop.

The maximum cut is $\overline{PP}$ where P is the set of labeled vertices and $\overline{P}$ is the set of unlabelled vertices.

$$P = \{a, b, g, d, h\}, \quad \overline{P} = \{z\}$$

Edges is cut $\overline{PP} = \{(d, z), (h, z)\}$

Capacity of the maximum cut = $C(d, z) + C(h, z) = 5 + 6 = 11$

The maximum flow in transport work is 11.

Q.16 Explain the following terms :

- i) Spanning trees (Refer section 8.7)
- ii) Properties of trees (Refer section 8.2, Theorem 1 to 4)
- iii) M-ary tree (Refer section 8.4.4)

Q.17 Give the stepwise construction of minimum spanning tree using Prim's algorithm for the following graph shown in Fig. 8.12.20. Obtain the total cost of minimum spanning tree. [7]

Ans. : Consider the following steps for the construction of the minimum spanning tree.

Step 1 : Select $A \in V(G)$ as a starting vertex.

$$T = \{\{A\}, \phi\}$$

Step 2 : Vertex A is adjacent to vertices B, C and D

Among all these edges minimum weight is

$$W(A, C) = 1$$

$$\therefore T = \{\{A, C\}, \{(A, C)\}\}$$

Step 3 : Vertex A is adjacent to B and D .

C is adjacent to B, D, F .

$$W\{A, B\} = 6, W\{A, D\} = 5, W\{C, B\} = 5, W\{C, D\} = 5, \\ W\{C, F\} = 4.$$

Among these minimum weight is $W(C, F) = 4$.

$$\therefore T = \{\{A, C, F\}, \{(A, C), (C, F)\}\}$$

Step 4 : Vertex A is adjacent to B and D .

C is adjacent to B and D .

F is adjacent to D and E .

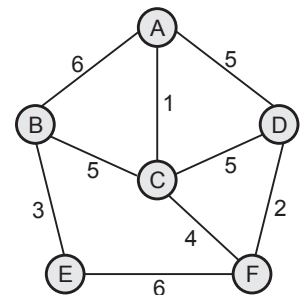


Fig. 8.12.20

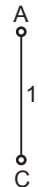


Fig. 8.12.21

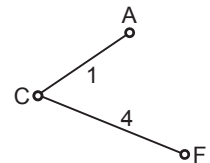


Fig. 8.12.22

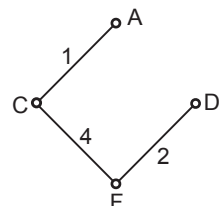


Fig. 8.12.23

Among all these minimum weight $W\{F, D\} = 2$.

$\therefore T = \{ \{A, C, F, D\}, \{(A, C), (C, F), (F, D)\} \}$

Step 5 : Vertex A is adjacent to B.

C is adjacent to B.

F is adjacent to E.

$\therefore W\{A, B\} = 6, W\{C, B\} = 5, W\{F, E\} = 6$.

$\therefore W(C, B) = 5$ is minimum

$T = \{ \{A, C, F, D, B\}, \{(A, C), (C, F), (F, D), (C, B)\} \}$

Step 6 : Vertex B is adjacent to E, $W(B, E) = 3$

Vertex F is adjacent to E, $W(F, E) = 6$

Minimum is $W(B, E) = 3$.

$T = \{ \{A, C, F, D, B, E\}, \{(A, C), (C, F), (F, D), (C, B), (B, E)\} \}$

The graph obtained is the minimum spanning tree of weight $= 1 + 5 + 3 + 4 + 2 = 15$.

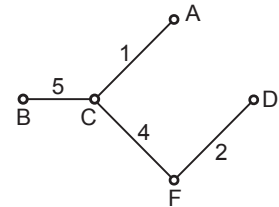


Fig. 8.12.24

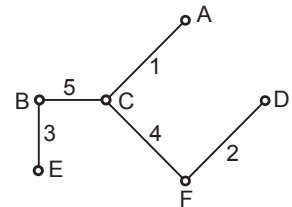


Fig. 8.12.25

- Q.18** Explain the following : i) Game tree (Refer section 8.10)
ii) Kruskal's algorithm (Refer example 8.7.3)

[6]

Dec. 2018

- Q.19** Find maximum flow in the transport network using labeling procedure. Determine the corresponding min cut : (Refer similar example 8.9.3)

[7]

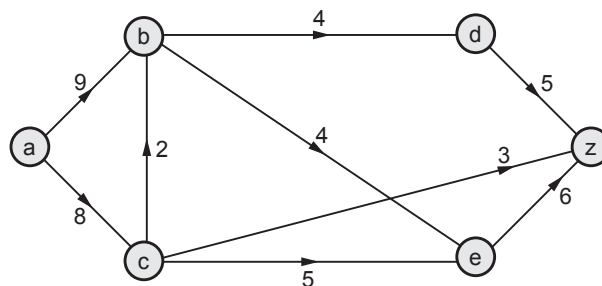


Fig. 8.12.26

- Q.20** Define the following terms :
(i) Level and height of a tree (Refer section 8.4.2)
(ii) Cut points (Refer section 8.3.3)
(iii) Eccentricity of a vertex (Refer section 8.3.1)

[6]

- Q.21** Find minimum spanning tree for the graph shown below using Kruskal's algorithm. (Refer section 8.7.4)

[7]

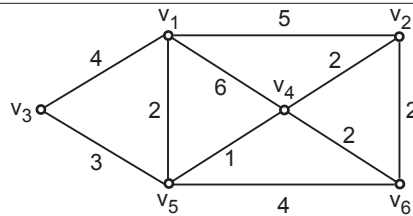


Fig. 8.12.27

- Q.22** Suppose data items A, B, C, D, E, F, G occur in the following frequencies respectively 10, 30, 5, 15, 20, 15, 5. Construct a Huffman code for the data. What is the minimum weighted path length? (Refer section 8.6.2) [6]

May 2019

- Q.23** Suppose data items A, B, C, D, E, F, G occurs in the following frequencies.

Data items	A	B	C	D	E	F	G
Weight	10	30	5	15	20	15	05

Construct a Huffman code for the data.

What is the minimum weighted path length. (Refer example 8.6.2) [6]

- Q.24** Using the labelling procedure to find maximum flow in the transport network in the following figure. Determine the corresponding minimum cut. [7]

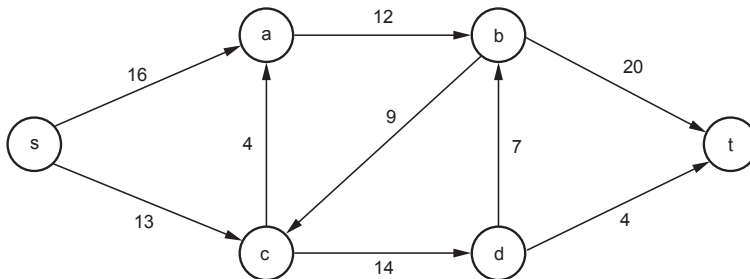


Fig. 8.12.28

Ans. : Assign the flow zero to each edge and $(-\infty)$ to the source s .

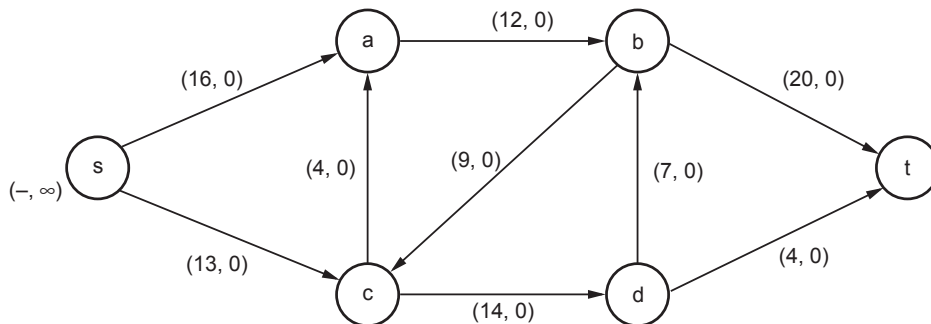


Fig. 8.12.29

Step 2 :

Since, vertices a and c are adjacent to source s. Therefore, label the vertices a and c as $(s^+, \Delta a)$ and $(s^+, \Delta c)$ respectively.

Where,

$$\Delta a = ((s-a) - f(s, a)) = 16$$

$$\Delta c = ((s-c) - f(s, c)) = 13$$

Label of a is $(s^+, 16)$ and label of c is $(s^+, 13)$

Now vertices b and d are adjacent to labelled vertices $(s^+, 16)$ and $(s^+, 13)$. Therefore, label the vertices b and d as $(a^+, \Delta b)$ and $(c^+, \Delta d)$.

Where,

$$\begin{aligned} \Delta b &= \min(\Delta a, c(a, b) - f(a, b)) \\ &= \min(16, 12 - 0) \\ &= 12 \end{aligned}$$

Similarly, $\Delta d = 13$

Label b is $(a^+, 12)$ and d is $(c^+, 13)$

Now t is the sink and adjacent to b and d, we can choose any vertex b or d to label the sink t.

Lets, choose vertex d, which is labelled as $(c^+, 13)$. The label of t will be $(d^+, \Delta t)$ where

$$\Delta t = \min(13, 4 - 0) = 4$$

Hence, label of t is $(d^+, 4)$. It is shown in the following diagram.

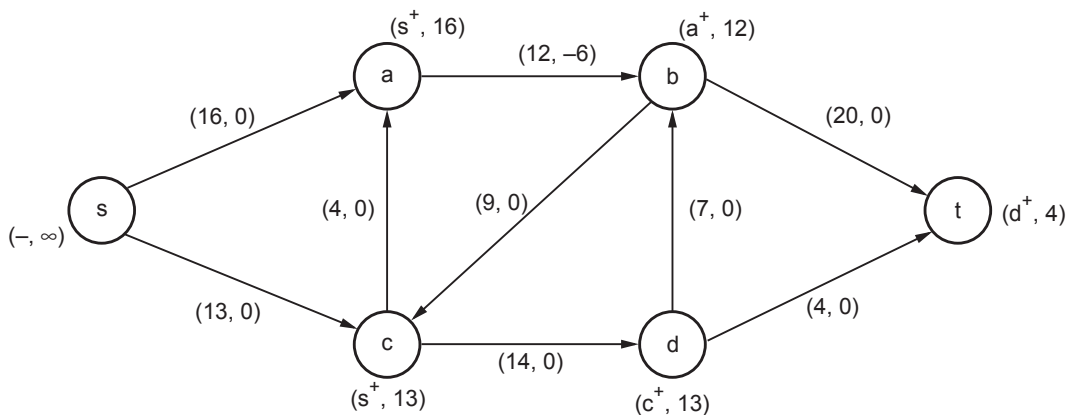


Fig. 8.12.30

Now adjust the flow in the edges

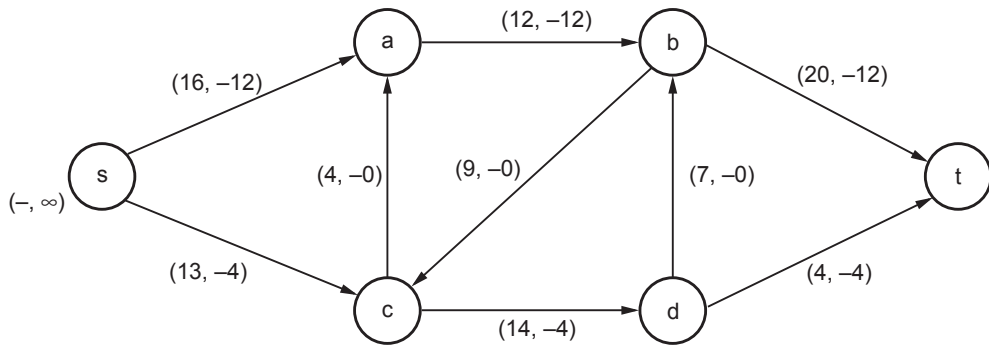


Fig. 8.12.30 (a)

Now adjust the flow in the edges

s-c, c-d, d-b and label the nodes

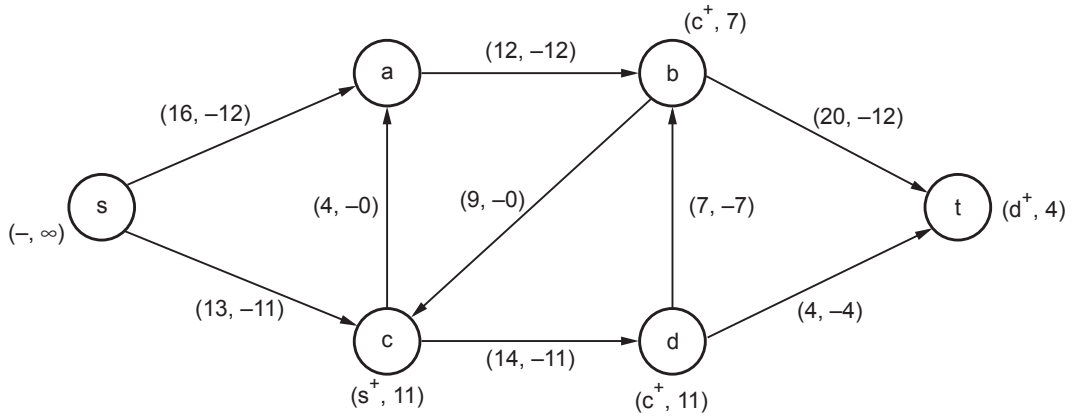


Fig. 8.12.30 (b)

Now adjust the flow in the edges b-t

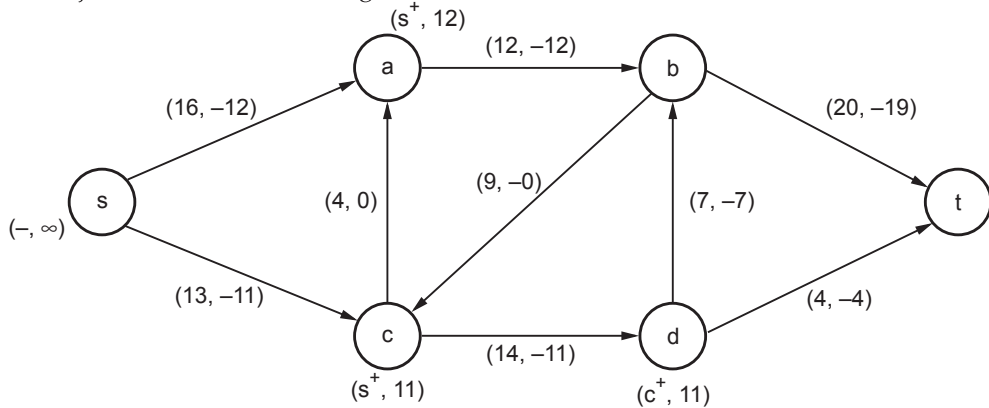


Fig. 8.12.30 (c)

We have to stop here.

P is set of labelled vertices

{s,a,c,d} and unlabelled vertices {b,t}

Capacity of cut is $c(a+b) + (d, b) + c(d, t) = 12 + 7 + 4 = 23$

Hence, maximum flow in the network = 23.

Q.25 Give the stepwise construction of minimum spanning tree using Prime algorithm for the following graph. Obtain the total cost of minimum spanning tree. [7]

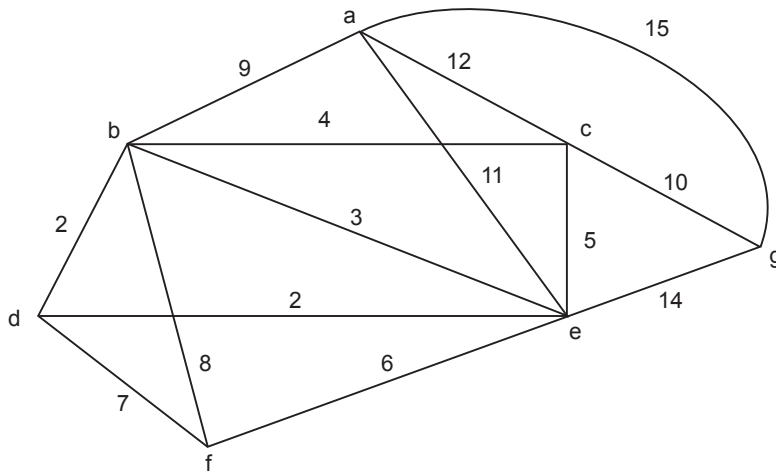


Fig. 8.12.31

Ans. :

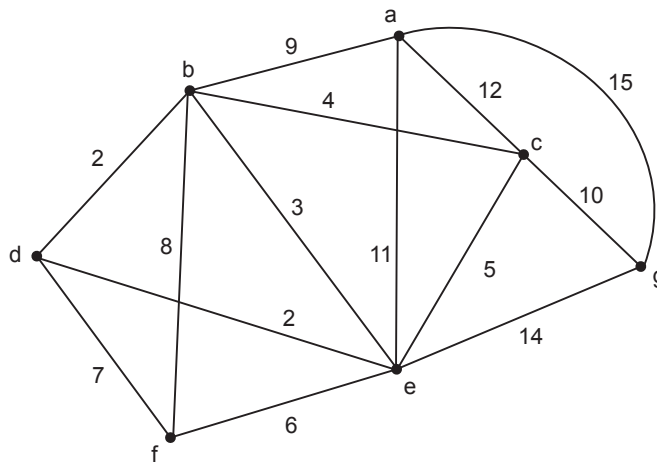


Fig. 8.12.31 (a)

i) Select $a \in G$ as a starting vertex

$$T = \{\{G\}, \phi\}$$

○ a
Fig. 8.12.31 (b)

ii) Select edge having minimum weight and one end is connected to a is $\{a, b\} = 9$.

$$T = \{\{a, b\}, e_1\}$$

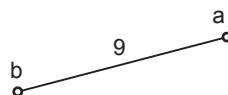


Fig. 8.12.31 (c)

iii) Select edge $\{b, d\} = 2$

$$T = \{\{a, b, d\}, \{e_1, e_2\}\}$$

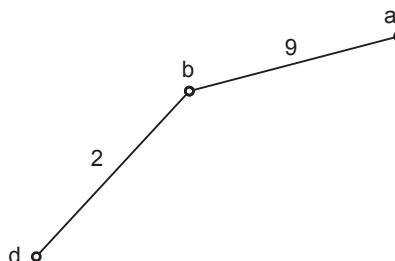


Fig. 8.12.31 (d)

iv) Select edge $\{d, e\} = 2$

$$T = \{\{a, b, d, e\}, \{e_1, e_2, e_3\}\}$$

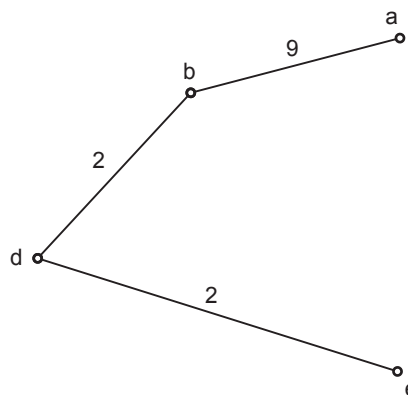


Fig. 8.12.31 (e)

v) Discard edge $\{b, e\} = 3$, since it creates cycle.

Select edge $\{b, c\} = 4$.

$$T = \{\{a, b, d, e, c\}, \{e_1, e_2, e_3, e_4\}\}$$

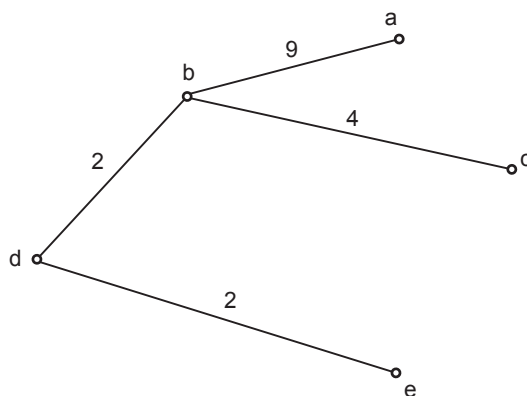


Fig. 8.12.31 (f)

vi) Discard edge $\{c,e\} = 5$, since it creates cycle. Select edge $\{e,f\} = 6$.

$$T = \{\{a,b,d,e,c,f\}, \{e_1, e_2, e_3, e_4, e_5\}\}$$

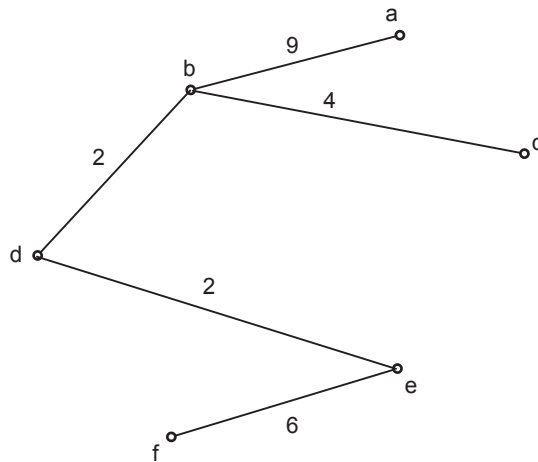


Fig. 8.12.31 (g)

vii) Discard edge $\{f,d\} = 7$, $\{f,b\} = 8$, since it creates cycle. Select edge $\{c,g\} = 10$.

$$T = \{\{a,b,d,e,c,f,g\}, \{e_1, e_2, e_3, e_4, e_5, e_6\}\}$$

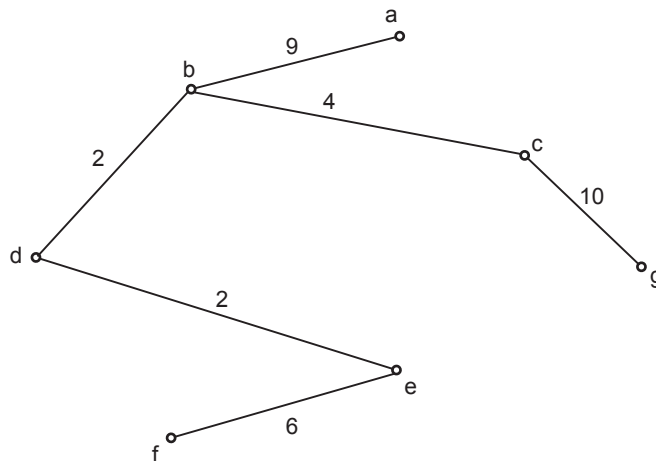


Fig. 8.12.31 (h)

This is a minimum spanning tree and cost of MST = $2 + 2 + 4 + 6 + 9 + 10 = 33$

Q.26 Define with example.

i) Level and height of tree (Refer section 8.4.2)

ii) Binary search tree (Refer section 8.6.4)

iii) Spanning tree. (Refer section 8.7)

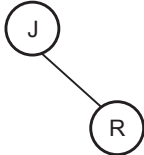
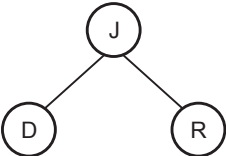
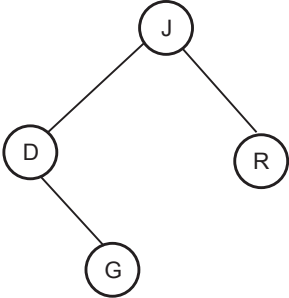
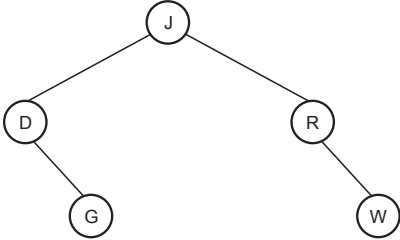
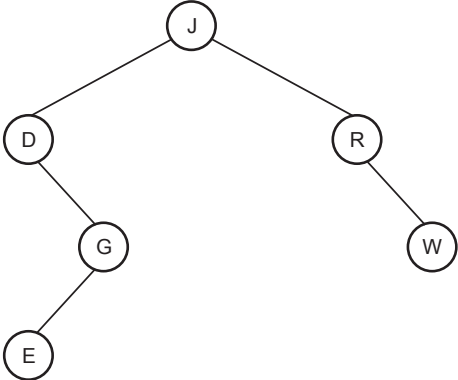
[6]

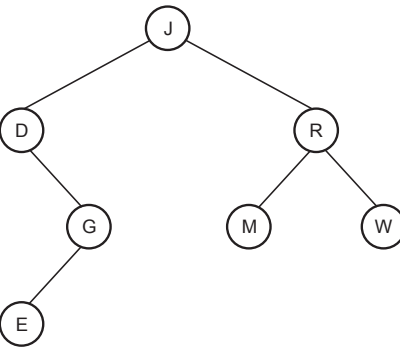
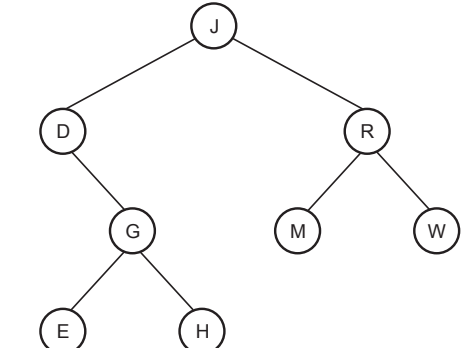
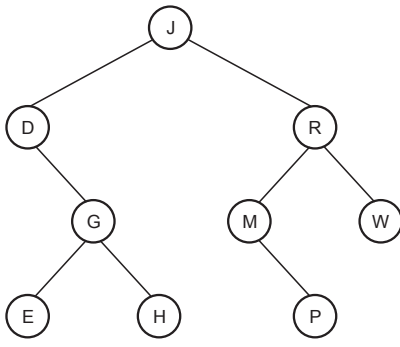
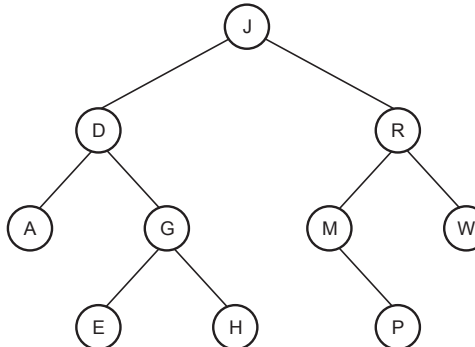
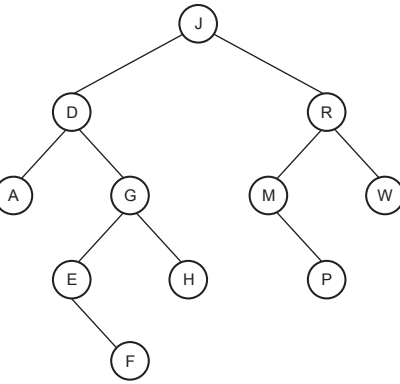
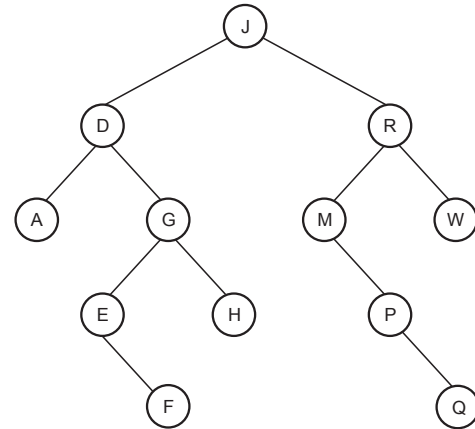
Dec. 2019

Q.27 Construct a binary search tree : J, R, D, G, W, E, M, H, P, A, F, Q.

[4]

Ans. : Given

i)	J 	ii)	R 
iii)	D 	iv)	G 
v)	W 	vi)	E 

vii)	M  <pre> graph TD J((J)) --- D((D)) J --- R((R)) D --- G((G)) G --- E((E)) R --- M((M)) R --- W((W)) </pre>	viii)	H  <pre> graph TD J((J)) --- D((D)) J --- R((R)) D --- G((G)) G --- E((E)) G --- H((H)) R --- M((M)) R --- W((W)) </pre>
ix)	P  <pre> graph TD J((J)) --- D((D)) J --- R((R)) D --- G((G)) G --- E((E)) G --- H((H)) R --- M((M)) R --- P((P)) </pre>	x)	A  <pre> graph TD J((J)) --- D((D)) J --- R((R)) D --- A((A)) D --- G((G)) G --- E((E)) G --- H((H)) R --- M((M)) R --- P((P)) </pre>
xi)	F  <pre> graph TD J((J)) --- D((D)) J --- R((R)) D --- A((A)) D --- G((G)) G --- E((E)) G --- H((H)) E --- F((F)) R --- M((M)) R --- P((P)) </pre>	xii)	Q  <pre> graph TD J((J)) --- D((D)) J --- R((R)) D --- A((A)) D --- G((G)) G --- E((E)) G --- H((H)) E --- F((F)) R --- M((M)) R --- P((P)) P --- Q((Q)) </pre> This is binary search tree

Q.28 Construct the binary tree with prefix codes representing :

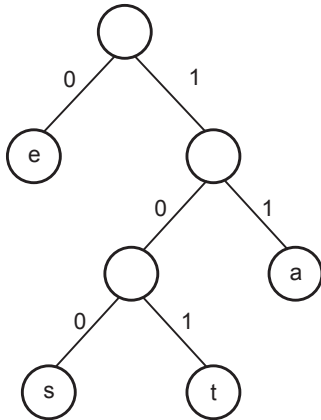
(i) $a : 11, e : 0, t : 101, s : 100$

(ii) $a : 1010, e : 0, t : 11, s : 1011, n : 1001, i : 10001$

[4]

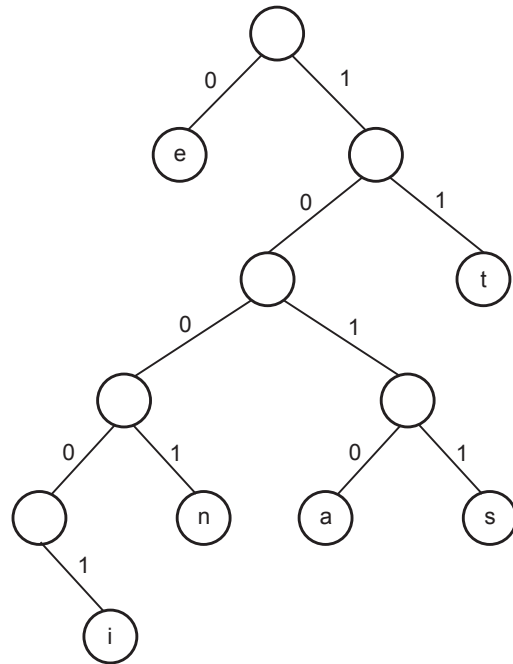
Ans. :

i)



Binary tree

ii)



Binary tree

Q.29 Give the stepwise construction of minimum spanning tree using Kruskal's algorithm for the following graph. Obtain the total cost of minimum spanning tree. [5]

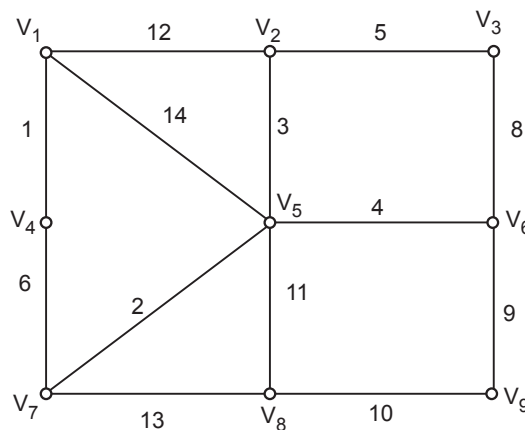


Fig. 8.12.32

Ans. :

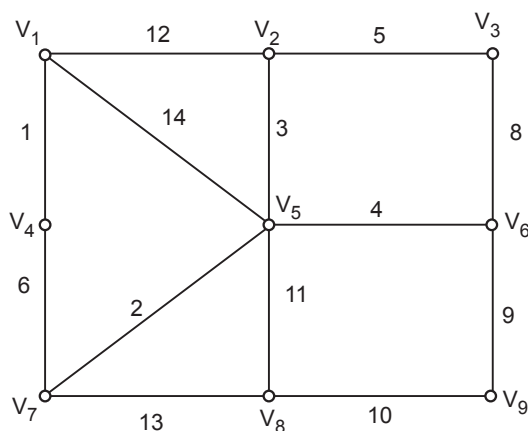
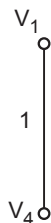


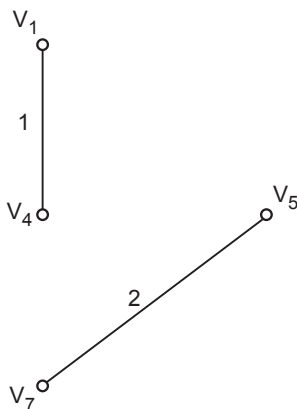
Fig. 8.12.32 (a)

Using Kruskal's algorithm minimum spanning tree is obtained as follows.

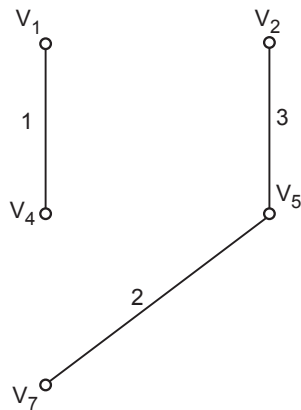
Step 1 : Select the edge ($V_1 - V_4$) with minimum weight = 1



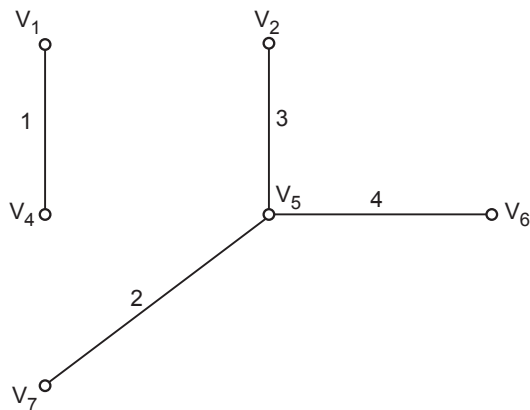
Step 2 : Select edge ($V_5 - V_7$) weight = 2



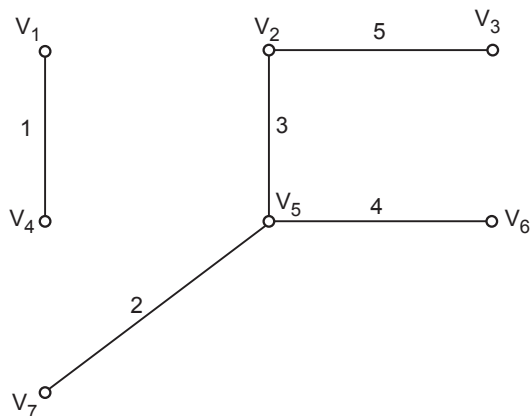
Step 3 : Select edge $(V_2 - V_5)$ weight = 3



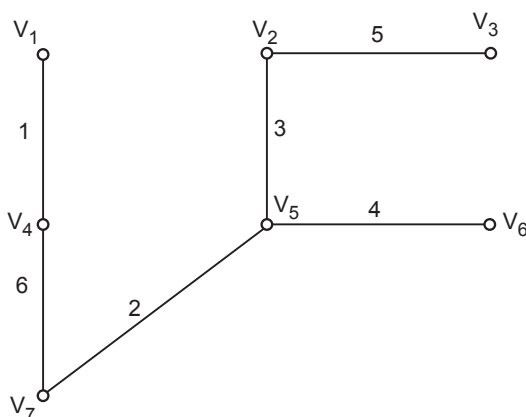
Step 4 : Select edge $(V_5 - V_6)$ weight = 4



Step 5 : Select edge $(V_2 - V_3)$ weight = 5

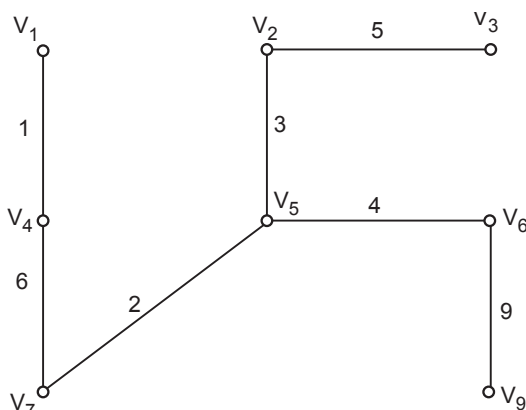


Step 6 : Select edge ($V_4 - V_7$) weight = 6

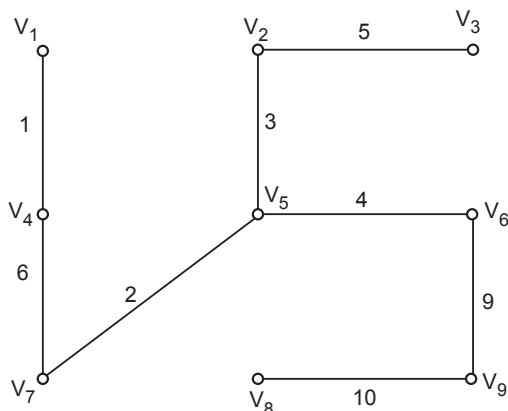


Step 7 : Select edge ($V_3 - V_6$) with weight = 8. But it will form the cycle so discard it.

Step 8 : Select edge ($V_6 - V_9$) with weight = 9

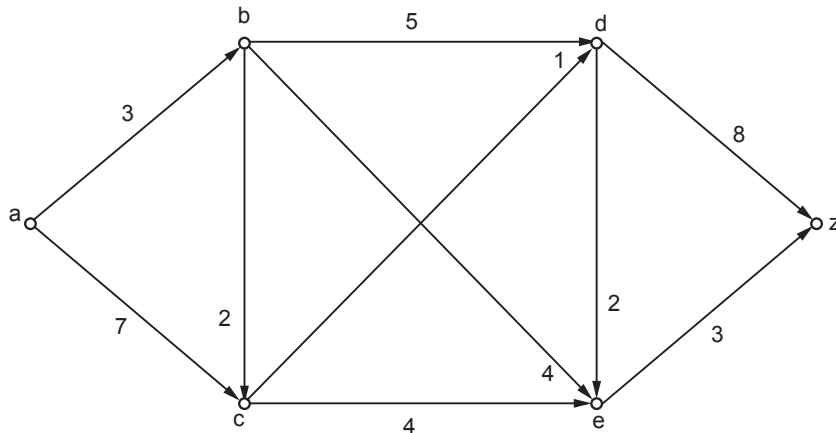


Step 9 : Select edge ($V_8 - V_9$) with weight = 10

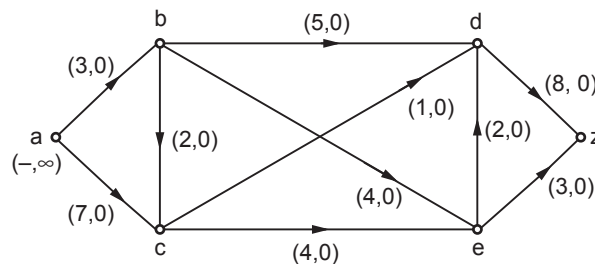


The obtained graph is minimum spanning tree of the given graph. The total cost is $(1 + 2 + 3 + 4 + 5 + 6 + 9 + 10) = 40$

Q.30 Using the labelling procedure to find maximum flow in the transport network in the following figure. Determine the corresponding minimum cut. [7]



Ans. : Step 1 : To find the maximum flow in the given transport network, assign the flow zero to each edge and label $(-, \infty)$ to the source a.



Step 2 : Since the vertices b and c are adjacent to the source a, therefore label the vertices b and c as $(a^+, \Delta b)$ and $(a^+, \Delta c)$ respectively.

$$\text{Where } \Delta b = ((a, b) - f(a, b)) = 3$$

$$\Delta c = ((a, c) - f(a, c)) = 7$$

$\therefore$ Label of b is $(a^+, 3)$ and label of c is $(a^+, 7)$

Now the vertices d and e are adjacent to labeled vertices b $(a^+, 3)$ and $(a^+, 7)$ therefore we label the vertices d and e as $(b^+, \cdot d)$ and $(c^+, \cdot e)$ where

$$\begin{aligned}
 \Delta d &= \min(\Delta b, c(b, d) - f(b, d)) \\
 &= \min(3, 5, -0) \\
 &= 3
 \end{aligned}$$

$$\begin{aligned}
 \text{Similarly } \Delta e &= \min(\Delta c, c(c, e) - f(c, e)) \\
 &= \min(7, 4, -0) \\
 &= 4
 \end{aligned}$$

$\therefore$ Label of d is $(b^+, 3)$ and label of e is $(c^+, 4)$

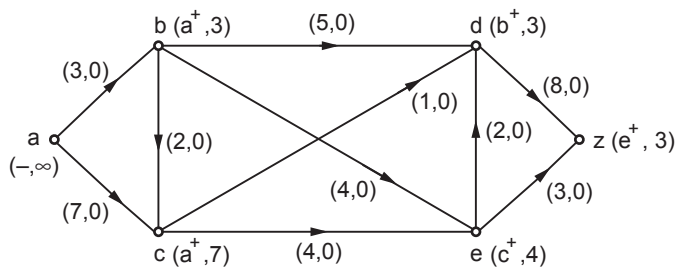
Now the sink z is adjacent to both the vertices d and e, we can choose any vertex d or e to label the sink z.

Let us choose the vertex e, which is labeled as $(c^+, 4)$

The label of z will be $(e^+, \Delta z)$ where

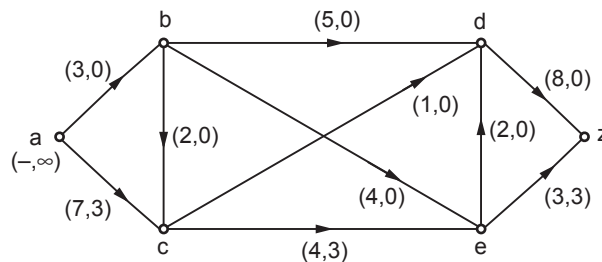
$$\begin{aligned}
 \Delta z &= \min(4, 3, -0) \\
 &= 3
 \end{aligned}$$

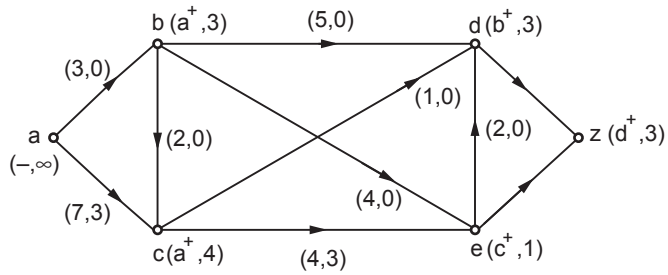
Hence label of z is $(e^+, 3)$



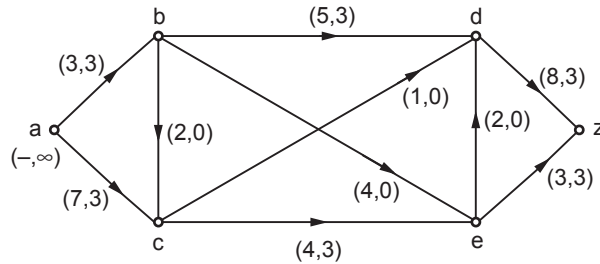
The labels of vertices can be shown as follows :

According to the label of the sink z, adjust the flow in the edges (e, z) , (c, e) and (a, c)

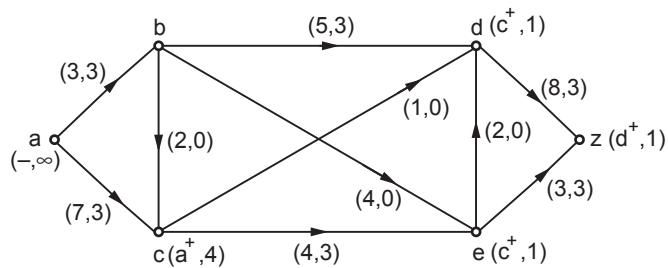




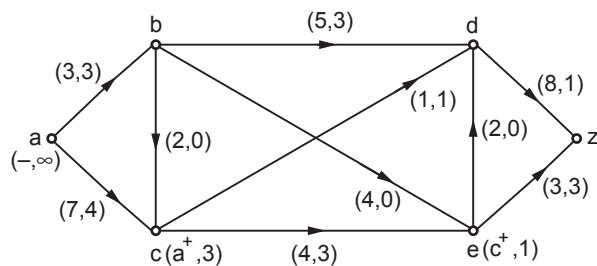
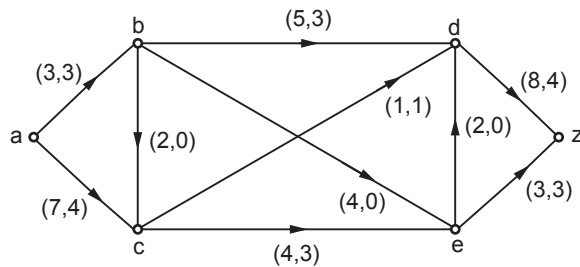
Repeat the step 2 in each pass, the new value of the flow is obtained.



After adjusting the flow we have



Step 4 :



Adjust the flow according to sink z label.

Since the vertex z (sink) cannot be labelled, we have to stop. The vertices b , d and z cannot be labelled because either the edges approaching to vertices are saturated or the edges in the opposite direction have flow zero. At this stage, we have minimum cut $(P, \bar{P})$.

P is the set of labelled vertices $P = \{a, c, e\}$. $\bar{P}$ is the set of unlabeled vertices, $\bar{P} = \{b, d, z\}$

The edges in cut $P \bar{P} \{(a, b), (c, d), (e, z)\}$ capacity of the cut is $c(a, b) + c(c, d) + c(e, z) = 3 + 1 + 3 = 7$.

Hence the maximum flow in the given transport network 7.

Q.31 Find fundamental cutsets and circuits of the following graph G_1 with respect to spanning tree T_1 [4]

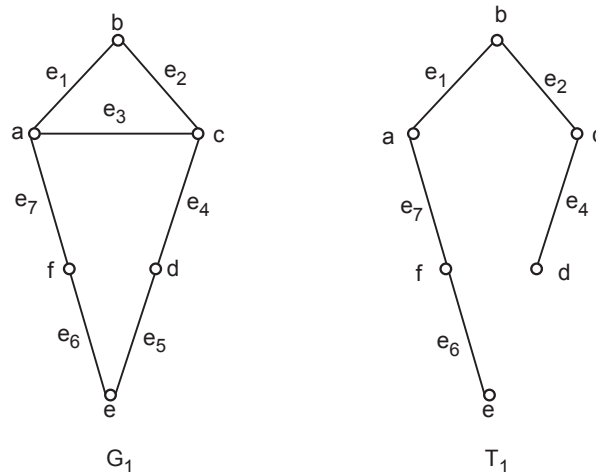


Fig. 8.12.33

Ans. :

Given spanning tree has 5 branches. These are e_1, e_2, e_4, e_6 and e_7

Chords of the T_1 are e_3, e_5

Fundamental circuits :

Consider the following table of chords and corresponding fundamental circuits

Chords	Corresponding fundamental circuits
e_3	$\{e_1, e_2, e_3\}$
e_5	$\{e_1, e_2, e_4, e_5, e_6, e_7\}$

Fundamental cut sets

Branches of T_1 are e_1, e_2, e_4, e_6, e_7

Branches	Corresponding fundamental cutset
e_1	$\{e_1, e_3, e_5\}$
e_2	$\{e_2, e_3, e_5\}$
e_4	$\{e_4, e_5\}$
e_4	$\{e_4, e_5\}$
e_7	$\{e_5, e_7\}$

Q.32 Define with example : Level and height of a tree. (Refer section 8.4.2)

[2]



UNIT - VI

9

Algebraic Structures and Coding Theory

Syllabus

The structure of algebra, Algebraic Systems, Semi Groups, Monoids, Groups, Homomorphism and Normal Subgroups, and Congruence relations, Rings, Integral Domains and Fields, Coding theory, Polynomial Rings and polynomial Codes, Galois Theory -Field Theory and Group Theory. Case studies : Cryptography used in world war - II.

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9.2	Algebraic Structures	Dec.-05, May-17,19 Marks 3
9.3	Special Algebraic Structures	Dec.-06, 13,17, May-18,19 · Marks 4
9.4	Group	Dec.-08, 09, 10, 12,17,18 May-14,17,19 · Marks 6
9.5	Modulo m	Dec.-11, May-08
9.6	Permutation Group	
9.7	Complexes and Subgroups	May-18 Marks 6
9.8	Cosets	
9.9	Order of an Element of a Group	
9.10	Cyclic Group	Dec.-18 Marks 6
9.11	Normal Subgroups	Dec.-16 Marks 5
9.12	Quotient Groups	Dec.-14, 15, May-15,17 · Marks 3
9.13	Homomorphism of Groups	Dec.-11, 12, 14, 15, May-15,17
9.14	Isomorphism of Groups	Dec.-10, 11, 12, 13,16, May-08,10,17,19 Marks 4
9.15	Rings, Integral Domains and Fields	Dec.-08, 10, 11, 12, 13, 14,16,17,18 May-14,17,18,19
9.16	Group Codes	Dec.-06, 11, 12, 15,18, May-18 Marks 5
9.17	Case Studies	
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9.1 Introduction

In this chapter, we shall deal with sets with additional structures, induced by one or more binary operations on the elements of the set. We will study some algebraic structures, groups, rings integral domains and fields.

9.2 Algebraic Structures

SPPU : Dec.-05, May-17,19

9.2.1 Binary Operation

I) Let A be any non empty set. A function $f : A \times A \rightarrow A$ is called the binary operation on the set A .

'*' is a binary operation on the set A iff $a * b \in A \forall a, b \in A$ and $a * b$ is unique.

II) An n -ary operation on a set $A \neq \emptyset$, is a function $f : A \times A \times \dots \times A$ (n times) $\rightarrow A$ i.e. $f : A^n \rightarrow A$

The n -ary operation is defined for each n -tuple $(a_1, a_2, \dots, a_n) \in A$ for $a_i \in A$

If $n = 1$ then f is called unary operation

If $n = 2$ then f is called binary operation

If $n = 3$ then f is called ternary operation

Examples

- 1) A function $f : \mathfrak{R} \rightarrow \mathfrak{R}$ such that $f(x) = x ; \forall x \in \mathfrak{R}$ then f is a unary operation.
- 2) A function $f : \mathfrak{R} \times \mathfrak{R} \rightarrow \mathfrak{R}$ such that $f(x, y) = x + y$ then f is a binary operation.
- 3) A function $f : \mathbb{Z} \times \mathbb{Z} \times \mathbb{Z} \rightarrow \mathbb{Z}$ such that $f(x, y, z) = x + y + z$. then f is a ternary operation.
- 4) If $A = \mathfrak{R}$ (or $\mathcal{C}$)

where $\mathfrak{R}$ = Real number set

$\mathcal{C}$ = Complex number set

then '+' '-' and ' $\times$ ' are binary operations as

$x \pm y$ and $x \cdot y \in \mathfrak{R}$ (or $\mathcal{C}$) ; $\forall x, y \in \mathfrak{R}$ (or $\mathcal{C}$)

5) For any $x, y \in \mathbb{N} = \{1, 2, 3, 4, \dots\}$, $x + y, x \times y \in \mathbb{N}$

$\therefore$ '+' and ' $\times$ ' are binary operations on $\mathbb{N}$

But '-' and $\div$ are not binary operations on $\mathbb{N}$

as $2 - 5 = -3 \notin \mathbb{N}$ and $\frac{2}{3} \notin \mathbb{N}$

- 6) Union, intersection and difference are binary operations in $P(A)$, for any non empty set A .

9.2.2 Properties of Binary Operations

I) Commutative Property

A binary operation '*' on A is said to be commutative if $a * b = b * a$, for all $a, b \in A$.

e.g. $x + y = y + x$ and $x \times y = y \times x$ for all $x, y \in \mathbb{R}$

$\therefore$ '+' and ' $\times$ ' are commutative binary operations on $\mathbb{R}$.

But '-' is not commutative on $\mathbb{R}$.

II) Associative Property :

A binary operation '*' on set A is said to be associative if $a * (b * c) = (a * b) * c$;
 $\forall a, b, c \in A$

e.g. '+' and ' $\times$ ' are associative on the set of real numbers '-' is not associative on $\mathbb{R}$.

III) Idempotent

A binary operation '*' on set A is said to be idempotent if $a * a = a$; for all $a \in A$

e.g.

- 1) 1 is idempotent element in $\mathbb{R}$ w.r.t. binary operation ' $\times$ '.
- 2) 0 is idempotent element in $\mathbb{R}$ w.r.t. '+'.
- 3) 1 is not idempotent element in $\mathbb{R}$ w.r.t. '+'.

Examples

Example 9.2.1 For each of the following, determine whether * is a binary operation.

- i) R is the set of real numbers and $a * b = ab$
- ii) z^+ is the set of positive integers and $a * b = a/b$.
- iii) On z^+ where $a * b = a - b$.
- iv) On R , where $a * b = \min \{a, b\}$
- v) On R , where $a * b = a \times |b|$
- vi) On z , where $a * b = a^b$.

Solution :

- i) Yes, since $f : R^2 \rightarrow R$ defined as $f(a, b) = ab$ is a function, with $a, b \in R$.
- ii) No, since $(a, b) \in z^+ \times z^+$ does not imply that $a * b = a/b \in z$
 $(1, 2) \in z^+ \times z^+$, but $1/2 \notin z^+$

- iii) No, since $(1, 2) \in \mathbb{Z}^+ \times \mathbb{Z}^+$ but
- iv) Yes, since, $*$ is a function with $\min\{a, b\} \in \mathbb{R}$
- v) Yes, since $*$ is a function with $a \times |b| \in \mathbb{R}$
- vi) No, since $2 * (-1) = 2^{-1} = \frac{1}{2} \notin \mathbb{Z}$

Example 9.2.2 Let $(A, *)$ be algebraic system where $*$ is a binary operation such that for any $a, b \in A$, $a * b = a$.

- i) Show that $*$ is an associative operation.
- ii) Can $*$ ever be a commutative operation ?

Solution :

$$\begin{aligned} \text{i) } a * (b * c) &= a * b = a \cdot (a * b) * c \\ &= a * c = a \end{aligned}$$

Hence $*$ is associative

- ii) $*$ is commutative only if $a * b = b * a$ i.e. $a = b$, for all $a, b \in A$. This is possible if A is the singleton set $\{a\}$ and $a * a = a$ i.e. $*$ is an idempotent operation on A .

Example 9.2.3 Determine whether or not following operations on the set of integers $\mathbb{Z}$ are associative.

- i) Division ii) Exponentiation

SPPU : Dec.-05, Marks 3

Solution :

- i) Division on the set of integers is not associative as
- $$(a/b)/c \neq a/(b/c)$$
- ii) Exponentiation on the set of integers is not associative as
- $$(a^b)^c \neq a^{(b^c)}$$

Example 9.2.4 Consider the binary operation $*$ defined on the set $A = \{a, b, c, d\}$ by the following table.

$*$	A	b	c	d
a	a	c	b	d
b	d	a	b	c
c	c	d	a	a
d	d	b	a	c

- Find i) $c * d$ and $d * c$ ii) $b * d$ and $d * b$
- iii) $a * (b * c)$ and $(a * b) * c$ iv) Is $*$ commutative, associative ?

Solution :

$$\begin{array}{ll}
 \text{i)} & c * d = a \qquad \qquad \qquad d * c = a \\
 \text{ii)} & b * d = c \qquad \qquad \qquad d * b = b \\
 \text{iii)} & b * c = b \qquad \qquad \qquad a * (b * c) = a * b = c \\
 & a * b = c \qquad \qquad \qquad \text{Hence } (a * b) * c = c * c = a
 \end{array}$$

iv) $*$ is not commutative, since $b * d \neq d * b$

$*$ is also not associative, since $a * (b * c) \neq (a * b) * c$

9.3 Special Algebraic Structures

SPPU : Dec.-06, 13,17, May-18,19

9.3.1 Groupoid

A non empty set k with binary operation ' $*$ ' is called groupoid if the binary operation ' $*$ ' satisfies $\forall a, b \in G, a * b \in G$

In other words, every algebraic structure is groupoid.

e.g. $(\mathbb{R}, +)$, $(\mathbb{R}, -)$, $(\mathbb{Z}, +)$, $(\mathbb{Z}, \times)$ are groupoids.

9.3.2 Semi Group

A non empty set G with binary operation ' $*$ ' is called a semigroup if it satisfies the following properties.

$$a * (b * c) = (a * b) * c ; \forall a, b, c \in G$$

i.e. ' $*$ ' is associative in G

A semigroup is said to be commutative if $*$ is commutative.

e.g. i) $(\mathbb{R}, +)$, $(\mathbb{N}, +)$, $(\mathbb{Z}, +)$ are commutative semigroups.

ii) $(\mathbb{R}, \times)$, $(\mathbb{Z}, \times)$ are commutative semigroups.

iii) $(\mathbb{Z}, -)$ is not a semigroup as ' $-$ ' is not associative.

9.3.3 Monoid

SPPU : Dec.-13

Let G be a non empty set and $*$ be a binary operation on G . $(G, *)$ is called monoid if.

i) Associative property : $a * (b * c) = (a * b) * c ; \forall a, b, c \in G$

ii) Existence of identity : $\exists$ an element $e \in G$ such that $e * a = a * e = a ; \forall a \in G$

The element ' e ' is called the identity element.

e.g. 1) $(\mathbb{R}, +)$ is a monoid as $a + b \in \mathbb{R}, \forall a, b \in \mathbb{R}$

i.e. $(\mathfrak{R}, +)$ is closed

$$\text{and } a + (b + c) = (a + b) + c$$

$$\text{and } 0 + a = a + 0 = a, \quad \forall a \in \mathfrak{R}$$

0 is the identity element in $\mathfrak{R}$ w.r.t. '+'.

Example 2

$(\mathbb{Z}, +)$, $(\mathbb{Z}, \times)$, $(\mathbb{Z}, +)$, $(\mathbb{Z}, \times)$, $(\mathbb{Q}, +)$ are monoids.

Example 3

$(\mathbb{N}, \times)$ is a monoid but $(\mathbb{N}, +)$ is not monoid as $0 \notin \mathbb{N}$

Example 9.3.1 Show that the algebraic system $(A, +)$ is a monoid, where A is the set of integers and '+' is a binary operation giving addition of two integers.

SPPU : Dec.-06, Marks 4

Solution : Let A be the set of all integers and '+' defined on A .

i) Closure property : $a + b \in A$; $\forall a, b, \in A$ as $a + b$ is an integer.

ii) Associative property :

$$a + (b + c) = (a + b) + c ; \forall a, b, c \in A$$

iii) Existence of identity element :

For any $a \in A$, $\exists 0 \in A$ such that

$$a + 0 = 0 + a = a$$

Therefore $(A, +)$ is a monoid.

9.4 Group

SPPU : Dec.-08, 09, 10, 12,17,18, May-14,17,19, Marks 4

Let G be a non empty set equipped with a binary operation '*'. $(G, *)$ is called a group if it satisfies the following postulates or axioms.

i) Associativity : $a * (b * c) = (a * b) * c$; $\forall a, b, c \in G$

ii) Existence of the identity : For any $a \in G$, $\exists e \in G$

$$\text{s.t.} \quad a * e = e * a = a;$$

An element e is called the identity element in $(G, *)$

iii) Existence of the inverse :

For all $a \in G$, $\exists b \in G$ such that

$$a * b = b * a = e$$

Then b is called the inverse of a in $(G, *)$

$$\therefore b = a^{-1}$$

9.4.1 Abelian Group or Commutative Group

A group $(G, *)$ is called an Abelian group if

$$a * b = b * a \quad ; \quad \forall a, b \in G.$$

i.e. $*$ is commutative in $(G, *)$

9.4.2 Properties of Group

SPPU : Dec.10, Marks 2

I) The identity element in a group is unique.

Suppose e_1 and e_2 are two identity elements in group G .

We have, $e_1 e_2 = e_1$; if e_2 is identity element in G .

and $e_1 e_2 = e_2$; If e_1 is identity element in G .

$\Rightarrow e_1 e_2 = e_1 = e_2$. Hence the identity element in group G is unique.

II) The inverse of each element in group G is unique.

SPPU : Dec.-10, Marks 4

Let a be any element of a group G and let e be identity element in group G .

Suppose b and c are two inverses of a in G .

$$\therefore ba = ab = e \text{ and } ac = ca = e$$

$$\text{We have, } b = be = b(ac)$$

$$b = (ba) c$$

$$b = ec$$

$$b = c$$

Hence, the inverse of each element is unique.

III) The inverse of an inverse of the element is the original element. i.e. If the inverse of a is a^{-1} then $(a^{-1})^{-1} = a$.

Let $e \in G$ be the identity element of the group G .

Let $a \in G$.

$$\text{We have, } a^{-1}a = e$$

$$[(a^{-1})^{-1} (a^{-1})]a = (a^{-1})^{-1} e \quad \dots \text{ (Multiplying by } (a^{-1})^{-1} \text{)}$$

$$[(a^{-1})^{-1} (a^{-1})] a = (a^{-1})^{-1} \quad \dots (\because \text{Associativity of } e \text{ identity})$$

$$e a = (a^{-1})^{-1}$$

$$\Rightarrow a = (a^{-1})^{-1} \quad \text{Hence, the proof}$$

IV) Prove that the inverse of the product of two elements of a group G is the product of the inverses taken in reverse order. i.e. $(ab)^{-1} = b^{-1} a^{-1}$, $\forall a, b \in G$.

Let a^{-1} and b^{-1} be the inverses of a and b in a group G respectively. Let e be the identity in G . Then $a^{-1} a = a a^{-1} = e$ and $b^{-1} b = b b^{-1} = e$

$$\begin{aligned}
 \text{Consider, } (ab) (b^{-1} a^{-1}) &= a (bb^{-1}) a^{-1} && (\because \text{Associativity}) \\
 &= a (e) a^{-1} && (\because bb^{-1} = e) \\
 &= (ae) a^{-1} \\
 &= aa^{-1} = e && \dots (1)
 \end{aligned}$$

$$\begin{aligned}
 \text{Similarly, } (b^{-1} a^{-1}) (ab) &= b^{-1} (a^{-1} a) b \\
 &= b^{-1} (e) b \\
 &= b^{-1} b = e && \dots (2)
 \end{aligned}$$

From equation (1) and equation (2), $(ab) (b^{-1} a^{-1}) = (b^{-1} a^{-1}) (ab)$

$b^{-1} a^{-1}$ is the inverse of ab

$$(ab)^{-1} = b^{-1} a^{-1}$$

V) Prove that the cancellation laws hold in a group. i.e. If $a, b, c \in G$ then $ab = ac \Rightarrow b = c$ and $ba = ca \Rightarrow b = c$.

Let a be any element in G and e be the identity element of a group G .

Now we have, $ab = ac$

Permuting by a^{-1} we get,

$$\begin{aligned}
 a^{-1} (ab) &= a^{-1} (ac) \\
 (a^{-1} a) b &= (a^{-1} a) c && \dots (\because \text{Associativity}) \\
 eb &= ec && \dots (\because aa^{-1} = e) \\
 b &= c
 \end{aligned}$$

Similarly, $ba = ca$

$$\Rightarrow (ba) a^{-1} = (ca) a^{-1} \Rightarrow be = ce$$

$$\Rightarrow b = c \quad \text{Hence, the proof}$$

VI) If a, b are any elements of a group G then equation $ax = b$ and $ya = b$ have unique solutions in G .

Let $a \in G$

$$\therefore \exists a^{-1} \in G \text{ such that } aa^{-1} = a^{-1}a = e$$

$$a, a^{-1} \in G \text{ and } b \in G \Rightarrow a^{-1}b \in G.$$

Now substituting $a^{-1}b$ for x in the L.H.S. given equation, we get,

$$ax = a(a^{-1}b) = (aa^{-1})b = eb = b$$

Thus, $x = a^{-1}b$ is the solution of $ax = b$

Let us suppose that x_1 and x_2 are two solutions of $ax = b$.

$$\therefore ax_1 = b \quad \text{and} \quad ax_2 = b$$

$$\Rightarrow b = ax_1 = ax_2 \Rightarrow x_1 = x_2$$

Hence solution is unique.

Similarly prove for $ya = b$.

Examples

Example 9.4.1 Let (A, x) be a monoid such that for every $x \in A$, $x * x = e$ where e is the identity element. Show that $(A, *)$ is an abelian group.

Solution : Given that, $(A, *)$ is a monoid. Therefore it satisfies closure property, associativity and the existence of identity.

We have $x * x = e$, $\forall x \in A$

$$x = x^{-1}, \forall x \in A$$

Thus inverse exists for all $x \in A$.

$\therefore (A, *)$ is a group.

Consider,

$$(a * b) * (b * a); a, b, \in A = a * (b * b) * a$$

... ($\because *$ is associative)

$$= a * e * a$$

... ($\because b * b = e$)

$$= a * a = e$$

$$\text{and } (b * a) * (a * b) = b * (a * a) * b$$

$$= b * e * b = b * b = e$$

Thus $b * a$ is the inverse of $a * b$

$$\text{But } x = x^{-1}, \forall x \text{ in } A$$

$$b * a = a * b$$

Thus, $(A, *)$ is an abelian group.

Example 9.4.2 Show that the set $G = \{a + b\sqrt{2} / a, b \in \mathbb{Q}\}$ is a group with respect to addition.

Solution : Given that, $G = \{a + b\sqrt{2} \mid a, b \in \mathbb{Q}\}$

(i) Closure property : Let $x, y \in G$.

$$x = a + b\sqrt{2}, y = c + d\sqrt{2} \quad a, b, c, d \in \mathbb{Q}.$$

$$\text{Consider, } x + y = (a + b\sqrt{2}) + (c + d\sqrt{2})$$

$$= (a + c) + (b + d)\sqrt{2}, \quad a + c \text{ and } b + d \in \mathbb{Q}$$

$$\Rightarrow x + y \in G$$

$\therefore G$ is closed w.r.t. $+$

(ii) Associativity : All elements of G are real numbers and associativity holds in $\mathbb{R}$.

$\therefore$ Associativity holds in G .

(iii) Existence of the identity : We have, $0 \in \mathbb{Q}$

$$0 + 0\sqrt{2} = 0 \in G$$

Consider

$$(0 + 0\sqrt{2}) + (a + b\sqrt{2}) = (0 + a) + (0 + b)\sqrt{2} = a + b\sqrt{2}$$

$$\text{and } (a + b\sqrt{2}) + (0 + 0\sqrt{2}) = a + b\sqrt{2}$$

$\therefore 0$ is the additive identity element in G .

(iv) Existence of the inverse : We have a, b and $-a, -b \in \mathbb{Q}$.

$$\therefore a + b\sqrt{2} \text{ and } -a - b\sqrt{2} \in G.$$

$$\text{Consider, } (a + b\sqrt{2}) + (-a - b\sqrt{2}) = (a - a) + (b - b)\sqrt{2}$$

$$= 0 + 0\sqrt{2} = 0$$

$$\text{and } (-a - b\sqrt{2}) + (a + b\sqrt{2}) = (-a + a) + (-b + b)\sqrt{2}$$

$$= 0 + 0\sqrt{2} = 0$$

$\therefore -a - b\sqrt{2}$ is the inverse of $a + b\sqrt{2}$. Hence G is a group with respect to addition.

Example 9.4.3 If set Q_1 of all rational numbers other than 1 with $a * b = a + b - ab$. Show that $(G, *)$ is a group.

SPPU : Dec.-09,18, Marks 4

Solution : We have, $a * b = a + b - ab, \forall a, b \in G$.

(i) Closure property : Let $a, b \in Q_1, a \neq 1, b \neq 1 \therefore ab \neq 1$.

$$\therefore a * b = a + b - ab \neq 1 \text{ and } a * b \in Q_1$$

Q_1 is closed w.r.t. $*$

(ii) Associativity : Let $a, b, c \in Q_1$.

$$\begin{aligned}
(a * b) * c &= (a + b - ab) * c \\
&= (a + b - ab + c - (a + b - ab) c) \\
&= a + b - ab + c - ac - bc - abc \quad \dots (1)
\end{aligned}$$

$$\begin{aligned}
a * (b * c) &= a * (b + c - bc) \\
&= a + b + c - bc \\
&= a (b + c - bc) \\
&= a + b + c - bc - ab - ac + abc \quad \dots (2)
\end{aligned}$$

∴ From equation (1) and equation (2)

$$(a * b) * c = a * (b * c)$$

∴ * is associative

(iii) Existence of the identity : Let e be the identity element in Q_1

$$a * e = a$$

$$\Rightarrow a + e = ae = a$$

$$\Rightarrow e - ae = 0$$

$$\Rightarrow e(1 - a) = 0 \quad (\text{But } a \neq 1)$$

$$\Rightarrow e = 0$$

∴ 0 is the identity element.

(iv) Existence of the inverse : Let $a \in Q_1$, $a \neq 1$.

Suppose b $\in Q_1$ is the inverse of a.

$$\therefore a * b = e$$

$$\Rightarrow a + b - ab = 0$$

$$\Rightarrow a + b(1 - a) = 0$$

$$b(1 - a) = -a$$

$$\Rightarrow b = \frac{-a}{1-a} = \frac{a}{a-1} \neq 1 \text{ and } b \in Q_1.$$

∴ The inverse exist for all a in Q_1 .

Thus, $(Q_1, *)$ is a group.

Example 9.4.4 If $S = \{(a,b) \mid a \neq 0, a, b, \in \mathbb{R}\}$ and $(a, b) * (c, d) = (ac, bc + d)$ then show that G is a group but not abelian group w.r.t.*

Solution : (i) Closure property : Let $(a, b), (c, d) \in S$; $a \neq 0$; $c \neq 0$

$$\therefore ac \neq 0.$$

$$(a, b) * (c, d) = (ac, bc + d) \in S$$

(ii) Associativity : Let $(a, b), (c, d)$ and $(e, f) \in S$

Consider

$$\begin{aligned} [(a, b) * (c, d)] * (e, f) &= [ac, bc + d] * (e, f) \\ &= ([ac] e, [bc + d] e + f) \\ &= (ace, bce + de + f) \end{aligned} \quad \dots (1)$$

$$\begin{aligned} \text{and } (a, b) * [(c, d) * (e, f)] &= (a, b) * [ce, de + f] \\ &= [ace, b(ce) + de + f] \\ &= (ace, bce + de + f) \end{aligned} \quad \dots (2)$$

From equation (1) and equation (2) $*$ is associative operation.

(iii) Existence of the identity : Let $(a, b) \in S$ and $(x, y) \in S$, $x \neq 0$

$$\text{Consider, } (a, b) * (x, y) = (a, b)$$

$$(ax, bx + y) = (a, b)$$

$$\Rightarrow ax = a \text{ and } bx + y = b$$

$$\Rightarrow x = 1 \text{ and } b + y = b$$

$$\Rightarrow y = 0$$

$$\text{Similarly, } (x, y) * (a, b) = (a, b)$$

$(1, 0)$ is the identity element in S .

(iv) Existence of the inverse : Let (a, b) and $(c, d) \in S$.

$$(a, b) * (c, d) = (1, 0)$$

$$(ac, bc + d) = (1, 0)$$

$$\Rightarrow ac = 1$$

$$\text{and } bc + d = 0$$

$$\therefore c = \frac{1}{a}$$

$$\text{and } d = -bc = -\frac{b}{a} \text{ and } c \neq 0$$

Thus $\left(\frac{1}{a}, -\frac{b}{a}\right)$ is the inverse of (a, b) in S .

Thus $(S, *)$ is a group.

Consider, $(a, b) * (c, d) = (ac, bc + d)$

and $(c, d) * (a, b) = (ca, da + b)$

$\Rightarrow (a, b) * (c, d) \neq (c, d) * (a, b)$

Thus, $(S, *)$ is not an abelian group.

Example 9.4.5 If $G = \left\{ \begin{bmatrix} a & 0 \\ 0 & 0 \end{bmatrix} / a \text{ is non zero real number} \right\}$ show that G is an abelian group w.r.t. matrix multiplication.

Solution : (i) **Closure property :** Let $A = \begin{bmatrix} a & 0 \\ 0 & 0 \end{bmatrix}$, $B = \begin{bmatrix} b & 0 \\ 0 & 0 \end{bmatrix}$ be any two elements of G where a, b are non-zero real numbers.

$$AB = \begin{bmatrix} ab & 0 \\ 0 & 0 \end{bmatrix}, ab \text{ is non zero real number.}$$

$\therefore AB \in G$

$\therefore G$ is closed w.r.t. multiplication.

(ii) **Associativity :** We know that matrix multiplication is associative.

(iii) **Existence of the identity :** Let $E = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix} \in G$

$$\text{Consider } AE = \begin{bmatrix} a & 0 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix} = \begin{bmatrix} a & 0 \\ 0 & 0 \end{bmatrix} = A$$

$$\text{and } EA = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} a & 0 \\ 0 & 0 \end{bmatrix} = \begin{bmatrix} a & 0 \\ 0 & 0 \end{bmatrix} = A$$

$\therefore E = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}$ is the identity element w.r.t. matrix multiplication.

(iv) **Existence of the inverse :**

$$\text{Let } A = \begin{bmatrix} a & 0 \\ 0 & 0 \end{bmatrix} \text{ and } B = \begin{bmatrix} b & 0 \\ 0 & 0 \end{bmatrix} \in G$$

$$\begin{aligned}
 \text{Consider } AB &= \begin{bmatrix} a & 0 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} b & 0 \\ 0 & 0 \end{bmatrix} \\
 &= \begin{bmatrix} ab & 0 \\ 0 & 0 \end{bmatrix}, ab \neq 0 \\
 &= \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}
 \end{aligned}$$

$$\Rightarrow ab = 1 \Rightarrow b = \frac{1}{a}$$

$$B = \begin{bmatrix} \frac{1}{a} & 0 \\ 0 & 0 \end{bmatrix} \in G$$

$$\text{And } AB = BA = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}$$

$\therefore B$ is the inverse of A in G .

(v) Commutativity :

$$AB = \begin{bmatrix} a & 0 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} b & 0 \\ 0 & 0 \end{bmatrix} = \begin{bmatrix} ab & 0 \\ 0 & 0 \end{bmatrix}$$

$$BA = \begin{bmatrix} b & 0 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} a & 0 \\ 0 & 0 \end{bmatrix} = \begin{bmatrix} ba & 0 \\ 0 & 0 \end{bmatrix}$$

$$\Rightarrow AB = BA$$

Hence G is an abelian group.

Example 9.4.6 If $G = \left\{ \begin{bmatrix} x & x \\ x & x \end{bmatrix} \middle/ x \text{ is non zero real number} \right\}$. Show that G is an abelian group w.r.t. matrix multiplication.

Solution : (i) Closure property :

$$\text{Let } A = \begin{bmatrix} x & x \\ x & x \end{bmatrix} \text{ and } B = \begin{bmatrix} y & y \\ y & y \end{bmatrix} \in G$$

$$\text{then } AB = \begin{bmatrix} x & x \\ x & x \end{bmatrix} \begin{bmatrix} y & y \\ y & y \end{bmatrix} = \begin{bmatrix} 2xy & 2xy \\ 2xy & 2xy \end{bmatrix} \in G$$

(ii) Associativity : We know that any matrix multiplication is associative.

(iii) Existence of the identity : Let $E = \begin{bmatrix} e & e \\ e & e \end{bmatrix}$ such that

$$AE = A \Rightarrow$$

$$\begin{bmatrix} x & x \\ x & x \end{bmatrix} \begin{bmatrix} e & e \\ e & e \end{bmatrix} = \begin{bmatrix} e & e \\ e & e \end{bmatrix}$$

$$\Rightarrow \begin{bmatrix} 2xe & 2xe \\ 2xe & 2xe \end{bmatrix} = \begin{bmatrix} e & e \\ e & e \end{bmatrix}$$

$$\Rightarrow 2xe = e$$

$$\Rightarrow 2x = 1$$

$$\Rightarrow x = \frac{1}{2}$$

$$\therefore E = \begin{bmatrix} \frac{1}{2} & \frac{1}{2} \\ \frac{1}{2} & \frac{1}{2} \end{bmatrix} \text{ is the identity element of } G.$$

(iv) Existence of the inverse : Let $A = \begin{bmatrix} x & x \\ x & x \end{bmatrix} \in G$ and

$$B = \begin{bmatrix} y & y \\ y & y \end{bmatrix} \in G \text{ such that } AB = E$$

$$\Rightarrow \begin{bmatrix} 2xy & 2xy \\ 2xy & 2xy \end{bmatrix} = \begin{bmatrix} \frac{1}{2} & \frac{1}{2} \\ \frac{1}{2} & \frac{1}{2} \end{bmatrix}$$

$$\Rightarrow 2xy = \frac{1}{2}$$

$$\Rightarrow xy = \frac{1}{4}$$

$$\Rightarrow y = \frac{1}{4x}$$

$$A^{-1} = \begin{bmatrix} \frac{1}{4x} & \frac{1}{4x} \\ \frac{1}{4x} & \frac{1}{4x} \end{bmatrix}$$

(v) Commutative property :

We have, $AB = BA = \begin{bmatrix} 2xy & 2xy \\ 2xy & 2xy \end{bmatrix} \in G$

Thus G is an abelian group w.r.t. matrix multiplication.

Example 9.4.7 If $G = \left\{ \begin{bmatrix} \cos \alpha & -\sin \alpha \\ \sin \alpha & \cos \alpha \end{bmatrix} / \alpha \text{ is non zero real number} \right\}$, then G is a group w.r.t. matrix multiplication.

Solution : (i) Closure property :

Let A and $B \in G$

$$A = \begin{bmatrix} \cos \alpha & -\sin \alpha \\ \sin \alpha & \cos \alpha \end{bmatrix} \text{ and } B = \begin{bmatrix} \cos \beta & -\sin \beta \\ \sin \beta & \cos \beta \end{bmatrix}$$

$$AB = \begin{bmatrix} \cos(\alpha + \beta) & -\sin(\alpha + \beta) \\ \sin(\alpha + \beta) & \cos(\alpha + \beta) \end{bmatrix} \in G$$

(ii) Matrix multiplication is associative.

(iii) Existence of the identity : As $0 \in \mathbb{R}$

$$E = \begin{bmatrix} \cos 0 & -\sin 0 \\ \sin 0 & \cos 0 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

$$AE = EA = E$$

$$E = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \text{ is the identity element in } G.$$

(iv) Existence of the inverse :

Let $A \in G$ and $A = \begin{bmatrix} \cos \alpha & -\sin \alpha \\ \sin \alpha & \cos \alpha \end{bmatrix}$

$$B = \begin{bmatrix} \cos \alpha & \sin \alpha \\ -\sin \alpha & \cos \alpha \end{bmatrix} \in G$$

Consider $AB = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = BA$

$\Rightarrow B$ is the inverse of A in G .

Thus G is a group w.r.t. matrix multiplication.

Example 9.4.8 Let G be the set of all non-zero real numbers and let $a * b = \frac{ab}{2}$. Show that $(G, *)$ is an abelian group.

SPPU : Dec.-08, May-17, Marks 6

Solution : (i) Closure property : Let $a, b \in G$.

$$a * b = \frac{ab}{2} \in G \text{ as } ab \neq 0$$

(ii) Associativity : Let $a, b, c \in G$

$$\text{Consider } a * (b * c) = a * \left(\frac{bc}{2} \right) = \frac{a(bc)}{4} = \frac{abc}{4}$$

$$(a * b) * c = \left(\frac{ab}{2} \right) * c = \frac{(ab)c}{4} = \frac{abc}{4}$$

$\Rightarrow *$ is associative in G .

(iii) Existence of the identity : Let $a \in G$ and $\exists e$ such that

$$a * e = \frac{ae}{2} = a$$

$$\Rightarrow ae = 2a$$

$$\Rightarrow e = 2$$

$\therefore 2$ is the identity element in G .

(iv) Existence of the inverse : Let $a \in G$ and $b \in G$ such that $a * b = e = 2$.

$$\Rightarrow \frac{ab}{2} = 2$$

$$\Rightarrow ab = 4$$

$$\Rightarrow b = \frac{4}{a}$$

$\therefore$ The inverse of a is $\frac{4}{a}, \forall a \in G$.

(v) Commutativity : Let $a, b \in G$

$$a * b = \frac{ab}{2}$$

$$\text{and } b * a = \frac{ba}{2} = \frac{ab}{2}$$

$\Rightarrow *$ is commutative

Thus, $(G, *)$ is an abelian group.

Example 9.4.9 Show that the four fourth roots of unity namely $1, -1, i, -i$ form a group w.r.t. multiplication.

Solution : Let $G = \{ 1, -1, i, -i \}$. Consider the following table.

	1	-1	i	-i
1	1	-1	i	-i
-1	-1	1	-i	i
i	i	-i	-1	1
-i	-i	i	1	-1

(i) **Closure property :** All elements of table belongs to G .

$\therefore G$ is closed w.r.t. multiplication.

(ii) **Associativity :** We know that multiplication of complex numbers is associative.

(iii) **Existence of the identity :** From table

$$(1)(1) = 1 \qquad (-1)(1) = -1$$

$$(i)(1) = i \qquad \text{and} \qquad (-i)(1) = -i$$

$\therefore 1$ is the identity element in G .

(iv) **Existence of the inverse :** From table,

$$1^{-1} = 1,$$

$$(-1)^{-1} = 1,$$

$$(i)^{-1} = -i,$$

and $(-i)^{-1} = i$

$\therefore$ The inverse exists for all elements of G . Thus, ' G ' is a group.

Example 9.4.10 Show that the set $G = \{1, w, w^2\}$ where w is the cube root of unity is a group with respect to multiplication.

SPPU : Dec.-12, Marks 6

Solution : Consider the following composition table of G .

	1	w	w ²
1	1	w	w ²
w	w	w ²	1
w ²	w ²	1	w

(i) Closure property : From table all elements belong to G .

$\therefore G$ is closed w.r.t. multiplication.

(ii) Associativity : As all elements of G are complex numbers and multiplication of complex numbers is associative.

$\therefore (G, \times)$ is associative.

(iii) Existence of the identity : From the composition table

$$1(1) = 1, 1(w) = 1, 1(w^2) = w^2$$

$\therefore 1$ is multiplicative identity in G .

(iv) Existence of the inverse : From table, the inverses of $1, w, w^2$ are $1, w^2, w$ respectively. Thus $(G, \times)$ is a group.

Example 9.4.11 Prove that $G = \{0, 1, 2, 3, 4, 5\} = \mathbb{Z}_6$ is an abelian group of order 6. w.r.t. addition modulo 6.

Solution : Let us consider the composition table

$+_6$	0	1	2	3	4	5
0	0	1	2	3	4	5
1	1	2	3	4	5	0
2	2	3	4	5	0	1
3	3	4	5	0	1	2
4	4	5	0	1	2	3
5	5	0	1	2	3	4

(i) Closure property : From the composition table all elements belong to $\mathbb{Z}_6$.

$\therefore \mathbb{Z}_6$ is closed w.r.t. $+_6$

(ii) Associativity : Let $a, b, c \in \mathbb{Z}_6$

$a +_6 (b +_6 c) =$ The smallest non zero remainder when $a + (b + c)$ is divided by 6.

$=$ The smallest non zero remainder when $(a + b) + c$ is divided by 6.

$= (a +_6 b) +_6 c$

$\therefore +_6$ is associative.

(iii) Existence of the identity : We have $0 \in \mathbb{Z}_6$

If $a \in \mathbb{Z}_6$ then $a +_6 0 = a = 0 +_6 a$

$\therefore 0$ is the additive identity.

(iv) **Existence of the inverse** : From the composition table, the inverses 0, 1, 2, 3, 4, 5 are 0, 5, 4, 3, 2, 1 respectively.

(v) **Commutativity** : Let $a, b \in \mathbb{Z}_6$

$$a +_6 b = b +_6 a \quad \dots (\because \text{From table})$$

Thus $(\mathbb{Z}_6, +_6)$ is an abelian group.

Example 9.4.12 Let $(A, *)$ be a group. Show that $(A, *)$ is abelian group

$$\text{iff } a^2 * b^2 = (a * b)^2$$

SPPU ; Dec.-12, Marks 4

Solution : Let $(A, *)$ be an abelian group.

$$a * b = b * a$$

$$\text{Consider, } a^2 * b^2 = (a * a) * (b * b)$$

$$= a * (a * b) * b \quad (* \text{ is associative})$$

$$= a * (b * a) * b$$

$$= (a * b) * (a * b)$$

$$a^2 * b^2 = (a * b)^2$$

Conversely suppose

$$a^2 * b^2 = (a * b)^2 \quad (1)$$

$$\text{L.H.S.} = a^2 * b^2 = (a * a) * (b * b) = a * (a * b) * b$$

$$\text{R.H.S.} = (a * b)^2 = (a * b) * (a * b) = a * (b * a) * b$$

Equation (1) $\Rightarrow$

$$a * (a * b) * b = a * (b * a) * b$$

By cancellation laws,

$$a * b = b * a$$

$\Rightarrow (A, *)$ is an abelian group.

Examples

Example 9.4.13 Show that the set of all idempotents in a commutative monoid S is a submonoid of S .

SPPU ; Dec.-12, Marks 4

Solution : We know that an element $x \in S$ is called an idempotent if $x * x = x$. Let T be the set of all idempotents in S .

i) For any $x, y \in T$, Consider

$$(x * y) * (x * y) = ((x * y) * x) * y$$

$$\begin{aligned}
&= (y * (x * x)) * y \\
&= (y * x) * y \\
&= (x * y) * y \\
&= x * (y * y) \\
&= x * y
\end{aligned}$$

$\therefore x * y \in T$ for all $x, y \in T$

T is closed w.r.t. ' $*$ '

ii) Associativity, Let $x, y, z \in T \subseteq S$

$$\therefore x * (y * z) = (x * y) * z \text{ in } T$$

$*$ is associative in T

iii) Existence of the identity :

Let e be the identity element in S

$$\text{As } e * e = e \therefore e \in T$$

$\therefore e$ is the identity element in T .

$\therefore T$ is a monoid

Thus T is a submonoid of S .

Example 9.4.14 Prove that the set $\mathbb{Z}$ of all integers with binary operation $*$ defined by

$a * b = a + b + 1$ such that $\forall a, b \in \mathbb{Z}$ is an abelian group :

SPPU : May-14, Marks 4

Solution :

We have $a * b = a + b + 1, \forall a, b \in \mathbb{Z}$

i) Closure property

$$\text{For } a, b \in \mathbb{Z} \Rightarrow a + b + 1 \in \mathbb{Z} \Rightarrow a * b \in \mathbb{Z}$$

$\therefore \mathbb{Z}$ is closed w.r.t. $*$

ii) Associative : Let $a, b, c \in \mathbb{Z}$

$$\begin{aligned}
(a * b) * c &= (a + b + 1) * c \\
&= a + b + 1 + c + 1 = a + b + c + 2 \quad \dots (1)
\end{aligned}$$

$$\begin{aligned}
\text{and } a * (b * c) &= a * (b + c + 1) \\
&= a + b + c + 1 + 1 = a + b + c + 2 \quad \dots (2)
\end{aligned}$$

From equation (1) and equation (2) $(a * b) * c = a * (b * c)$

$\therefore *$ is associative in $\mathbb{Z}$

iii) Existence of the identity :

Let e be the identity in $\mathbb{Z}$

$\therefore$ For any $a \in \mathbb{Z}$, $a * e = e * a = a$

$$\Rightarrow a + e + 1 = a$$

$$\Rightarrow e + 1 = a$$

$$\Rightarrow e = -1 \text{ is the identity element in } \mathbb{Z}$$

iv) Existence of the inverse :

Let $a \in \mathbb{Z}$. Suppose $b \in \mathbb{Z}$ is the inverse of a in $\mathbb{Z}$

$$\therefore a * b = e$$

$$a + b + 1 = -1$$

$$a + b = -2$$

$$b = -a - 2 \in \mathbb{Z}$$

$\therefore$ The inverse exists for all $a \in \mathbb{Z}$

Thus $(\mathbb{Z}, *)$ is a group.

Now consider

$$\begin{aligned} a * b &= a + b + 1 & (\because a + b = b + a) \\ &= b + a + 1 \\ &= b * a, \quad \forall a, b \in \mathbb{Z} \end{aligned}$$

$\therefore *$ is commutative in $\mathbb{Z}$

$\therefore (\mathbb{Z}, *)$ is an abelian group.

9.5 Modulo m

SPPU : Dec.-11, May-08

I) Let a and b are any integers and m is a fixed positive integer then the addition modulo m denoted by $a +_m b$ and defined as $a +_m b = r$; $0 \leq r \leq m$

Where r is the least non negative remainder when $a + b$ is divided by m .

$$\begin{aligned} \text{e.g.} \quad 5 +_3 9 &= 2 & \text{as} \quad 5 + 9 = 14 \text{ and } 14 = 3 \times 4 + 2 \\ 15 +_5 25 &= 0 & \text{as} \quad 15 + 25 = 40 \text{ and } 40 = 5 \times 8 + 0 \end{aligned}$$

II) Let a and b be any integers and m is a fixed positive integer. Then the multiplication modulo m is denoted by $a \times_m b$ and defined as

$$a \times_m b = r ; 0 \leq r \leq m$$

Where r is the least non negative remainder when $a \times b$ is divided by m .

e.g. $3 \times_4 5 = 3$ as $3 \times 5 = 15$ and $15 = 4 \times 3 + 3$
 $9 \times_6 4 = 0$ as $9 \times 4 = 36$ and $36 = 6 \times 6 + 0$

III) If a and b are any two integers such that $a - b$ is divisible by a fixed positive integer m , is called " a congruent to b modulo m ".

It is denoted by $a \equiv b \pmod{m}$

Note :

1) If $a \equiv b \pmod{m}$ iff $a - b$ is divisible by m

i.e. $a - b = pm$; for some integer p .

$$a = pm + b$$

2) If $a \equiv b \pmod{m}$ then $b \equiv a \pmod{m}$

3) For any $a \in \mathbb{Z}$, $a \equiv a \pmod{m}$

e.g.

i) $5 \mid (12 - 2) \Rightarrow 12 \equiv 2 \pmod{5}$

$$5 \mid (12 - 2) \Rightarrow 2 \equiv 12 \pmod{5}$$

ii) $8 \mid (30 - 6) \Rightarrow 30 \equiv 6 \pmod{8}$

iii) $5 \mid (9 - 9) \Rightarrow 9 \equiv 9 \pmod{5}$

Examples

Example 9.5.1 Show that $(G, +_8)$ is an abelian group where $G = \{0, 1, 2, 3, 4, 5, 6, 7\}$

Solution : Consider the following table for $+_8$

$+_8$	0	1	2	3	4	5	6	7
0	0	1	2	3	4	5	6	7
1	1	2	3	4	5	6	7	0
2	2	3	4	5	6	7	0	1
3	3	4	5	6	7	0	1	2
4	4	5	6	7	0	1	2	3
5	5	6	7	0	1	2	3	4
6	6	7	0	1	2	3	4	5
7	7	0	1	2	3	4	5	6

1) Closure property : Every element of the table belongs to G

i.e. $a +_8 b \in G$; $\forall a, b \in G$

$\therefore +_8$ is closed in G .

2) Associativity : For any $a, b, c \in G$

$$a +_g (b +_g c) = (a +_g b) +_g c$$

$\therefore +_g$ is associative in G .

3) Existence of the identity

By observing first row and first column,

For any $a \in G$, $\exists 0 \in G$ such that

$$a +_g 0 = 0 +_g a = a$$

Hence 0 is the identity element for '+_g' in G

4) Existence of the inverse

Identify the identity element 0 from table.

$\therefore$ By table ; As $0 +_g 0 = 0$; 0 is the inverse of 0

As $1 +_g 7 = 0$; 1 is the inverse of 7

As $7 +_g 1 = 0$; 7 is the inverse of 1

As $2 +_g 6 = 0$; 2 is the inverse of 6

As $6 +_g 2 = 0$; 6 is the inverse of 2

As $3 +_g 5 = 0$; 5 is the inverse of 3

As $5 +_g 3 = 0$; 3 is the inverse of 4

As $4 +_g 4 = 0$; 4 is the inverse of 4

Hence, inverse exists in G for all $a \in G$

5) Commutative property : From table, $\forall a, b \in G$

$$a +_g b = b +_g a$$

$\therefore$ '+_g' is commutative in G .

Hence, $(G, +_g)$ is an abelian group.

Example 9.5.2 Show that $(G, \times_7)$ is an abelian group, where $G = \{1, 2, 3, 4, 5, 6\}$

Solution : Consider the following table

$\times_7$	1	2	3	4	5	6
1	1	2	3	4	5	6
2	2	4	6	1	3	5
3	3	6	2	5	1	4
4	4	1	5	2	6	3
5	5	3	1	6	4	2
6	6	5	4	3	2	1

i) Closure Property

Every element of table belongs to G

i.e. $a \times_7 b \in G$ for all $a, b \in G$

$\therefore \times_7$ is closed in G.

ii) Associativity

For any $a, b, c \in G$

$$a \times_7 (b \times_7 c) = (a \times_7 b) \times_7 c$$

$\therefore \times_7$ is associative in G.

iii) Existence of the Identity

By observing the first row and the first column,

For any $a \in G, \exists 1 \in G$ such that

$$a \times_7 1 = 1 \times_7 a = a$$

$\therefore 1$ is the identity element in G.

iv) Existence of the Inverse

Identify the identity elements from table

$\therefore$ By table,

As $1 \times_7 1 = 1$, 1 is the inverse of 1

As $2 \times_7 4 = 4 \times_7 2 = 1$, $\therefore 2$ is the inverse of 4 and 4 is the inverse of 2

As $3 \times_7 5 = 5 \times_7 3 = 1$, $\therefore 3$ is the inverse of 5 and 5 is the inverse of 3

As $6 \times_7 6 = 1$, $\therefore 6$ is the inverse of 6

Hence, inverse exists in G for all $a \in G$

v) Commutative Property

From table, for all $a, b \in G$

$$a \times_7 b = b \times_7 a$$

$\therefore \times_7$ is commutative in G .

Hence $(G, \times_7)$ is an abelian group.

Example 9.5.3 Is $(G, \times_6)$ an abelian group? Where $G = \{1, 2, 3, 4, 5\}$

Solution : Here $2, 3 \in G$ but $2 \times_6 3 = 0 \notin G$

$\therefore \times_6$ is not binary operation in G

i.e. $\times_6$ is not closed in G .

Hence $(G, \times_6)$ is not a group.

Thus $(G, \times_6)$ is not a abelian group.

Example 9.5.4 Show that $(z_6, +)$ is an abelian group.

SPPU : May-08

Solution : Let z_6 be the set of residue classes mod 6.

$$\therefore z_6 = \{[0], [1], [2], [3], [4], [5]\}$$

$$\text{Or } z_6 = \{\bar{0}, \bar{1}, \bar{2}, \bar{3}, \bar{4}, \bar{5}\}$$

Consider the following table

+	$\bar{0}$	$\bar{1}$	$\bar{2}$	$\bar{3}$	$\bar{4}$	$\bar{5}$
$\bar{0}$	$\bar{0}$	$\bar{1}$	$\bar{2}$	$\bar{3}$	$\bar{4}$	$\bar{5}$
$\bar{1}$	$\bar{1}$	$\bar{2}$	$\bar{3}$	$\bar{4}$	$\bar{5}$	$\bar{0}$
$\bar{2}$	$\bar{2}$	$\bar{3}$	$\bar{4}$	$\bar{5}$	$\bar{0}$	$\bar{1}$
$\bar{3}$	$\bar{3}$	$\bar{4}$	$\bar{5}$	$\bar{0}$	$\bar{1}$	$\bar{2}$
$\bar{4}$	$\bar{4}$	$\bar{5}$	$\bar{0}$	$\bar{1}$	$\bar{2}$	$\bar{3}$
$\bar{5}$	$\bar{5}$	$\bar{0}$	$\bar{1}$	$\bar{2}$	$\bar{3}$	$\bar{4}$

i) Closure Property

Every element of a table belongs to z_6

$$\text{i.e. } \bar{a} + \bar{b} \in z_6 \quad \forall \bar{a}, \bar{b} \in z_6$$

$\therefore +$ is closed in z_6

ii) Associativity

By table, for any $\bar{a}, \bar{b}, \bar{c} \in z_6$

$$\bar{a} + (\bar{b} + \bar{c}) = (\bar{a} + \bar{b}) + \bar{c}$$

$\therefore '+'$ is associative in z_6

iii) Existence of the Identity

By observing the first row and the first column,

For any $\bar{a} \in z_6, \exists \bar{0} \in z_6$ such that

$$\bar{a} + \bar{0} = \bar{0} + \bar{a} = \bar{a}$$

$\therefore \bar{0}$ is the identity element in z_6 .

iv) Existence of the Inverse

From table, identify the identity elements

As $\bar{0} + \bar{0} = \bar{0},$ $\bar{0}$ is the inverse of $\bar{0}$

As $\bar{1} + \bar{5} = \bar{5} + \bar{1} = \bar{0},$ $\bar{1}$ is the inverse of $\bar{5}$ and $\bar{5}$ is the inverse of $\bar{1}$

As $\bar{2} + \bar{4} = \bar{4} + \bar{2} = \bar{0},$ $\bar{2}$ is the inverse of $\bar{4}$ and $\bar{4}$ is the inverse of $\bar{2}$

As $\bar{3} + \bar{3} = \bar{0},$ $\bar{3}$ is the inverse of $\bar{3}$

$\therefore$ Every element of z_6 has inverse in z_6 .

v) Commutative Property

From table, For all $\bar{a}, \bar{b} \in z_6$

$$\bar{a} + \bar{b} = \bar{b} + \bar{a}$$

$\therefore$ '+' is commutative in z_6 .

Hence $(z_6, +)$ is an abelian group.

Example 9.5.5 Let $z = \{0, 1, 2, 3, 4, \dots, (n-1)\}$ and ' $\diamond$ ' be a binary operation such that $a \diamond b$ = remainder of abelian when divided by n. Construct a table for $n = 4$,
Is $(z_4, \diamond)$ is groupoid, monoid, semigroup and abelian group ?

SPPU : Dec.-11

Solution :

We have $z_4 = \{0, 1, 2, 3\}$

Table of z_4 is

$\diamond$	0	1	2	3
0	0	0	0	0
1	0	1	2	3
2	0	2	0	2
3	0	3	2	1

i) From table, for any $a, b \in z_4, a \diamond b \in z_4$

$\therefore (z_4, \diamond)$ is a groupoid and $\diamond$ is a binary operation in z_4 i.e. $\diamond$ is closed in z_4 .

ii) Semigroup : By table for any $a, b, c, \in \mathbb{Z}_4$

$$a \diamond (b \diamond c) = (a \diamond b) \diamond c$$

$\therefore \diamond$ is associative in $\mathbb{Z}_4$

OR Let $a \diamond b = r$ and $b \diamond c = t$

$$\therefore ab = 4p + r \quad \text{and} \quad bc = 4l + t$$

$$\therefore (a \diamond b) \diamond c = r \diamond c = s \quad \text{where} \quad rc = 4q + s$$

$$a \diamond (b \diamond c) = a \diamond t = k \quad \text{where} \quad at = 4m + k$$

Prove that $s = k$

$$a(bc) = 4a l + at = 4a l + 4m + k \quad \dots (1)$$

$$(ab)c = (4p+r)c = 4pc + rc = 4pc + 4q + s \quad \dots (2)$$

Equations (1) and equation (2) are equal

$$\Rightarrow 4a l + 4m + k = 4pc + 4q + s$$

$$\Rightarrow k = s$$

Hence $(a \diamond b) \diamond c = a \diamond (b \diamond c)$

Thus $(\mathbb{Z}_4, \diamond)$ is a semigroup

iii) Monoid

By observing the first row and the first column of table

For any $a \in \mathbb{Z}_4, \exists 0 \in \mathbb{Z}_4$ such that

$$a + 0 = 0 + a = a$$

$\therefore 0$ is the identity element in $\mathbb{Z}_4$

Thus from (i), (ii) and (iii), $(\mathbb{Z}_4, +)$ is a monoid.

iv) Existence of the inverse

From table, identify the identity elements.

x is the inverse of 3 in $\mathbb{Z}_4$

$$\therefore x \diamond 3 = 0 \quad \text{and} \quad x \diamond 3 = 4 \text{ if } 3x = 4m + r$$

$$\therefore 3x - 4m = 0$$

$$\Rightarrow 3x = 4m$$

$$\Rightarrow x = \frac{4m}{3} \quad \text{which is an integer}$$

For $m = 1, 2,$ x is not an integer

For $m = 3,$ $x = 4$ which is an integer

But $x = 4 \notin \mathbb{Z}_4$

Thus 3 does not have inverse in $\mathbb{Z}_4$

$\therefore (\mathbb{Z}_4, \diamond)$ is not a group

Hence $(\mathbb{Z}_4, \diamond)$ is not an abelian group.

9.6 Permutation Group

I) Let $S = \{1, 2, 3, \dots, n\}$ be a finite set with n distinct elements. If $f : S \rightarrow S$ is a bijective function then f is called a permutation of degree n .

Let $f(a_1) = b_1, \quad f(a_2) = b_2, \quad f(a_3) = b_3, \quad \dots, f(a_n) = b_n$

Then the permutation is denoted by

$$f = \begin{pmatrix} a_1 & a_2 & a_3 & \dots & a_n \\ b_1 & b_2 & b_3 & \dots & b_n \end{pmatrix} \begin{matrix} \rightarrow \text{Elements in Domain} \\ \rightarrow \text{Elements in Co-domain} \end{matrix}$$

II) The permutation corresponding to the identity function is called the identity permutation

$$I = \begin{pmatrix} a_1 & a_2 & a_3 & \dots & a_n \\ b_1 & b_2 & b_3 & \dots & b_n \end{pmatrix}$$

III) If $f_1 = \begin{pmatrix} a_1 & a_2 & a_3 & \dots & a_n \\ b_1 & b_2 & b_3 & \dots & b_n \end{pmatrix}$ and $f_2 = \begin{pmatrix} b_1 & b_2 & b_3 & \dots & b_n \\ c_1 & c_2 & c_3 & \dots & c_n \end{pmatrix}$

Then the product of two permutations is given by

$$f_1 f_2 = \begin{pmatrix} a_1 & a_2 & a_3 & \dots & a_n \\ c_1 & c_2 & c_3 & \dots & c_n \end{pmatrix}$$

(To write product use $a_1 \rightarrow b_1 \rightarrow c_1 \Rightarrow a_1 \rightarrow c_1$ and so on)

Example 9.6.1 Show that the set of all permutations on $S = \{1, 2, 3\}$ forms a group w.r.t. permutation multiplication.

Solution :

Let $P_3 = \{f/f \text{ is a permutation degree } 3\}$

$$P_3 = \begin{matrix} \begin{Bmatrix} (123) & (123) & (123) & (123) & (123) & (123) \\ (123) & (132) & (213) & (231) & (312) & (321) \end{Bmatrix} \\ \downarrow & \downarrow & \downarrow & \downarrow & \downarrow & \downarrow \\ I & f_1 & f_2 & f_3 & f_4 & f_5 \end{matrix}$$

Consider the following table

0	I	f_1	f_2	f_3	f_4	f_5
I	$\textcircled{I}$	f_1	f_2	f_3	f_4	f_5
f_1	f_1	$\textcircled{I}$	f_3	f_2	f_5	f_4
f_2	f_2	f_4	$\textcircled{I}$	f_5	f_1	f_3
f_3	f_3	f_5	f_1	f_4	$\textcircled{I}$	f_2
f_4	f_4	f_2	f_5	$\textcircled{I}$	f_3	f_1
f_5	f_5	f_3	f_4	f_1	f_2	$\textcircled{I}$

i) Closure Property : Each element of a table belongs to P_3

e.g. for any $f, g \in P_3$, $f \circ g \in P_3$

$\therefore P$ is closed w.r.t. composition of permutations.

ii) Associativity

For any $f, g, h \in P_3$

$$f \circ (g \circ h) = (f \circ g) \cdot h$$

Hence it is associative

iii) Existence of the Identity Element

By observing, first row and the first column of table

For any $f \in P_3$, $\exists I \in P_3$

such that $f \circ I = I \circ f = f$

$\therefore I$ is the identity permutation.

iv) Existence of the Inverse

From table, identify the identity element I ,

As $I \circ I = I \Rightarrow I^{-1} = I$

$$f_1 \circ f_1 = I \Rightarrow f_1^{-1} = f_1$$

$$f_2 \circ f_2 = I \Rightarrow f_2^{-1} = f_2$$

$$f_3 \circ f_4 = I = f_4 \circ f_3 \Rightarrow f_3^{-1} = f_4 \text{ and } f_4^{-1} = f_3$$

$$f_5 \circ f_5 = I \Rightarrow f_5^{-1} = f_5$$

$\therefore$ Every element of P has inverse in P_3 .

Hence $(P_3, \cdot)$ is a group.

But $f_1 \circ f_2 \neq f_3$ and $f_2 \circ f_1 = f_4$

$\Rightarrow f_1 \circ f_2 \neq f_2 \circ f_1 \Rightarrow$ It is not commutative.

$\Rightarrow (P_3, \circ)$ is not abelian group.

$(P_3, \circ)$ is the smallest non abelian group.

9.7 Complexes and Subgroups

SPPU : May-18

Depending upon the nature of subset of a group there are two types of subsets, which are given below.

9.7.1 Complex of a Group

Let $(G, *)$ be a group. Any non empty subset of a group G is called a complex of the group.

e.g. $H_1 = \{1, 2, 3, 4, 5\}, \quad H_2 = \{1, 2, 3, \dots\}, \quad H_3 = \mathbb{Z}$

are complexes of a group $(\mathbb{R}, +)$

9.7.2 Subgroup

Let $(G, *)$ be a group. A non empty subset H of a group G , is said to be subgroup of G if $(H, *)$ itself is a group.

Examples

- 1) $(\mathbb{Z}, +)$ is a subgroup of $(\mathbb{R}, +)$
- 2) $(\mathbb{R}, +)$ is a subgroup of $(\mathbb{C}, +)$
- 3) $(\mathbb{N}, +)$ is not a subgroup of $(\mathbb{Z}, +)$
- 4) $(\mathbb{R}_0, \cdot)$ is a subgroup of $(\mathbb{C}_0, \cdot)$
- 5) $(\{1\}, \cdot)$ is a subgroup of $(\mathbb{Q}_0, \cdot), (\mathbb{R}_0, \cdot), (\mathbb{C}, \cdot)$
- 6) $(\mathbb{R}, +)$ is a subgroup of $(\mathbb{R}, +)$

Note :

- 1) $\{e\}$ and $\{G\}$ are subgroups of group G : These subgroups are called improper subgroups.
- 2) Subgroups which are not improper are called proper subgroups.

9.7.3 Properties of Subgroup

Theorem I) : Prove that the identity of a subgroup is the same as that of the group.

Let H be a subgroup of the group G .

Let e and e' be the identities of G and H respectively.

Now, $a \in H \Rightarrow e' a = a$... [$\because e'$ is identity of H]

also $a \in H \Rightarrow a \in G \Rightarrow ea = a$ [$\because e$ is identity of G]

$$e'a = ea$$

$\therefore$ in G we have,

$\Rightarrow e' = e$... [by right cancellation law in G]

Theorem II) : Prove that the inverse of any element of a subgroup is the same as the inverse of the same regarded as an element of the group.

Let H be a subgroup of the group G .

Let e be the identity of G as well as of H .

Let $a \in H$, suppose b is the inverse of a in H and c is the inverse of a in G . Then we have,

$$ba = e$$

and $ca = e$

$\therefore$ in G we have, $ba = ca \Rightarrow b = c$

Theorem III) : Prove that the order of any element of a subgroup is the same as the order of the element regarded as a member of the group.

Ans. : Let H be a subgroup of the group G .

Let $a \in H$, But $a \in G$ and $a^n = e$ in G .

$$a^n = e \text{ in } H \text{ also.}$$

Hence the proof.

Theorem IV) : A necessary and sufficient condition for a non-empty subset H of a group G to be a subgroup is that $a \in H, b \in H \Rightarrow ab^{-1} \in H$ where b^{-1} is the inverse of b in G .

Suppose H is a subgroup of G .

Let $a \in H, b \in H$. Now each element of H must possess inverse because H itself is a group.

$$b \in H \Rightarrow b^{-1} \in H$$

Further H must be closed with respect to multiplication i.e. the composition in G .

$$\therefore a \in H, b^{-1} \in H \Rightarrow ab^{-1} \in H$$

(i) The condition is sufficient. Now it is given that $a \in H, b \in H \Rightarrow ab^{-1} \in H$. We have to prove that H is a subgroup of G .

(ii) Existence of identity :

We have $a \in H, a \in H \Rightarrow aa^{-1} \in H$

... By given condition

$$\Rightarrow e \in H$$

Thus the identity e is an element of H .

(iii) Existence of inverse :

Let a be any element of H . Then by the given condition, we have,

$$e \in H, a \in H \Rightarrow ea^{-1} \in H \Rightarrow a^{-1} \in H.$$

Thus each element of H possesses inverse.

(iv) Closure property : Let $a, b \in H$. Then as shown above $b \in H \Rightarrow b^{-1} \in H$. Therefore applying the given condition we have,

$$a \in H, b^{-1} \in H \Rightarrow a(b^{-1})^{-1} \in H \Rightarrow ab \in H.$$

(v) Associativity : The elements of H are also the elements of G . The composition in G is associative. Therefore, it must also be associative in H .

Hence H itself is a group for the composition in G . Therefore, H is a subgroup of G .

Theorem V) : If H_1 and H_2 are two subgroups of a group G , then $H_1 \cap H_2$ is also a subgroup of G .

Let H_1 and H_2 be any two subgroups of G . Then $H_1 \cap H_2 \neq \emptyset$ Since at least the identity element e is common to both H_1 and H_2 .

In order to prove that $H_1 \cap H_2$ is a subgroup it is sufficient to prove that

$$a \in H_1 \cap H_2, b \in H_1 \cap H_2 \Rightarrow ab^{-1} \in H_1 \cap H_2$$

Now, $a \in H_1 \cap H_2 \Rightarrow a \in H_1$ and $a \in H_2$

$$a \in H_1 \cap H_2 \Rightarrow b \in H_1 \text{ and } b \in H_2$$

But H_1, H_2 are subgroups.

$$\therefore a \in H_1, b \in H_1 \Rightarrow ab^{-1} \in H_1$$

$$a \in H_2, b \in H_2 \Rightarrow ab^{-1} \in H_2$$

$$\text{Finally } ab^{-1} \in H_1, ab^{-1} \in H_2 \Rightarrow ab^{-1} \in H_1 \cap H_2$$

Thus, we have shown that, $a \in H_1 \cap H_2$, $b \in H_1 \cap H_2 \Rightarrow ab^{-1} \in H_1 \cap H_2$. Hence $H_1 \cap H_2$ is a subgroup of G .

Theorem VI) : Show that the union of two subgroups is a subgroup if and only if one is contained in the other.

Suppose H_1 and H_2 are two subgroups of a group G .

Let $H_1 \subseteq H_2$ or $H_2 \subseteq H_1$

Then $H_1 \cup H_2 = H_2$ or H_1 .

But H_1, H_2 are subgroups and therefore, $H_1 \cup H_2$ is also a subgroup. Conversely suppose $H_1 \cup H_2$ is a subgroup. To prove that $H_1 \subseteq H_2$ or $H_2 \subseteq H_1$.

Let us assume that H_1 is not a subset of H_2 and H_2 is also not a subset of H_1 .

Now H_1 is not a subset of $H_2 \Rightarrow \exists a \in H_1$ and $a \notin H_2$... (1)

and H_2 is not a subset of $H_1 \Rightarrow \exists b \in H_2$ and $b \notin H_1$... (2)

From equation (1) and (2).

We have, $a \in H_1 \cup H_2$ and $b \in H_1 \cup H_2$

Since $H_1 \cup H_2$ is a subgroup, therefore $ab = c$ (say) is also an element of $H_1 \cup H_2$.

But $ab = c \in H_1 \cup H_2 \Rightarrow ab = c \in H_1$ or H_2

Suppose $ab = c \in H_1$

Then $b = a^{-1}c \in H_1$, ... [$\because H_1$ is a subgroup. $\therefore a \in H_1 \Rightarrow a^{-1} \in H_1$]

But from (2), we have $b \notin H_1$. Thus we get the contradiction.

Again suppose $ab = c \in H_2$

Then, $a = cb^{-1} \in H_2$... [$\because H_2$ is a subgroup, therefore $b \in H_2 \Rightarrow b^{-1} \in H_2$]

But from (1), we have $a \notin H_2$. Thus here also we get a contradiction.

Hence either $H_1 \subseteq H_2$ or $H_2 \subseteq H_1$

Example 9.7.1 Is union of two subgroups is a subgroup? If not, give example.

Solution : The union of two subgroups is not necessarily a subgroup.

Let $H_1 = \{..., -4, -2, 0, 2, 4, 6, ...\}$

$H_2 = \{..., -6, -3, 0, 3, 6, ...\}$ are subgroup of $(\mathbb{Z}, +)$

Now $H_1 \cup H_2 = \{-6, -4, -3, -2, 0, 2, 3, 4, 6, ...\}$

$2, 3 \in H_1 \cup H_2$ but $2 + 3 = 5 \notin H_1 \cup H_2$

$\therefore +$ is not binary operation on $H_1 \cup H_2$

$\Rightarrow H_1 \cup H_2$ is not a subgroup of G .

9.8 Cosets

Let $(G, *)$ be a group and H be any subgroup of G .

Let $a \in G$ be any element, then the set

$H * a = \{h * a \mid \forall h \in H\}$ is called a right coset of H in G .

and $a * H = \{a * h \mid \forall h \in H\}$ is called a left coset of H in G .

Note :

- 1) $H * a$ and $a * H$ are subsets of G .
- 2) If $(G, *)$ is an abelian group then $H * a = a * H$ in G .

e.g. 1) Let $(\mathbb{Z}, +)$ is a group and

$H = \{\dots, -10, -5, 0, 5, 10, \dots\}$ is a subgroup of $G = \mathbb{Z}$

$\therefore$ For $1 \in \mathbb{Z}$, $H+1 = \{\dots, -9, -4, 1, 6, 11, \dots\}$

$3 \in \mathbb{Z}$, $H+3 = \{\dots, -7, -2, 3, 8, 13, \dots\}$

$5 \in \mathbb{Z}$, $H+5 = \{\dots, -5, 0, 5, 10, \dots\} = H$

are right cosets of H in G .

9.9 Order of an Element of a Group

Let $(G, \cdot)$ be a group. The smallest positive integer is called the order of an element $a \in G$ if

$$a^n = e(\text{identity element in } G)$$

If is denoted by $o(a) = n$.

If no such positive number exists, then we say that a is of infinite order or zero order.

Note :

- 1) For the order of the group is the number of distinct elements in G .
- 2) The order of the identity element is 1 i.e. $o(e) = 1$
- 3) In a group G , $o(a) = o(a^{-1})$; $\forall a \in G$
- 4) In a group G , $o(a) \leq o(G)$

Examples

1) $G = \{1, -1, i, -i\}$ is a multiplicative group

1 = Identity element in G and $o(G) = 4$

$$o(1) = 1$$

$$o(-1) = 2 \quad \text{as} \quad (-1)^2 = 1$$

$$o(-i) = 4 \quad \text{as} \quad (i)^4 = 1$$

$$o(-i) = 4 \quad \text{as} \quad (-i)^4 = 1$$

2) $(\mathbb{Z}, +)$ is a group of infinite order

0 is the identity element, $o(0) = 1$

We have $5 \in \mathbb{Z}$ and $5^n = 5 + 5 + \dots$ n times $= n5 \neq 0$

$\therefore o(5) = \text{infinity or zero}$

9.10 Cyclic Group**SPPU : Dec.-18**

A group G is called a cyclic group if $\exists$ at least one element $a \in G$ such that every element $x \in G$ can be written as $x = a^m$ where m is some integer.

The element a is called the generator of G and denoted by $G = \langle a \rangle$

Example 1 : Four fourth roots of unity form a cyclic group with respect to multiplication.

$$G = \{1, -1, i, -i\}$$

$$\text{We have} \quad (i)^1 = i, \quad (i)^2 = -1, \quad (i)^3 = -i, \quad (i)^4 = 1$$

$\therefore i$ is the generator of $G \Rightarrow G$ is a cyclic group.

$$\text{Moreover,} \quad (-i)^1 = -i, \quad (-i)^2 = -1, \quad (-i)^3 = i, \quad (-i)^4 = 1$$

$\therefore -i$ is also generator of G

$$\therefore G = \langle i \rangle = \langle -i \rangle$$

9.11 Normal Subgroups**SPPU : Dec.-16**

A subgroup H of a group $(G, *)$ is said to be a normal subgroup of G if for all $g \in G$ and for all $h \in H$

$$g * h * g^{-1} \in H. \quad (\text{We may write } g h g^{-1} \in H)$$

Every group G possesses at least two normal subgroups namely $\{e\}$ and G . These groups are called improper normal subgroups.

Simple Group : A group G is said to be simple group if it has only two normal subgroups, $\{e\}$ and G .

Notes :

- 1) Every subgroup of an abelian group is normal.
- 2) The intersection of normal subgroups is a normal subgroup.

Example 1 : $(\mathbb{Z}, +)$ is an abelian group.

$\therefore (2\mathbb{Z}, +), (3\mathbb{Z}, +)$ are normal subgroups of $(\mathbb{Z}, +)$

9.12 Quotient Groups

SPPU : Dec.-14, 15, May-15,17

Let $(G, *)$ be a group and N be a normal subgroup of G . Let G/N be the collection of all cosets of N in G .

$$G/N = \{N * a / a \in G\}$$

$(G/N, *)$ is called the quotient group or factor group.

Theorem 1) Prove that $(G/N, *)$ is a group.

SPPU : Dec.-15, May-15

Solution : Let $(G, *)$ be a group and N is the normal subgroup at G

$$\therefore G/N = \{N * a / \forall a \in G\}$$

Theorem :

1) Closure Property : Let $a, b \in G, \therefore N * a, N * b \in G/N$

$$(N * a) * (N * b) = N * (a * N) * b \quad (\because N \text{ is normal})$$

$$= N * (N * a) * b$$

$$= (N * N) * (a * b)$$

$$= N * c \quad (\because N * N = N \text{ and } a * b = c \in G)$$

$$\therefore (N * a) * (N * b) \in G/N$$

$\therefore G/N$ is closed w.r.t. $*$

2) Associativity : Let $a, b, c \in G$

and $N * a, N * b$ and $N * c \in G/N$

$$\therefore (N * a) * [(N * b) * (N * c)] = N * a * [N * (b * c)] \quad \text{by (1)}$$

$$= N * a * (b * c) \quad \text{by (1)}$$

$$\begin{aligned}
&= N * (a * b) * c) && (* \text{ is associative in } G) \\
&= (N * (a * b)) * (N * c) \\
&= [(N * a) * (N * b)] * (N * c) \\
&= [(N * a) * (N * b)] * (N * c)
\end{aligned}$$

Thus $*$ is associative in G/N

3) Existence of the identity :

We have $N = N * e \in G/N$ and for any $N * a \in G/N$

$$\begin{aligned}
(N * a) * (N * e) &= N * (a * e) \quad (\text{by (1)}) \\
&= N * a
\end{aligned}$$

$\therefore N * e = N$ is the identity element in G/N .

4) Existence of the inverse : Let $a \in G$, $N * a \in G/N$

$\therefore \exists a^{-1} \in G$ and $N * a^{-1} \in G/N$ such that

$$(N * a) * (N * a^{-1}) = N * (a * a^{-1}) = N * e = N$$

Hence $N * a^{-1}$ is the inverse of $N * a$ in G/N

Thus $(G/N, *)$ is a group, known as quotient group.

Theorem 1 :

Lagranges theorem : The order of each subgroup of a finite group is a divisor of the order of the group.

Ans. : Let G be a group of finite order n . Let H be a subgroup of G and let $O(H) = m$. Suppose $h_1, h_2 \dots h_m$ are the m members of H .

Let $a \in G$. Then Ha is a right coset of H in G and we have

$$Ha = \{h_1a, h_2a, \dots, h_ma\}$$

Ha has m distinct members, since $h_ia = h_ja \Rightarrow h_i = h_j$.

Therefore each right coset of H in G has m distinct members. Any two distinct right cosets of H in G are disjoint i.e. they have no element in common. Since G is a finite group, the number of distinct right cosets of H in G will be finite, say equal to K . The union of these K distinct right cosets of H in G is equal to G .

Thus, if $Ha_1, Ha_2, \dots, Ha_k$ are the K distinct right cosets of H in G , then

$$G = Ha_1 \cup Ha_2 \cup \dots \cup Ha_k$$

$\Rightarrow$ The number of elements in G = The number of elements in Ha_1 + The number of elements in Ha_2 + + The number of elements in Ha_k

... [$\because$ two distinct right cosets are mutually disjoint]

$$\Rightarrow O(G) = Km \Rightarrow n = Km$$

$$\Rightarrow K = \frac{n}{m} \Rightarrow m \text{ is a divisor of } n$$

$$\Rightarrow O(H) \text{ is a divisor of } O(G).$$

Theorem 2 :

The order of every element of a finite group is a divisor of the order of the group.

Suppose G is a finite group of order n .

Let $a \in G$ and $O(a) = m$. Prove that m is a divisor of n .

Let $H = \{ \dots, a^{-3}, a^{-2}, a^{-1}, a^0, a^1, a^2, a^3, \dots \}$ be the subset of G consisting of all integral powers of a .

We know that H is a subgroup of G .

We have to show that H contains only m distinct elements and that they are $a, a^2, a^3, \dots, a^m = e$.

Let $1 \leq r \leq m, 1 \leq s \leq m$, and $r > s$.

$$\text{then } a^r = a^s \Rightarrow a^r a^{-s} = a^s a^{-s} \Rightarrow a^{r-s} = a^0 \Rightarrow a^{r-s} = e$$

Thus there exists a positive integer $r - s$ less than m such that $a^{r-s} = e$. m is the least positive integer such that $a^m = e$.

$$\therefore a^r \neq a^s$$

$\therefore a, a^2, a^3, \dots, a^m = a^0 = e$ are all distinct elements of H .

Now suppose a^t is any element of H where t is any integer.

By division algorithm,

$$\text{We have, } t = mp + q \quad \dots [p \text{ and } q \text{ are some integers, } 0 \leq q < m]$$

$$\begin{aligned} \text{We have, } a^t &= a^{mp+q} \\ &= a^{mp} a^q \\ &= (a^m)^p a^q = a^q \quad \dots (0 \leq q < m) \end{aligned}$$

$\therefore a^q$ is one of the m element $a, a^2, \dots, a^m = a^0$

Hence, H has only m distinct elements. Thus order of H is m .

By Lagrange's theorem m is a divisor of n .

Theorem 3 :

If G is a finite group of order n and $a \in G$, then $a^n = e$.

In a finite group, the order of each element is finite. Let $O(a) = m$. The subset H of G consisting of all integral powers of a is a subgroup of G and the order of H is m .

By Lagrange's theorem, m is a divisor of n .

Let $k = \frac{n}{m}$ then $n = mk$

Now, $a^n = a^{mk} = (a^m)^k = e^k = e$... $O(a) = m \Rightarrow am = e$

Theorem 4 :

Every cyclic group is an abelian group.

SPPU : Dec.-14, Marks 3

Let $G = \{a\}$ be a cyclic group generated by a .

Let x, y be any two elements of G . Then there exist integers r and s such that $x = a^r$, $y = a^s$.

Now, $xy = a^r a^s = a^{r+s} = a^{s+r} = a^s a^r = yx$

Thus, we have, $xy = yx \forall x, y \in G$.

Therefore G is abelian.

Theorem 5 :

If a is a generator of a cyclic group G , then a^{-1} is also a generator of G .

Let $G = \{a\}$ be a cyclic group generated by a .

Let a^r be any element of G , where r is some integer.

as $a^r = (a^{-1})^{-r}$

$\therefore -r$ is also some integer.

$\therefore$ Each element of G , is generated by a^{-1}

Thus, a^{-1} is also a generator of G .

Theorem 6 :

Every group of order 3 is cyclic.

OR

Every group of prime order is cyclic.

Suppose G is a finite group whose order is a prime number P , then to prove that G is a cyclic group. As an integer P is said to be a prime number if $P \neq 0$, $P \neq \pm 1$, and if the only divisors of P are $\pm 1, \pm P$.

$\therefore G$ is a group of prime order, therefore G must contain at least 2 elements.

As 2 is the least positive prime integer.

There must exist an element $a \in G$ such that $a \neq e$, the identity element e .

Since a is not the identity element, therefore $O(a)$ is definitely ≥ 2 .

Let $O(a) = m$, If H is the cyclic subgroup of G generated by a then

$$O(H) = O(a) = m.$$

By Lagrange's theorem m must be divisor of P .

But P is prime and $m \geq 2$. Hence $m = P$

$\therefore H = G$. Since H is cyclic. Therefore G is cyclic and a is a generator of G .

7) Every subgroup of a cyclic group is cyclic.

Suppose $G = \{a\}$ is a cyclic group generated by a . If $H = G$ or $\{e\}$, then obviously H is cyclic. So let H be a proper subgroup of G . The elements of H are integral powers of a . If $a^s \in H$, then the inverse of a^s .

i.e. $a^{-s} \in H$.

$\therefore H$ contains elements which are positive as well as negative integral powers of a .

Let m be the least positive integer such that $a^m \in H$.

Then we shall prove that $H = \{a^m\}$

i.e. H is cyclic and is generated by a^m .

Let a^t be any arbitrary element H .

By division algorithm,

there exist integers q and r such that $t = mq + r$, $0 \leq r < m$.

Now, $a^m \in H \Rightarrow (a^m)^q \in H$... by closure property

$$\Rightarrow a^{mq} \in H \Rightarrow (a^{mq})^{-1} \in H$$

$$\Rightarrow a^{-mq} \in H$$

Also, $a^t \in H$ $a^{-mq} \in H \Rightarrow a^t a^{-mq} \in H$

$$\Rightarrow a^{t-mq} \in H$$

$$\Rightarrow a^r \in H.$$

Now m is the least positive integer such that $a^m \in H$ and $0 \leq r < m$.

Therefore r must be equal to 0. Hence $t = mq$.

$\therefore a^t = a^{mq} = (a^m)^q$

Thus every element $a^t \in H$ is of the form $(a^m)^q$.

Therefore H is cyclic and a^m is a generator of H .

Examples :

Example 9.12.1 Give an example of a finite abelian group which is not cyclic.

Solution : Let G be the set of the four real matrices.

$$I = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}, A = \begin{bmatrix} -1 & 0 \\ 0 & 1 \end{bmatrix}, B = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}, C = \begin{bmatrix} -1 & 0 \\ 0 & -1 \end{bmatrix}$$

It can be easily seen that G is an abelian group with respect to multiplication of matrices.

The identity element of this group is the identity matrix I .

Let us find the order of each element of G .

We have, $O(I) = 1$

$$\text{Also, } A^2 = \begin{bmatrix} -1 & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} -1 & 0 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = I \quad [\because O(A) = 2]$$

$$B^2 = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = I \quad [\because O(B) = 2]$$

$$C^2 = \begin{bmatrix} -1 & 0 \\ 0 & -1 \end{bmatrix} \begin{bmatrix} -1 & 0 \\ 0 & -1 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = I \quad [\because O(C) = 2]$$

Now G is a group of order 4 and G contains no element of order 4.

Therefore G is not a cyclic group. Hence G is a finite abelian group which is not cyclic.

Example 9.12.2 Show that the multiplicative group $\{1, -1\}$ is a subgroup of the multiplicative group $\{1, -1, i, -i\}$

Solution : Let $H = \{1, -1\}$ and $G = \{1, -1, i, -i\}$, $H \subset G$.

Consider the composition table

X	1	-1
1	1	-1
-1	-1	1

(i) **Closure property :** All elements of a table belong to H . Hence it is closed w.r.t. multiplication.

(ii) **Multiplication** of complex numbers is associative.

(iii) **Existence of the identity :** $1 \in H$, is the identity element.

(iv) **Existence of the inverse :** We have, $(1)^{-1} = 1$ and $(-1)^{-1} = -1$.

Therefore, $(H, \times)$ is a subgroup of a group $(G, \times)$.

Example 9.12.3 Define the subgroup of a group. Let $(G, 0)$ be a group. Let $H = \{a \mid a \in G \text{ and } aob = boa \text{ for all } b \text{ in } G\}$. Show that H is normal subgroup of G .

Solution : Let (G, x) be a group. A non empty subset H of a group G is said to be a subgroup of G if $(H, *)$ itself is a group.

Given that, $H = \{a \mid a \in G \text{ and } a o b = b o a ; \forall b \in G\}$

Let $a, b \in H \Rightarrow a o x = x o a \text{ and } b o x = x o b, \forall x \in G.$

$$\begin{aligned} \Rightarrow (b o x)^{-1} &= (x o b)^{-1} \\ \Rightarrow x^{-1} o b^{-1} &= b^{-1} o x^{-1} \quad \dots (1) \\ \Rightarrow b^{-1} &\in H. \end{aligned}$$

$$\begin{aligned} \text{Now, } (a o b^{-1}) o x &= a o (b^{-1} o x) && [\because o \text{ is associative}] \\ &= a o (x o b^{-1}) && [\because \text{use (1) or } b^{-1} \in H] \\ &= (a o x) o b^{-1} && [\because a \in H] \\ &= (x o a) o b^{-1} \\ &= x o (a o b^{-1}) \end{aligned}$$

$$\Rightarrow a o b^{-1} \in H$$

Therefore H is a subgroup of group G .

Let $h \in H$ and $g \in G$ and any x in G .

Consider,

$$\begin{aligned} (g o h o g^{-1}) o x &= (g o g^{-1} o h) o x && [\because h \in H] \\ &= (e o h) o x = h o x \\ &= x o h && (\because h \in H) \\ &= x o (h o g o g^{-1}) \\ &= x o (g o h o g^{-1}) && (\because h \in H) \end{aligned}$$

$$\Rightarrow g o h o g^{-1} \in H \text{ for any } g \in G$$

$\therefore H$ is a normal subgroup of G .

Example 9.12.4 Show that the set $\{1, w, w^2\}$ is a cyclic group of order 3 with respect to multiplication, w is the cube root of unity.

Solution : Let $G = \{1, w, w^2\}$ and $w^3 = 1$

We have, $1 = 1^1 = w^3 = (w^2)^3$

$$w = w^1 = (w^2)^2$$

$$w^2 = w^2 = (w^2)^1$$

Thus every element of G can be expressed in the powers of w or w^2 . Therefore G is cyclic group and its generators are w and w^2 .

Example 9.12.5 Let $G = \{1, -1, i, -i\}$ where $i^2 = -1$ and multiplication is the binary operation in G .

- (I) Determine whether G is an abelian group.
 (II) If G is a cyclic group then find its generators.
 (III) Find the order of each element of G .

Solution :

(I) Let $G = \{1, -1, i, -i\}$.

The composition table of G with respect to multiplication is given below.

X	1	-1	i	-i
1	①	-1	i	-i
-1	-1	①	-i	i
i	i	-i	-1	①
-i	-i	i	①	-1

(i) **Closure property :** All elements of the table belong to G . Therefore G is closed w.r.t. X .

(ii) **Associativity :** The multiplication of complex numbers is associative. $\therefore (G, X)$ is associative.

(iii) **Existence of the identity :** $1 \in G$ is the identity element.

(iv) **Existence of the inverse :** From table

$$(1)^{-1} = 1, (i)^{-1} = -i, (-1)^{-1} = -1, (-i)^{-1} = i$$

(v) **Commutativity :** The composition table is symmetric about main diagonal.

$\therefore (G, X)$ is commutative.

Thus (G, X) is an abelian group.

(II) We have, $(i)^4 = 1$, $(i)^1 = i$, $(i)^2 = -1$, $(i)^3 = -i$

$$(-i)^4 = 1, (-i)^1 = -i, (-i)^2 = -1, (-i)^3 = i$$

Thus every element of G can be expressed as i^n or $(-i)^n$

$\therefore (G, X)$ is a cyclic group with i and $-i$ generators.

(III) We have, $1^1 = 1$, $(-1)^2 = 1$, $(i)^4 = 1$ and $(-i)^4 = 1$.

$\therefore$ The orders of $1, -1, i, -i$ are $1, 2, 4, 4$ respectively.

Example 9.12.6 Show that the four permutations $I, (a\ b), (c\ d), (a\ b)(c\ d)$ on four symbols a, b, c, d form a finite abelian group with respect to the multiplication.

Solution : Let $f_1 = I = \begin{pmatrix} a & b & c & d \\ a & b & c & d \end{pmatrix}$

$$f_2 = (a \ b) = \begin{pmatrix} a & b & c & d \\ b & a & c & d \end{pmatrix}$$

$$f_3 = (c \ d) = \begin{pmatrix} a & b & c & d \\ a & b & d & c \end{pmatrix}$$

$$f_4 = (a \ b) (c \ d) = \begin{pmatrix} a & b & c & d \\ b & a & d & c \end{pmatrix}$$

Let $G = \{f_1, f_2, f_3, f_4\}$

Consider the multiplication table

X	f_1	f_2	f_3	f_4
f_1	$\textcircled{f_1}$	f_2	f_3	f_4
f_2	f_2	$\textcircled{f_1}$	f_4	f_3
f_3	f_3	f_4	$\textcircled{f_1}$	f_2
f_4	f_4	f_3	f_2	$\textcircled{f_1}$

(i) **Closure property** : All elements of table belong to G. (G, x) is closed w.r.t. x.

(ii) **Multiplication** of permutations is associative.

(iii) $f_1 = I$ is the identity element in G.

(iv) In G, $f_1^{-1} = f_1$, $f_2^{-1} = f_2$, $f_3^{-1} = f_3$ and $f_4^{-1} = f_4$

v) Multiplication table is symmetric about main diagonal $\therefore (G, x)$ is commutative.

Hence (G, x) is an abelian group of order 4.

9.13 Homomorphism of Groups

SPPU : Dec.-11, 12, 14, 15, May-15,17

Let $(G_1, *)$ and (G_2, o) be two groups. A function

$f : (G_1, *) \rightarrow (G_2, o)$ is said to be homomorphism.

If $f(a * b) = f(a) o f(b)$ for all $a, b \in G_1$.

i.e. $a * b$ in $G_1 \rightarrow f(a) o f(b)$ in G_2

A homomorphism from G to itself is called as endomorphism.

Properties of Group Homomorphism.

Let $f : G_1 \rightarrow G_2$ be group homomorphism and $((G_1, *)$ and (G_2, o) are groups then

$$\text{i) } f(e_1) = (e_2)$$

$$\text{ii) } f(a^{-1}) = [f(a)]^{-1}$$

Proof : i) Let $a \in G_1$ and $f(a) \in G_1$ and e_2 is the identity element in G_2 .

$$\therefore f(a) \circ e_2 = f(a)$$

$$= f(a * e_1)$$

$$f(a) \circ e_2 = f(a) \circ f(e_1)$$

$$\Rightarrow f(e_1) = e_2$$

2) Let $a \in G_1$ then $a^{-1} \in G$

$$\text{and } e_2 = f(e_1)$$

$$= f(a * a^{-1})$$

(f is homomorphism)

$$e_2 = f(a) \circ f(a^{-1})$$

$$\Rightarrow f(a^{-1}) = [f(a)]^{-1}$$

9.14 Isomorphism of Groups

SPPU : Dec.-10,11,12,13,16, May-08,10,17,19

Let $(G_1, *)$ and (G_2, o) be two groups. A function

$f : (G_1, *) \rightarrow (G_2, o)$ is said to be isomorphism.

If i) f is a homomorphism from $G_1 \rightarrow G_2$

ii) f is bijective function.

If $f : G_1 \rightarrow G_2$ is an isomorphism of groups then G_1 and G_2 are called as isomorphic groups and denoted by $G_1 \cong G_2$.

An isomorphism from G to itself is called as automorphism of group G .

Examples :

Example 9.14.1 Let G be a group with identity e show that a function $f : G \rightarrow G$ defined by $f(a) \forall a \in G$ is a homomorphism (Endomorphism).

SPPU : May-08,10,19, Marks 3

Solution : We have $f : G \rightarrow G$ and $f(a) = e, \forall a \in G$.

Let $a, b \in G \Rightarrow f(a), f(b) \in G$

$$\therefore f(a * b) = e$$

(as $a * b \in G$)

$$= e * e$$

$$= f(a) * f(b)$$

$\therefore f$ is a homomorphism.

Example 9.14.2 Explain homomorphism and automorphism of groups with examples.

SPPU : Dec.-11, 12, 13, Marks 3

Solution : Refer 9.12 and 9.13 for definition.

e.g. 1) The homomorphism $f : (\mathbb{Z}, +), (\mathbb{Z}, +)$ such that

$$f(n) = -n \text{ is an automorphism of group.}$$

2) The homomorphism $f : (\mathbb{R}_0, o) \rightarrow (\mathbb{R}_0, o)$ such that

$$f(x) = x; \forall x \in \mathbb{R}_0 \text{ is an automorphism of groups.}$$

Example 9.14.3 Let $\mathbb{R}$ be the additive group of real numbers and $\mathbb{R}_+$ be the multiplicative group of positive real numbers. Prove that the mapping $f : (\mathbb{R}, +) \rightarrow (\mathbb{R}_+, \times)$ defined by $f(x) = e^x, \forall x \in \mathbb{R}$ is an isomorphism of $\mathbb{R}$ onto $\mathbb{R}_+$.

SPPU : Dec.-10, Marks 4

Solution : If x is any real number, then e^x is always positive real number and e^x is unique. Therefore, $f : \mathbb{R} \rightarrow \mathbb{R}_+$ is a function such that $f(x) = e^x$.

$$\text{Let } x_1, x_2 \in \mathbb{R} \text{ then } f(x_1) = f(x_2)$$

$$\Rightarrow e^{x_1} = e^{x_2} \Rightarrow x_1 = x_2$$

$\therefore f$ is one to one mapping.

For any $y \in \mathbb{R}_+$ then $\log y \in \mathbb{R}$ such that

$$f(\log y) = e^{\log y} = y$$

$\therefore f$ is onto.

Now for any $x_1, x_2 \in \mathbb{R}$

$$\text{Consider, } f(x_1 + x_2) = e^{x_1 + x_2}$$

$$= e^{x_1} \times e^{x_2}$$

$$= f(x_1) \times f(x_2)$$

$\therefore f$ preserves compositions in $\mathbb{R}$ and $\mathbb{R}_+$

$\therefore f$ is an isomorphism of $\mathbb{R}$ onto $\mathbb{R}_+$.

$$\text{Hence } \mathbb{R} \cong \mathbb{R}_+$$

Example 9.14.4 Show that the additive group of integers.

$G = \{\dots, -3, -2, -1, 0, 1, 2, 3, \dots\}$ is isomorphic to the additive group

$G' = \{\dots, -3m, -2m, -1m, 0, 1m, 2m, 3m, \dots\}$

where m is any fixed integer not equal to zero.

Solution : If $x \in G$, then obviously $mx \in G'$

Let $f : G \rightarrow G'$ be defined by $f(x) = mx \forall x \in G$

(i) f is one to one. Let $x_1, x_2 \in G$.

$$\text{Then } f(x_1) = f(x_2)$$

$$\Rightarrow mx_1 = mx_2$$

$$\Rightarrow x_1 = x_2$$

$\therefore f$ is one to one.

(ii) f is onto : Suppose y is any element of G' .

Then obviously $\frac{y}{m} \in G$. Also $f\left(\frac{y}{m}\right) = m\left(\frac{y}{m}\right) = y$

Thus $y \in G' \Rightarrow$ that there exists $\frac{y}{m} \in G$ such that $f\left(\frac{y}{m}\right) = y$.

$\therefore$ Each element of G' is the f image of some element of G . Hence f is onto. Again if x_1 and x_2 are any two elements of G , then

$$\begin{aligned} f(x_1 + x_2) &= m(x_1 + x_2) && \dots \text{by definition of } f. \\ &= mx_1 + mx_2 && \dots \text{by distributive law for integers} \\ &= f(x_1) + f(x_2) \end{aligned}$$

Thus f preserves compositions in G and G' . Therefore, f is an isomorphic mapping of G onto G' .

Hence G is isomorphic to G' .

9.15 Rings, Integral Domains and Fields

SPPU : Dec.-08, 10, 11, 12, 13, 14,16,17,18, May-14,17,18,19

In previous sections, we have discussed an algebraic structures with a single binary operation. Now, we will study an algebraic structures with two binary operations such as rings, integral domains and fields.

9.15.1 Rings

SPPU : Dec.-13, May-14

Let R be a non empty set equipped with two binary operations called addition and multiplication and denoted by '+' and '.' respectively.

An algebraic structure $(R, +, \cdot)$ is called a ring if it satisfies following axioms.

1) $(R, +)$ is an abelian group i.e.

i) **Closure property** : for $a, b \in R$, $a + b \in R$

ii) **Associativity** : for $a, b, c \in R$, $a + (b + c) = (a + b) + c$

iii) **Existence of the identity** : For any $a \in R$, $\exists 0 \in R$ s.t., $a + 0 = 0 + a = a$.

$\therefore 0$ is called as the additive identity element of ring.

iv) **Existence of the inverse** : for each $a \in R$, $\exists -a \in R$

Such that $a + (-a) = -a + a = 0$

$-a$ is called the additive inverse of a

v) **Commutative property** : For $a, b \in R$

$$a + b = b + a$$

2) $(R, \cdot)$ is semigroup i.e.

i) **Closure property** : $\forall a, b \in R$, $a \cdot b \in R$

ii) **Associativity** : for $a, b, c \in R$,

$$a \cdot (b \cdot c) = (a \cdot b) \cdot c$$

3) Multiplication distributes over addition $\forall a, b, c \in R$

$$\text{i) } a \cdot (b + c) = a \cdot b + a \cdot c \quad \text{(Right distributive law)}$$

$$\text{ii) } (a + b) \cdot c = a \cdot c + b \cdot c \quad \text{(Left distributive law)}$$

9.15.2 Commutative Ring

A ring $(R, +, \cdot)$ is said to be commutative ring if $\forall b \in R$, $a \cdot b = b \cdot a$

9.15.3 Ring with Unity

A ring $(R, +, \cdot)$ is said to be ring with unity if $\forall a \in R$, $\exists 1 \in R$ such that $a \cdot 1 = 1 \cdot a = a$.

Examples :

1) $(\mathbb{Z}, +, \cdot)$ is a commutative ring with unity.

2) $(2\mathbb{Z}, +, \cdot)$ is a commutative ring without unity where $2\mathbb{Z}$ = set of even integers.

3) The set of $n \times n$ matrices over real numbers with respect to usual matrix addition and multiplication is a non commutative ring with unity.

9.15.4 Properties of a Ring

If $(R, +, \cdot)$ is a ring with identity 0 and unit element 1 then following are true for all $a, b, c \in R$.

$$\text{i) } a \cdot 0 = 0 \cdot a = 0$$

$$\text{ii) } a \cdot (-b) = (-a) \cdot b = -(a \cdot b)$$

$$\text{iii) } (-a) \cdot (-b) = a \cdot b$$

iv) Unit element is unique

9.15.5 Subring

Let $(R, +, \cdot)$ be a ring. A non empty subset S of R is said to be subring of R if $(S, +, \cdot)$ is a ring. e.g. $(\mathbb{Z}, +, \cdot)$ is a subring of $(\mathbb{R}, +, \cdot)$.

9.15.6 Zero Divisors

Let $(R, +, \cdot)$ be a commutative ring. An element $a \neq 0$ in R is said to be zero divisor if $\exists b \neq 0$ in R such that $a \cdot b = 0$.

A ring $(R, +, \cdot)$ is said to be without zero divisors.

if $a \cdot b = 0 \Rightarrow a = 0$ or $b = 0, \forall a, b \in R$.

e.g. 1) $\bar{2}$ is a zero divisor in $(\mathbb{Z}_4, +, \cdot)$ as $\bar{2} \cdot \bar{2} = \bar{4} = 0$

2) $(M_{2 \times 2} \mathbb{R}, +, \cdot, 1)$ is a ring with zero divisors.

as
$$A \cdot B = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix} = 0 \text{ but } A \neq 0 \text{ and } B \neq 0$$

3) $(\mathbb{Z}, +, \cdot)$ is a ring without zero divisors i.e. $a \cdot b = 0 \Rightarrow a = 0$ or $b = 0$.

9.15.7 Integral Domain**SPPU : Dec.-13, 11, 10, May-14**

A commutative ring with zero divisors is called an integral domain.

e.g.1) $(\mathbb{R}, +, \cdot)$, $(\mathbb{Z}, +, \cdot)$, $(\mathbb{Q}, +, \cdot)$ are integral domains.

2) $(\mathbb{Z}_4, +, \cdot)$ is a ring with zero divisors

$\therefore$ It is not integral domain.

9.15.8 Field**SPPU : Dec.-13, 12, 11, 10, May-14**

A commutative ring with unity in which every non zero element possesses their multiplicative inverse, is called as field.

A field is an integral domain.

e.g. 1) $(\mathbb{R}, +, \cdot)$, $(\mathbb{Q}, +, \cdot)$, $(\mathbb{Q}, +, \cdot)$ are fields.

2) $(\mathbb{Z}, +, \cdot)$ is integral domain but not field.

9.15.9 Ring Homomorphism

Let $(R, +, *)$ and $(S, +', \xi *')$ be two rings.

A function $\phi : R \rightarrow S$ is called a ring homomorphism

If for any $a, b \in R$

- i) $\phi(a+b) = \phi(a) + \phi(b)$
 ii) $\phi(a * b) = \phi(a) *' \phi(b)$

If ϕ is bijective then it is called as a ring isomorphism.

The kernel of ring homomorphism is defined as the set $\{a \in R \mid \phi(x) = 0'\}$.

It is denoted by $\ker(\phi)$ or $\ker \phi$.

Examples :

Example 9.15.1 Let $R = \left\{ \begin{bmatrix} a & b \\ b & a \end{bmatrix} : a, b \in \mathbb{Z} \right\}$ f is the mapping that takes $\begin{bmatrix} a & b \\ b & a \end{bmatrix}$ to $a - b$.

i) Show that f is a homomorphism. ii) Find Kernel of f .

SPPU : Dec.-08

Solution :

$$\begin{aligned} f &= \left(\begin{bmatrix} a & b \\ b & a \end{bmatrix} + \begin{bmatrix} c & d \\ d & c \end{bmatrix} \right) = f \left(\begin{bmatrix} a+c & b+d \\ b+d & a+c \end{bmatrix} \right) \\ &= (a+c) - (b+d) \\ &= (a-b) + (c-d) \\ &= f \left(\begin{bmatrix} a & b \\ b & a \end{bmatrix} \right) + f \left(\begin{bmatrix} c & d \\ d & c \end{bmatrix} \right) \end{aligned}$$

$$\begin{aligned} \text{Also } f \left(\begin{bmatrix} a & b \\ b & a \end{bmatrix} \right) \cdot f \left(\begin{bmatrix} c & d \\ d & c \end{bmatrix} \right) &= f \begin{bmatrix} ac+bd & ad+bc \\ bc+ad & bd+ac \end{bmatrix} \\ &= (ac+bd) - (ad+bc) \\ &= (ac-bc) + (bd-ad) \\ &= (a-b)(c-d) \\ &= f \left(\begin{bmatrix} a & b \\ b & a \end{bmatrix} \right) \cdot f \left(\begin{bmatrix} c & d \\ d & c \end{bmatrix} \right) \end{aligned}$$

$$\text{ii) } \ker f = \left\{ \begin{bmatrix} a & b \\ b & a \end{bmatrix} : a-b=0 \right\} \quad \text{i.e. } \left\{ \begin{bmatrix} a & a \\ a & a \end{bmatrix} ; a \in \mathbb{Z} \right\}$$

Example 9.15.2 Show that $S = \{a+b\sqrt{2} ; a, b \in \mathbb{Z}\}$ for the operations $+$, $\times$ is an integral domain but not a field.

SPPU : Dec.-14,16, May-17,19

Solution : We have,

$$(a+b\sqrt{2}) + (c+d\sqrt{2}) = (a+c) + (b+d)\sqrt{2}$$

$$(a + b\sqrt{2}) + (c + d\sqrt{2}) = (ac + 2bd) + (bc + ad)\sqrt{2}$$

Clearly S is commutative ring with unit element 1.

We have to prove S is an integral domain.

$$\text{Let } (a + b\sqrt{2})(c + d\sqrt{2}) = 0$$

$$\therefore ac + 2bd = 0 \quad \dots (1)$$

$$\text{and } bc + ad = 0 \quad \dots (2)$$

Suppose $a = 0$; then $bd = bc = 0$

either $b = 0$ or both $d = c = 0$

Hence, if $a = 0$, $a + b\sqrt{2} = 0$

$$\text{or } c + d\sqrt{2} = 0$$

Assume $a \neq 0$. Multiplying equation (1) by d

$$\text{we have, } acd + 2bd^2 = 0 \quad \dots (3)$$

From equation (2)

$$ad = -bc$$

Hence substituting this value in equation (3)

$$\text{We have, } -bc^2 + 2bd^2 = 0$$

$$\Rightarrow b(2d^2 - c^2) = 0$$

$$\therefore b = 0 \text{ or } c^2 = 2d^2, \text{ i.e. } c = \sqrt{2}d$$

Since c is an integer, $c^2 = 2d^2$ is true only if $c = d = 0$

Hence if $c^2 \neq 2d^2$, $b = 0$. But $b = 0$ implies $a = 0$

Hence, in any case either $a + b\sqrt{2} = 0$ or $c + d\sqrt{2} = 0$

Hence, S is an integral domain.

To show that S is not a field consider the element $2 + \sqrt{2}$. It's multiplicative inverse does not exist in S , for $(2 + \sqrt{2})(c + d\sqrt{2}) = 1$

$$\Rightarrow 2c + 2d = 1 \Rightarrow c + d = \frac{1}{2}$$

Absurd, since $c, d \in \mathbb{Z}$.

Example 9.15.3 If R is a ring such that $a^2 = a, \forall a \in R$ then prove that $a + a = 0, \forall a \in R$.
 R is a commutative ring.

Solution : (i) Let $b = -a$ is the inverse of a ,

Now prove that, $a = b$

Consider, $a - b = (a - b)(a - b)$

$$\begin{aligned}\therefore a - b &= a^2 - ba - ab + b^2 \\ &= a - ba - ab + b \\ &= (a + b) - ba - ab\end{aligned}$$

$$\therefore a - b = -ba - ab = -(ba + ab) \quad \dots (1)$$

Now, $a + b = (a + b)(a + b)$

$$\begin{aligned}&= a^2 + ab + ba + b^2 \\ &= a + ab + ba + b \\ &= a + b + (ab + ba) \\ &= ab + ba\end{aligned}$$

$$\dots (2)$$

But, $a + b = 0$

$$\Rightarrow ab + ba = 0$$

$$\therefore \text{Equation (1)} \Rightarrow a - b = 0$$

$$\Rightarrow a = b$$

Example 9.15.4 Consider a ring $(R, +, *)$ defined by $a * a = a$. Determine whether ring is commutative or not ?

Solution : Let $a, b \in R$

$$\text{and } (a + b)^2 = a + b$$

$$(a + b)(a + b) = a + b$$

$$a^2 + ab + ba + b^2 = a + b$$

$$a + ab + ba + b = a + b$$

$$\dots (\because aa = a)$$

$$(a + b) + (ab + ba) = a + b$$

$$ab + ba = 0$$

$$\Rightarrow a + b = 0$$

$$\Rightarrow a + b = a + a$$

$$\Rightarrow b = a$$

$$\Rightarrow \quad ab = ba$$

$\therefore R$ is commutative ring.

Example 9.15.5 Prove that the set E of all even integers is a commutative ring with respect to usual addition and multiplication, but it has no unit element.

Solution : Let $2\mathbb{Z}$, be the set of all even integers, then

(i) For any two elements $2m, 2n \in 2\mathbb{Z}$, $m, n \in \mathbb{Z}$;

$$2m + 2n = 2(m + n) \in 2\mathbb{Z} \text{ and}$$

$$2m \cdot 2n = (2mn) \in 2\mathbb{Z} \text{ and}$$

$$2m \cdot 2n = (2mn) \in 2\mathbb{Z},$$

Since $m + n \in \mathbb{Z}$ and $2mn \in \mathbb{Z}$.

(ii) Since $(\mathbb{Z}, +)$ $(\mathbb{Z}, \cdot)$ are associative and commutative, $2\mathbb{Z} \in \mathbb{Z}$, so $(2\mathbb{Z}, +)$ and $(2\mathbb{Z}, \cdot)$ are also associative and commutative.

(iii) $0 \in 2\mathbb{Z}$ such that $a + 0 = 0 + a = a \quad \forall a \in 2\mathbb{Z}$.

(iv) For any $2m \in 2\mathbb{Z} - 2m = (-m) \in 2\mathbb{Z} \in \mathbb{Z}$, so " $\cdot$ " is distributive on " $+$ " over $2\mathbb{Z}$.

Hence $(2\mathbb{Z}, +, \cdot)$ is a commutative ring but it is not a ring with unity as it does not contain $1 \in E$. (Being the set of all even integers).

Example 9.15.6 Prove that $(R, +, *)$ is a ring with zero divisors, where R is 2×2 matrix and $+$ and $*$ are usual addition and multiplication operations.

Solution : $R = \{ \text{a set of } 2 \times 2 \text{ matrices} \}$

First we show that $(R, +)$ is an abelian group.

(i) **Closure :** Since the sum of two 2×2 matrices is also a 2×2 matrix closure is satisfied.

(ii) **Associative :** If A, B and C are 2×2 matrices then

$$(A_{2 \times 2} + B_{2 \times 2}) + C_{2 \times 2} = A_{2 \times 2} + (B_{2 \times 2} + C_{2 \times 2}) \text{ always exists.}$$

(iii) **Identity :** The identity element $\begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}_{2 \times 2}$ belong to R and

$$A + \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix} = A. \text{ Identity is satisfied.}$$

(iv) **Inverse :** The inverse of a 2×2 matrix say A is always $-A$ which is also 2×2 matrix.

(v) **Commutative :** The sum of two matrices is always commutative.

$\therefore (R, +)$ is an abelian group. $(R, \cdot)$ is also a semigroup w.r.t. multiplication and also multiplication is distributive with respect to addition. Hence $(R, \cdot, +)$ is a ring.

Now, Let $A = \begin{bmatrix} 2 & 0 \\ 0 & 0 \end{bmatrix}$, $B = \begin{bmatrix} 0 & 0 \\ 0 & 3 \end{bmatrix}$ be two non zero elements of this ring.

Then, we have $AB = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$

Thus the product of two non-zero elements of the ring is equal to the zero element of the ring.

$\therefore AB$ is a ring with zero divisors.

Example 9.15.7 Write out the operation table for $[z_2^2, +_2, \times_2]$ Is z_2^2 a ring ? Is an integral domain ? Is a field ? Explain.

Solution : The operation tables are as follows :

We have, $z_2 = \{0, 1\}$

+	0	1
0	0	1
1	1	1

$\times$	0	1
0	0	0
1	0	1

Since $(z_2, +_2, \times_2)$ satisfies the following properties :

(i) **Closure axiom :** All the entries in both the tables belong to z_2 . Hence closure is satisfied.

(ii) **Commutative :** In both the tables all the entries about the main diagonal are same, therefore commutativity is satisfied.

(iii) **Associative law :** The associative law for addition and multiplication are also satisfied.

(iv) **Identity :** Here 0 is the additive identity and 1 is the multiplicative identity. Identity property is satisfied.

(v) **Inverse :** Inverse exists in both the tables. The additive inverse of 0, 1 are 1, 0 and the multiplicative inverse of non-zero element of z_2 is 1.

(vi) **Multiplication :** It is distributive over addition. Hence, $(z_2, +_2, \times_2)$ is a ring as well as field. Since we know that every field is an integral domain, therefore it is also an integral domain.

Example 9.15.8 Let $z_8 = \{0, 1, 2, 3, 4, 5, 6, 7\}$. Let R is a relation under the operations addition modulo 7 and multiplication modulo 7. Does this system form a ring ? It is a commutative ring ?

SPPU : Dec.-11

Solution : Consider the following tables,

Table 1 :

$\times_7$	0	1	2	3	4	5	6	7
0	0	0	0	0	0	0	0	0
1	0	1	2	3	4	5	6	0
2	0	2	4	6	1	3	5	0
3	0	3	6	2	5	1	4	0
4	0	4	1	5	2	6	3	0
5	0	5	3	1	6	4	2	0
6	0	6	5	4	3	2	1	0
7	0	0	0	0	0	0	0	0

Table 2 :

$+_7$	0	1	2	3	4	5	6	7
0	0	1	2	3	4	5	6	0
1	1	2	3	4	5	6	0	1
2	2	3	4	5	6	0	1	2
3	3	4	5	6	0	1	2	3
4	4	5	6	0	1	2	3	4
5	5	6	0	1	2	3	4	5
6	6	0	1	2	3	4	5	6
7	0	1	2	3	4	5	6	0

I) S.T. $(z_8, +_7)$ is an abelian group.

i) From table 2, all element are in $z_8 \therefore z_8$ is closed w.r.t. $+_7$

ii) Associativity : for all $a, b, c \in z_8$.

$$a +_7 (b +_7 c) = (a +_7 b) +_7 c$$

iii) By observing the first row of table 2. 0 is the additive identity in z_8 .

9.15.10 Polynomial Ring

Let R be an arbitrary ring and x an indeterminate. The set of all polynomials $f(x)$

$$f(x) = \sum_{i=0}^{\infty} a_i x^i$$

Where the a_i 's are elements of the ring R and only a finite number of them are equal to zero is called $R[x]$.

$R[x]$ from a ring with respect to addition and multiplication of polynomials and hence $R[x]$ is known as polynomial ring.

Note : 1) If $f(x) = \sum_{i=0}^{\infty} a_i x^i$ and $g(x) = \sum_{i=0}^{\infty} b_i x^i$

and if $f(x) = g(x)$ then $a_i = b_i \forall$ non negative integer i .

Thus, two polynomials are equal iff their corresponding coefficients are equal.

$$2) f(x) + g(x) = c_0 x^0 + c_1 x^1 + \dots$$

$$= \sum_{i=0}^{\infty} c_i x^i$$

Where $c_i = a_i + b_i, \forall_i$ thus in order to add two polynomials we should add the coefficients of like power of x

$$3) f(x) g(x) = d_0 x^0 + d_1 x^1 + \dots d_n x^n$$

$$\text{Where, } c_i = d_0 x^0 + a_1 b_{n-1} + a_2 b_{n-2} + \dots + a_n b_0$$

$$\text{Thus } d_i = \sum_{i+j=n} a_i b_j$$

Examples :

Example 9.15.9 Find the deg of $[f(x)+g(x)], f(x)g(x)h(x)h(x)h(x)$ where the polynomials are on the integer (mod 8) and operation are addition and multiplication and $f(x) = 2x+4x^2$, $g(x) = 2+6x+4x^2$, $h(x) = 2+4x$.

$$\begin{aligned} \text{Solution : } f(x) + g(x) &= (2x+4x^2) + [2+6x+4x^2] \\ &= 2 + 8x + 8x^2 \\ &= 2 \pmod{8} \end{aligned}$$

$$\text{Hence deg } f(x) + g(x) = 0$$

$$\begin{aligned} f(x)g(x) &= (2x+4x^2) \cdot (2+6x+4x^2) \\ &= 0 + (2 \cdot 2 + 0 \cdot 6) + (2 \cdot 4 + 6 \cdot 2 + 4 \cdot 0)x^2 + (2 \cdot 4 + 4 \cdot 6)x^3 + (4 \cdot 4)x^4 \\ &= 4x + 20x^2 + 32x^3 + 16x^4 \\ &= 4x + 4x^2 + 0x^3 + 0x^4 \pmod{8} \end{aligned}$$

$$= 4x + 4x^2 \pmod{8}$$

$$\text{Hence } \deg f(x) + g(x) = 2$$

$$\text{Also } h(x)h(x)h(x) = (2 + 4x) [(2 + 4x) (2 + 4x)]$$

$$= (2 + 4x) [2 \cdot 2 + (2 \cdot 4 + 4 \cdot 2)x + 4 \cdot 4x^2]$$

$$= (2 + 4x) (4 + 16x + 16x^2)$$

$$[\text{Now } 4 + 16x + 16x^2 = 4 \pmod{8}]$$

$$= (2 + 4x) \cdot 4$$

$$= (2 \cdot 4) + 4 \cdot 4x$$

$$= 8 + 16x$$

$$= 0 \pmod{8}$$

$$\text{Hence } \deg [h(x)h(x)h(x)] = 0.$$

9.16 Group Codes

SPPU : Dec.-06, 11, 12, 15, 18, May-18

In communication, a code is a rule for converting a data (information or message) into another form or representation, not necessarily of the same type. In communication and information processing, encoding is the process by which data from a source is converted into some symbols to be communicated. Decoding is the reverse process of encoding. In decoding process we convert these code symbols back into information or data which is understandable to receiver.

For a example, in the Chapter 2 of In sem (Propositional Calculus), we convert the information (statement) into mathematical symbol (quantifiers) and vice versa. This is the process of encoding and decoding.

One of the reason for coding to enable communication in places where ordinary spoken or written language in difficult to understand.

The coding problem is that of representing distinct messages or statements by distinct sequence of letters from a given symbols e.g. alphabets, mathematical symbols or binary alphabets $\{0,1\}$.

In this contest, we assume that symbols are only binary alphabets $\{0,1\}$.

- A sequence of letters from an alphabets is also called as word.
- Code is the collection of words which are used to represent messages. A word in code is also known as code word.
- A block code is consisting of words of the same length.
- The number of bits (digits) in the binary word is called the length of binary word.

One of the reason for choosing a block code to represent set of messages is its ability to correct errors. Suppose some code word is transmitted from its origin to its destination. In the course of transmission due to some problem, some 1's are received as 0's the and some 0's are received as 1's. Due to this error the received word is different from the transmitted word.

Therefore to find the exact error and correct it is essential to get correct message. It is known as error detection and correction.

From above discussion, it is clear that the error detection and correction both depend upon the words in the set of possible transmitted words being sufficiently different from one another. The codes with error detecting and correcting properties have a mathematical structure known as group structure we now present this group structure of codes with some definitions.

9.16.1 Group Code

Let S_n be the set of all binary words of length n . Let $\oplus$ be a binary operation on S_n such that for all $x, y \in S_n$ where $x = (x_1, x_2, \dots, x_n)$, $y = (y_1, y_2, \dots, y_n)$, $x \oplus y = (x_1 \oplus y_1, x_2 \oplus y_2, \dots, x_n \oplus y_n)$

Where the operation $\oplus$ denotes the addition modulo 2 on $\{0, 1\}$ and is given by the following table.

$\oplus$	0	1
0	0	1
1	1	0

The algebraic structure $(S_n, \oplus)$ forms a group in which n tuple of 0's $(0, 0, 0, \dots, 0)$ is the identity and each element is its own inverse. In general, any code which is a group under the operation $\oplus$ is called a group code. Group code was first introduced by Hamming and it is very useful in binary encoding techniques.

e.g. if $x = (1\ 0\ 1\ 0\ 1)$ and $y = (0\ 1\ 1\ 0\ 0)$

then $x \oplus y = (1\ 1\ 0\ 0\ 1)$

9.16.2 Hamming Distance

SPPU : Dec.-11, Dec.-15

Let x be a word in S_n . The weight of x is denoted by $w(x)$ and defined as

$$w(x) = \text{Number of one's in } x$$

e.g. $w(0\ 0\ 1\ 0\ 1) = 2$, $w(0\ 0\ 0) = 0$, $w(1\ 1\ 1\ 1) = 4$

Let $x = (x_1, x_2, \dots, x_n)$, $y = (y_1, y_2, \dots, y_n)$ be any two elements in $(S_n, \oplus)$. The Hamming distance between x and y is denoted by $d(x, y)$ and defined as

$d(x, y)$ = The number of co-ordinates at which x_i and y_i are different.

e.g. If $x = (1 \ 0 \ 1 \ 1 \ 0 \ 1)$
 $y = (0 \ 1 \ 1 \ 1 \ 1 \ 0)$
 $\downarrow \downarrow \quad \downarrow \downarrow$
different

$\therefore d(x, y) = 4$, as x and y have first second, fifth, sixth positions different

Now $x \oplus y = (1 \ 1 \ 0 \ 0 \ 1 \ 1)$

$w(x \oplus y) = 4 = d(x, y)$

Thus for any $x, y \in S_n$, $d(x, y) = w(x \oplus y)$

Properties of Hamming Distance :

Let $x, y, z \in (S_n, \oplus)$ then

- i) $d(x, y) \geq 0$
- ii) $d(x, y) = 0$ iff $x = y$
- iii) $d(x, y) = d(y, x)$
- iv) $d(x, z) \leq d(x, y) + d(y, z)$

The minimum distance of a code is the minimum of all the distances between distinct pair of code words.

e.g. Let $x = (1 \ 1 \ 0 \ 1 \ 1 \ 0)$, $y = (0 \ 0 \ 1 \ 1 \ 1 \ 1)$, $z = (1 \ 0 \ 1 \ 0 \ 1 \ 0)$

$\therefore d(x, y) = 4, d(y, z) = 3, d(x, z) = 3$

Out of 4, 3, 3, minimum is 3.

Therefore the minimum distance between the words x, y, z is 3

By using the weight and minimum distance, a combination of errors can be detected and corrected.

Theorem 1) The minimum weight of the non zero code words in a group code is equal to its minimum distance.

Theorem 2) A code can detect all combinations of k or fewer errors iff the minimum distance between any two code words is at least $k+1$,

Theorem 3) A code can correct all combinations of k or fewer errors iff the minimum distance between any two code words is at least $2k+1$.

For example, Let a code containing words of length 3

$c = \{(0, 0, 0), (1, 1, 1)\}$, c contains only two words.

The minimum weight of non zero code word is 3.

∴ By theorem 1, the minimum distance is also 3.

By theorem 2, this code can detect any combination of two errors or one error. And by theorem 3, this code can correct only one error.

9.16.3 Generation of Codes By Using Parity Checks

In 1950, Hamming developed the first complete error detecting and error correcting encoding procedure. This procedure has been frequently used in computer systems.

Hamming constructed the codes, called Hamming codes by introducing redundant digit called parity digits. In a message, that is n digits long, m digits ($m < n$) are used to represent the information part of the message and the remaining $k = n - m$ digits are used for the detection and correction of errors. These K digits are called parity checks.

Hamming single error detecting codes can be described as follows.

i) The actual message is contained in the first $(n-1)$ digits of a code word of length n and the last digit position is set to 0 or 1 so as to make the entire message contain an even numbers of 1's. such an encoding procedure is called a even parity check.

ii) Odd parity check can be used by making the entire message containing an odd no. of 1's. e.g. In even parity check {00, 10, 11} becomes {000, 101, 110} In odd parity check

{ 00, 01, 10, 11} becomes {001, 010, 100, 111}

↑ ↑ ↑

↑ ↑ ↑ ↑

The code words of length n in which information is contained m digits ($m < n$) and remaining $k = n - m$ digits are parity checks, can be generated by using $k \times n$ matrix H . This matrix is called parity check matrix, where elements are from set $\{0, 1\}$ A single error correcting code of length n generated by H with k parity check is given by

$$2^k \geq n + 1 = (k + m) + 1$$

$$\therefore m \leq 2^k - k - 1$$

The number of code words generated by H is $2^m = 2^{n-k}$ and the code generated in this way called the Hamming code.

Theorem 4 : Let H be a parity check matrix which consists of k rows and n columns. Then the set of words $x = (x_1, x_2, x_3, \dots, x_n)$ which belongs to the following set.

$C = \{ x \mid xH^t = 0 \pmod{2} \}$ is a group code under the operation $\oplus$ (addition modulo 2) where H^t = Transpose of matrix H .

Theorem 5 : The parity check matrix H generates a code word of weight q iff $\exists$ a set of columns of H such that their k - tuple sum is zero.

For the parity check matrix H of order $k \times n$.

Where n = Length of code word

k = Number of parity check bits

The information digits will be $m = n - k$. The number of code words generated is $2^m = 2^{n-k}$. Consider the following steps for the generation of code words.

Step 1 : Find the system of equations from $x \cdot H^t = 0$.

Step 2 : Find the values of parity check bits interms of information digits from the equations obtained in step 1.

Step 3 : Give values 0 or 1 to information digits and calculate the values of parity check bits according to step 2. The binary words obtained in this way will be the required code words generated by H .

To find the number of errors detected and corrected first find the columns of H whose sum (addition modulo 2) is zero. The number of theses columns is equal to the minimum weight of the code and which is equal to the minimum distance of the code.

Examples :

Example 9.16.1 Find the minimum distance of an encoding function $e : B^2 \rightarrow B^5$ given as .

$e(00) = 00000$, $e(01) = 10011$, $e(10) = 01110$, $e(11) = 11111$.

SPPU : Dec.-06

Solution :

$$d[e(00), e(01)] = 3$$

$$d[e(00), e(11)] = 5$$

$$d[e(00), e(10)] = 3$$

$$d[e(01), e(10)] = 3$$

$$d[e(01), e(11)] = 4$$

$$d[e(10), e(11)] = 2$$

$$d[e(10), e(11)] = 2$$

The minimum along all distances is 2.

Thus the minimum distance of an encoding is 2.

Example 9.16.2 What is Hamming function (distance) ? Find the distance between the code words of $C = \{ (0\ 0\ 0\ 0), (0\ 1\ 0\ 1), (1\ 0\ 1\ 1), (0\ 1\ 1\ 1) \}$ Rewrite the message by adding even parity check bit and parity check bit.

SPPU : Dec.-11, Marks 5

Solution : Refer 9.16.2 for the definition.

$$\text{We have } d[e(0\ 0\ 0\ 0), e(0\ 1\ 0\ 1)] = 2$$

$$d[e(0\ 0\ 0\ 0), e(1\ 0\ 1\ 1)] = 3$$

$$d[e(0\ 0\ 0\ 0), e(0\ 1\ 1\ 1)] = 3$$

$$d[e(0\ 1\ 0\ 1), e(1\ 0\ 1\ 1)] = 3$$

$$d[e(0\ 1\ 0\ 1), e(0\ 1\ 1\ 1)] = 1$$

$$d[e(1\ 0\ 1\ 1), e(0\ 1\ 1\ 1)] = 2$$

Even parity check bit of c is $\{(0\ 0\ 0\ 0\ 0), (0\ 1\ 0\ 1\ 0), (1\ 0\ 1\ 1\ 1), (0\ 1\ 1\ 1\ 1)\}$

Odd parity check bit of c is $\{(0\ 0\ 0\ 0\ 1), (0\ 1\ 0\ 1\ 1), (1\ 0\ 1\ 1\ 0), (0\ 1\ 1\ 1\ 0)\}$

Example 9.16.3

Given the parity check matrix $H = \begin{bmatrix} 1 & 1 & 0 & 1 & 0 & 0 \\ 0 & 1 & 1 & 0 & 1 & 0 \\ 1 & 0 & 1 & 0 & 0 & 1 \end{bmatrix}$ find the minimum

distance of code generated by H . How many errors can it detect and correct.

SPPU : Dec.-15, Marks 5

Solution : From the given matrix H , select minimum number of column whose sum is zero.

$$\therefore \text{Consider columns } h_1 = \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}, h_2 = \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}, h_3 = \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix}$$

The sum of these columns

$$h_1 \oplus h_2 \oplus h_3 = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

Hence the minimum weight of the code is 3 which is equal to its minimum distance.

We know that the code can detect all combinations of k or fewer errors iff the minimum distance between any two code words is at least $k+1$.

Here minimum distance is $3 = k+1 \Rightarrow k = 2$.

$\therefore$ This code can detect 2 or less errors.

Also it can correct k errors if the minimum distance is $2k+1$.

$$\therefore 2k + 1 = 3 \Rightarrow k = 1$$

This code can correct only one error.

Therefore this is a single error correcting code.

Example 9.16.4 Find the number of code words generated by the parity check matrix H given

$$\text{by } H = \begin{bmatrix} 1 & 1 & 0 & 1 & 0 & 0 \\ 0 & 1 & 1 & 0 & 1 & 0 \\ 1 & 0 & 1 & 0 & 0 & 1 \end{bmatrix} \text{ find all the code words generated.}$$

SPPU : Dec.-12, Marks 5

Solution : The parity check matrix is

$$H = \begin{bmatrix} 1 & 1 & 0 & 1 & 0 & 0 \\ 0 & 1 & 1 & 0 & 1 & 0 \\ 1 & 0 & 1 & 0 & 0 & 1 \end{bmatrix}_{3 \times 6}$$

H is of order 3×6 . Hence the length of the code word is 6 in which last three digits are parity check bits. The information digits are

$$M = n - k = 6 - 3 = 3$$

$\therefore$ The matrix H will generate. $2^m = 8$ code words.

They are the solution of $x \cdot H^t = 0$.

$$[x_1 \ x_2 \ x_3 \ x_4 \ x_5 \ x_6] \begin{bmatrix} 1 & 0 & 1 \\ 1 & 1 & 0 \\ 0 & 1 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} = [0 \ 0 \ 0]$$

$$\Rightarrow x_1 + x_2 + \quad + x_4 + \quad = 0$$

$$x_2 + x_3 + \quad + x_5 + \quad = 0$$

$$x_1 + \quad + x_3 + \quad + x_6 = 0$$

$$x_4 = x_1 + x_2$$

$$\Rightarrow x_5 = \quad x_2 + x_3$$

$$x_6 = x_1 + \quad + x_3$$

By giving different combinations of 0 and 1 to x_1, x_2, x_3 , we get the following code words.

x_1	x_2	x_3	x_4	x_5	x_6
0	0	0	0	0	0
0	0	1	0	1	1
0	1	0	1	1	0
0	1	1	1	0	1
1	0	0	1	0	1
1	0	1	1	1	0
1	1	0	0	1	1
1	1	1	0	0	0

Hence the code is

$$C = \{ (0\ 0\ 0\ 0\ 0) (0\ 0\ 1\ 0\ 1\ 1) (0\ 1\ 0\ 1\ 1\ 0) (0\ 1\ 1\ 1\ 0\ 1) (1\ 0\ 0\ 1\ 0\ 1) \\ (1\ 0\ 1\ 1\ 1\ 0) (1\ 1\ 0\ 0\ 1\ 1) (1\ 1\ 1\ 0\ 0\ 0) \}$$

9.16.4 Decoding

For group codes, there is an efficient way to determine the transmitted word corresponding to the word received according to minimum distance criteria. Basic group theory is involved in this process.

Let $(C, \oplus)$ be a group code. Suppose the codewords transmitted contain n binary digits in which the first n digits contain the information part and the last k digits are parity check bits.

Let $(S_n, \oplus)$ be a set of all n -tuples, where $(C, \oplus)$ is a subgroup of $(S_n, \oplus)$. Suppose the codeword $x \in C$ being transmitted through the noisy channel and the received word is y . Let e be the error in the received word y where e consists of one's in digit position in which an error occurs.

Thus $x \oplus e = y$, where $y \in S_n$

Since the operation $\oplus$ is addition modulo 2 therefore $x = e \oplus y$. This shows that the transmitted codeword x is closed to y in terms of Hamming distance i.e. the error e should be of least weight.

To find error with least weight, we use the concept of cosets of subgroup C of S_n . We know that set of all cosets of C in S_n forms a partition in S_n .

Consider the element $y \in S_n$, clearly y is in the left coset $y \oplus C$ among all the elements $y \oplus C$ we can determine the element 'a' with least weight.

Such an element is known as coset leader. Thus for each coset, we have a coset leader. For the coset of C itself, 0 is the coset leader, since C has 2^m elements, the number of cosets will be

$$r = 2^{n-m}$$

Hence all together the coset leaders are $0, a_1, a_2, \dots, a_{r-1}$ if a_1 is coset leader, then elements of coset $a_1 \oplus C$ are given as $a_1 \oplus x$ where $x \in C$. Further the element of C which is closed to $a_1 \oplus x$. Thus, if the received word is $a_1 \oplus x$, the transmitted code word is x .

Consider the following steps for constructing a decoding table.

Step i : List the codewords of C , starting with 0 as the first elements.

Step ii : Choose any word $a_1 \in S_n$ not in the first row of minimum weight. List the coset $a_1 \oplus C$ as the second row by putting a_1 under 0 and $a_1 \oplus x$ under x for each $x \in C$

Step iii : From these words of S_n , not in first and second row, choose a_2 of minimum weight and list the coset $a_2 \oplus C$ as in step 2 to get third row.

Step iv : Continue in this way untill all the cosets are listed and every element of S_n appears exactly once.

The decoding table has the form

0	x	x^1	x^n	$\dots$
a_1	$a_1 \oplus x$	$a_1 \oplus x^1$	$a_1 \oplus x^n$	$\dots$
a_2	$a_2 \oplus x$	$a_2 \oplus x^1$	$a_2 \oplus x^n$	$\dots$
$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\dots$

In the table, the left most column denotes the coset leader and the top row denotes the codewords generated. A received word 'y' can be denoted by first finding y in a row of the decoding table. Suppose y is in $(i+1)^{\text{th}}$ row of the decoding table. Then the decoded word is given by $x = a_i \oplus y$ where a_i is the coset leader of $(i+1)^{\text{th}}$ row. Also if no error has occurred in transmission, then the coset leader will be 0 .

9.16.5 Cyclic Code

Let C be a linear code over a finite field F of **block length** n . C is called a cyclic code, if for every codeword $C = (c_1, c_2, \dots, c_n)$ from C , the word $(c_n, c_1, \dots, c_{n-1})$ in F^n obtained by a cyclic right shift of components is also codeword. If C is the smallest

cyclic code containing c , or, in other words C is the linear code generated by c and all codeword's obtained by cyclic shifts of its components.

I) Algebraic Structure

Cyclic codes can be linked to ideals in certain rings.

Consider the polynomial code over $G = \{0, 1\}$ with $n = 5$, $m = 2$ and generator polynomial $g(x) = x^2 + x + 1$. This code consists of the following code words,

$$0, x^2 + x + 1, x^3 + x^2 + x, x^3 + 1, x^4 + x^3 + x^2, x^4 + x^3 + x + 1, x^4 + x, x^4 + x^2 + 1$$

equivalently expressed as strings of binary digits of the codewords are,

$$00000, 00111, 01110, 01001, 11100, 11011, 10010, 10101$$

A block code is called a cyclic code if for every codeword $(a_0 \ a_1 \ a_2 \ \dots \ a_{n-2} \ a_{n-1})$ the sequence $(a_{n-1} \ a_0 \ a_1 \ a_2 \ \dots \ a_{n-2})$ is also a codeword.

In other words, for a cyclic code, a cyclic shift of a codeword is also a codeword.

To describe cyclic codes,

let $(F, +, \cdot)$ be a field of integer modulo 2.

Now the binary sequence $a_0 \ a_1 \ a_2 \ \dots \ a_{n-1}$ can be expressed as a polynomial.

$$a_0 + a_1 x + \dots + a_{n-1} x^{n-1} \text{ in the polynomial ring } (F[x], +, \cdot)$$

Let $(F_n[x], \oplus, \odot)$ be the ring of polynomials modulo $1 + x^n$.

Let I be an ideal or $(F_n[x], \oplus, \odot)$

Now to show that polynomial in I constitute a cyclic code.

As $(I, \oplus)$ form an abelian group, the polynomial in I constitute group codes.

Let $f(x) \in I$

then $\forall x \in F[x], f(x) \in I$

$x \odot f(x) \in I$ [by definition of ideals $\forall r \in R, s \in S, r s \in S$ also $s r \in S$ where R is ring and S is subring of R]

$$\text{Let } f(x) = a_0 + a_1 x + \dots + a_{n-1} x^{n-1}$$

$$\text{then } x \odot f(x) = \text{remainder of } \left(\frac{a_0 x + a_1 x^2 + \dots + a_{n-1} x^n}{1 + x^n} \right)$$

$$= \frac{a_0 x + a_1 x^2 + \dots + a_{n-1} + a_{n-1}(1 + x^n)}{1 + x^n}$$

$$= a_{n-1} + a_0 x + a_1 x^2 + \dots + a_{n-2} x^{n-1}$$

Therefore polynomial in I constitute a cyclic code.

On the other hand, let I denote a set of polynomials corresponding to the codeword in a cyclic code.

$(I, \oplus)$ is an abelian group.

If $f(x)$ is in I .

$x \odot f(x)$ is in I .

Hence $x \odot (x \odot f(x)) = x^2 \odot f(x)$ is in I .

and this way

$x^i \odot f(x)$ is in I .

and $b_j x^j \odot f(x)$ is in I .

Because the polynomials in I form a cyclic code, it follows that $B(x) \odot f(x)$ is in I for $B(x)$ in $F_n[x]$.

9.17 Case Studies

Case Study : Cryptography used in a world-war II -

- Cryptography is a technique of securing information and communication by converting it to non-readable format through the use of code. This information can understand the person for whom the information is intended. Thus, preventing unauthorized access to information.
- Cryptography contains two phases encryption and decryption. In encryption original message (plain text) is converted into a non-readable format (cipher text). It is done at the sender's end. In decryption, again cipher text is converted into plain text. It is done at receiver's end. There are different algorithms used for cryptography some wellknown algorithms are RSA, DES, AES, etc. There are two types of algorithms public key cryptography and private key cryptography. RSA belongs to public key cryptography and DES, AES belongs to private key cryptography.
- One of the most famous example is Engima machine used by German Government and military during world-war II. The standard military Engima used there 26-point wired metal and black-plastic rotors selecting from a set of 5 to 8. Each rotor was a cylinder with a large, movable, notched wheel on one end with an alphabet or numbers around its circumference. One face of the cylinder had 26 spring-loaded copper pin protruding from it and the other face had 26 flush copper contacts. Inside each cylinder was a wired 'maze' connecting the contacts to the pins.

- Engima used 3 rotors, each one wired differently and a reflector. When a letter was input, the rightmost wheel would turn one space forward and electrical pulse would rotate through each wheel in turn.
- It uses a monoalphabetic substitution cipher, rotation cipher and rotating substitution cipher. A alphabet encrypts to many letter. Repeated letters will not encrypt the same way in different positions.

9.18 University Questions with Answers

Dec. - 2016

Q.1 Consider binary relation $*$ defined on the set :

$A = \{a, b, c, d\}$ by the following table. Fill the empty cell.

[2]

*	e	a	b	c
e	e	a	b	c
a	a	b	c	e
b				
c				

Ans. : Given that,

$$A = \{a, b, c, d\}$$

Given table :

x	e	a	b	c
e	e	a	b	c
a	a	b	c	e
b				
c				

a and $c \in A$ but $a * c = e$ which is not in A

$\therefore$ The relation is not binary relation on A .

Hence this equation is wrong.

Q.2 Prove that :

$((a + b\sqrt{2}), +, *)$ where $a, b \in \mathbb{R}$ is integral domain. (Refer example 9.15.2)

[6]

Q.3 Define normal subgroup and rings with example.

(Refer sections 9.11 and 9.15.1)

[5]

Q.4 Let $R = \{0^\circ, 60^\circ, 120^\circ, 180^\circ, 240^\circ, 300^\circ\}$ and $*$ is a binary operation so that a and b in R , $a * b$ is overall angular rotation corresponding to successive rotations by a and then b . Show that $(R, *)$ is a group. [6]

Ans. : Given that,

$$R = \{0^\circ, 60^\circ, 120^\circ, 180^\circ, 240^\circ, 300^\circ\}$$

Consider the composition table,

*	0°	60	120	180	240	300
0	0	60	120	180	240	300
60	60	120	180	240	300	0
120	120	180	140	300	0	60
180	180	240	300	0	60	120
240	240	300	0	60	120	180
300	300	0	60	120	180	240

i) Every element of table is present in R

$\therefore R$ is closed w.r.t. $*$

ii) Rotation of objects is associative

$\therefore *$ is associative in R .

iii) From the composition table, 0° is the identity element in R .

iv) From table, the inverses of 0, 60, 120, 180, 240, 300 are 0, 300, 240, 180, 120, 60 respectively.

Thus $(R, *)$ is a group.

Q.5 Let $A = \{0, 1\}$. Is A closed under : i) Multiplication ii) Addition [3]

Ans. : Given that

$A = \{0, 1\}$ consider the following tables

$\times$	0	1
0	0	0
1	0	1

+	0	1
0	0	1
1	1	2

By multiplication table, 'A' is closed w.r.t. $\times$.

By addition table, 'A' is not closed w.r.t. $+$ as $1 + 1 = 2 \notin A$.

Q.6 Explain isomorphism and automorphism. (Refer section 9.14)

[4]

May - 2017

Q.7 Prove that : $(a + b\sqrt{2}, +, \times)$ where $a, b \in \mathbb{R}$ is integral domain.
(Refer example 9.15.2)

[6]

Q.8 Explain isomorphism and homomorphism of two semigroups.
(Refer sections 9.13 and 9.14)

[3]

Q.9 Prove that every cyclic group is an abelian group.
(Refer section 9.12 theorem 4)

[4]

Q.10 Let G be set of all non-zero real numbers and let : $a * b = \frac{ab}{2}$. Show that $(G, *)$ is an abelian group. (Refer example 9.4.8)

[6]

Q.11 Explain Galois theory.

[3]

Ans. : • Galois used permutating graphs to describe how the various roots of a given polynomial equations are related to each other.

- The central idea of Galois theory is to consider permutations of roots such that any algebraic equation satisfied by the roots is still satisfied after the roots have been permuted.
- The permutations together form a permutation group, also called Galois group of polynomial.

Q.12 Explain properties of binary operations. (Refer section 9.2.2)

[4]

Dec. - 2017

Q.13 Consider the set $A = \{1, 3, 5, 7, 9, \dots\}$ i.e. a set of odd positive integers. Determine whether A is closed under : i) Addition ii) Multiplication

[4]

Ans. : Given that $A = \{1, 3, 5, 7, 9, \dots\}$

i) **Addition :** We know that the addition of two odd numbers is an even number

i.e. $x, y \in A \Rightarrow x + y = \text{even number} \notin A$.

In particular, $1, 3 \in A$ but $1 + 3 = 4 \notin A$

Addition is not closed on A .

ii) **Multiplication :** We know that the multiplication of two odd numbers is an odd number

$\therefore$ For any $x, y \in A$, $x \times y$ is an odd number

$\therefore x \times y \in A$

$\therefore$ Multiplication is closed on A .

Q.14 Check whether the algebraic system $(A, *)$ whose table is given below is a SEMI group. [4]

$*$	a	b	c
a	a	b	c
b	a	c	b
c	a	b	c

Ans. : An algebraic structure $(A, *)$ is a semi group if $a * (b * c) = (a * c) * c$

$\forall a, b, c, \in A$

i.e. $*$ is associative on A .

Consider

$$(a * c) * b = c * b = b$$

$$a * (c * b) = a * b = b$$

Now, $(b * a) * c = a * c = c$

$$b * (a * c) = b * c = b$$

$\therefore (b * a) * c \neq b * (a * c) \Rightarrow *$ is not associative

$\therefore (A, *)$ is not a semigroup

Q.15 Discuss in brief about the Galois theory - Field theory and group theory (Refer Q.11). [5]

Q.16 Define algebraic system. Explain the steps to identify the following with suitable example.

i) Monoid (Refer section 9.3.3)

ii) Abelian group (Refer section 9.4.1)

iii) Ring. (Refer section 9.15.1) [13]

May - 2018

Q.17 Let $R = \{0, 60, 120, 180, 240, 300\}$ and $*$ = binary operation so that for a and b in R , $a * b$ is overall angular rotation corresponding to successive rotation by a and by b . Show $(R, *)$ is a group. (Refer Q.4) [7]

Q.18 Explain the following terms with examples :

i) Ring (Refer section 9.15.1) ii) Integral domain (Refer section 9.15.7)

iii) Field (Refer 9.15.8) [6]

OR

Q.19 Show that $(I, \oplus, \square)$ is a commutative ring with identity where $\oplus$ and $\square$ are defined as : $a \oplus b = a + b - 1$ and $a \square b = a + b$ [7]

Ans. : We have $(I, \oplus, \square)$

We have, $a \oplus b = a + b - 1$ and $a \square b = a + b$

I) $(R, \oplus)$ is an abelian group

i) Closure property : for any $a, b \in I$

$$a \oplus b = a + b - 1 \in I$$

ii) Associativity : for any $a, b, c \in I$

$$(a \oplus b) \oplus c = (a + b - 1) \oplus c = a + b - 1 + c - 1 = a + b + c - 2$$

$$a \oplus (b \oplus c) = a \oplus (b + c - 1) = a + b + c - 1 - 1 = a + b + c - 2$$

$$\therefore (a \oplus b) \oplus c = a \oplus (b \oplus c)$$

iii) Existence of the Identity :

for any $a \in I, \exists 1 \in I$ S.t.

$$a \oplus 1 = a + 1 - 1 = a = 1 \oplus a$$

$\therefore 1$ is called the identify element.

iv) Existence of the inverse : For any $a \in I$

$$\exists 2 - a \in I \text{ such that } a \oplus (2 - a) = a + 2 - a - 1 = 1$$

$\therefore 2 - a$ is the inverse of a

v) Commutative property : For any $a, b \in I$

$$a \oplus b = a + b - 1 = b + a - 1 = b \oplus a$$

II) $(I, \square)$ is semigroup

i) Closure property : $a \square b = a + b \in I$

ii) Associative property : $(a \square b) \square c = a \square (b \square c) \forall a, b, c \in I$

III) Multiplication distributes over addition :

$$a \square (b \oplus c) = a \square [b + c - 1] = a + b + c - 1$$

$$(a \square b) \oplus (a \square c) = (a + b) \oplus (a + c) = a + b + a + c - 1$$

$$a \square (b \oplus c) \neq (a \square b) \oplus (a \square c)$$

Multiplication is not distribute over addition.

$\therefore (I, \oplus, \square)$ is not a ring, Hence. it is not commutative ring.

Q.20 Explain the following terms :

i) Monoids (Refer section 9.3.3)

ii) Sub-group (Refer section 9.7.2)

iii) Group codes (Refer section 9.16)

[6]

Dec. - 2018

Q.21 Let $Z_n = \{0, 1, 2, \dots, n-1\}$. In Z_{13} what is the order of 3, 6 and 8. [3]

Ans. : We have $Z_{12} = \{0, 1, 2, 3, \dots, 11\}$

In Z_{12} , $+_{12}$ is the binary operation.

n is the order of x in Z_{12} if $nx = 0$

We have,

$$4(3) = 0 \therefore 0(3) = 4$$

$$2(6) = 0 = 12, \therefore 0(6) = 2$$

$$3(8) = 24 = 0, \therefore 0(8) = 3$$

Q.22 Let $(Q, *)$ is an Algebraic system. $*$ is a binary operation defined as $a * b = a + b - ab$ $\forall a, b \in Q$. Determine whether $(Q, *)$ is a group. [4]

(Refer similar example 9.4.3)

Q.23 Define :

i) Rings (Refer section 9.15.1)

ii) Integral domain (Refer section 9.15.7)

iii) Field (Refer section 9.15.8) [6]

Q.24 Let $Z_n = \{0, 1, 2, \dots, n-1\}$. Let $*$ be a binary operation such that $a * b =$ remainder of $(a + b)$ divided by n . Construct a table for $n = 4$. Is $(Z_4, *)$ a monoid, semigroup, group and abelian group. [7]

Ans. : Let $Z_4 = \{0, 1, 2, 3\}$

In, Z_4 , $a * b =$ Remainder of $(a + b)$ divided consider the following table

*	0	1	2	3
0	0	1	2	3
1	1	2	3	0
2	2	3	0	1
3	3	0	1	2

i) All elements of table are in $Z_4 \therefore *$ is closed on Z_4

ii) As $*$ is defined in terms of addition

$\therefore *$ is associative on Z_4

iii) 0 is the identity element w.r.t $*$

iv) The inverse of 0 is 0 $\therefore 0^{-1} = 0$

$$(1)^{-1} = 3, (2)^{-1} = 2, (3)^{-1} = 1$$

$\therefore$ Every element in Z_4 has inverse in Z_4 w.r.t. $*$

v) Table is symmetric $\therefore a * b = b * a$

Hence $(Z, *)$ is a monoid, semigroup, group and abelian group.

Q.25 Define :

(i) Group code (Refer section 9.16)

(ii) Galois theory (Refer Q.11)

(iii) Cyclic group (Refer section 9.10)

[6]

May - 2019

Q.26 What is Monoid ? Show that the algebraic structure $(A, +)$ is a monoid, where A is set of integers and $+$ is a binary operation giving addition of two integers.

(Refer section 9.3.3 and example 9.3.1)

[3]

Q.27 Define the following terms.

i) Ring (Refer section 9.15.1)

ii) Field (Refer section 9.15.8)

iii) Integral domain. (Refer section 9.15.7)

[3]

Q.28 Show that $R = \{a + b\sqrt{2} ; b \in I\}$ for the operation $+, *$ is integral domain but not a field.

(Refer example 9.15.2)

[7]

Q.29 Let $A = \{0, 1\}$. Is A closed under

1) Multiplication 2) Addition (Refer Q.5)

[4]

Q.30 Define

1) Properties of binary operations

2) Ring with unity (Refer section 9.15.1)

[4]

Q.31 Let $R = \{0, 60, 120, 180, 240, 300\}$ and $*$ = binary operation so that for a and b in R $a * b$ is overall angular rotation corresponding to successive rotations by a and by b show $(R, *)$ is a group. (Refer Q.4)

[5]

Q.32 Write properties of binary operations. (Refer section 9.2.2)

[5]

Q.33 Prove that the set 2 of all integers with binary operation $*$ defined by :

[5]

$a * b = a + b + 1 \quad \forall a, b \in 2$ is an abelian group. (Refer section 9.4.14)

Q.34 Let $A = \{0, 1\}$. Is A closed under : 1) Multiplication 2) Addition.

[3]

Ans. : 1) Multiplication

*	0	1
0	0	0
1	0	1

Every element of table belongs A. i.e. $0, 1 \in A$

Hence A is closed w.r.t multiplication

2) Addition

+	0	1
0	0	1
1	1	2

Here $2 \notin A$. Hence A is not closed w.r.t addition.

Q.35 $f : G \rightarrow G$, G is group with identity 'e' such that $f(a) \in e$ for all $a \in G$ prove that function f is homomorphism. (Refer example 9.14.1) **[5]**

Q.36 In the set R of real number. Decide whether the following composition is associative $a, b, c \in R : (1) a * b = a + 2b$ (2) $a * b = a$. **[4]**

Ans. : Associate property : A binary operation '*' on set A is said to be associative if $a * (b * c) = (a * b) * c \forall a, b, c \in A$

$$1) \quad a * b = a + 2b$$

$$\begin{aligned} a * (b * c) &= a * (b + 2c) \\ &= a + 2(b + 2c) \\ &= a + 2b + 4c \end{aligned}$$

$$\begin{aligned} (a * b) * c &= (a + 2b) * c \\ &= (a + 2b) + 2c \\ &= a + 2b + 2c \end{aligned}$$

$$a * (b * c) \neq (a * b) * c$$

Hence * is not associative

$$2) \quad a * b = a$$

$$a * (b * c) = a * b = a$$

$$(a * b) * c = a * c = a$$

Hence $*$ is associative

Q.37 Prove that the set : $A = \{0, 2, 4, 6, 8\}$ with t_{10} and x_{10} operation i.e.
 $R = \{A, t_{10}, x_{10}\}$ is a ring. **[4]**

Ans. :

$+_{10}$ = Addition modulo 10

$*_{10}$ = Multiplication modulo 10

Consider the table for $+_{10}$

I) $(A, +_{10})$

i) From Table 9.18.1, all elements are in A.

i.e. A is closed with respect to $+_{10}$

ii) Associativity : For all $a, b, c \in A$

$$a +_{10} (b +_{10} c) = (a +_{10} b) +_{10} c$$

iii) 0 is the identity element

iv) Each element of A has inverse element

i.e. 0 is the inverse of 0, 2 is the inverse of 8, 4 is the inverse of 6.

v) Commutativity : For all $a, b \in A$

$$a +_{10} b = b +_{10} a$$

$+_{10}$	0	2	4	6	8
0	0	2	4	6	8
2	2	4	6	8	0
4	4	6	8	0	2
6	6	8	0	2	4
8	8	0	2	4	6

Table 9.18.1

$*_{10}$	0	2	4	6	8
0	0	0	0	0	0
2	0	4	8	2	6
4	0	8	6	4	2
6	0	2	4	6	8
8	0	6	2	8	4

Table 9.18.2

Hence $(A, +_{10})$ is an abelian group

II) $(A, +_{10})$

i) From Table 9.18.2, all elements are in A

i.e. A is closed w.r.t $*_{10}$

ii) Associativity

For all $a, b, c \in A$

$$a *_{10} (b *_{10} c) = (a *_{10} b) *_{10} c$$

Hence $(A, *_{10})$ is a semigroup

$\therefore (A, +_{10}, *_{10})$ is a ring.



SOLVED MODEL QUESTION PAPER

(As Per 2019 Pattern)

Discrete Mathematics

S.E. (Comp) Sem - I (End Sem)

Time : 2 $\frac{1}{2}$ Hours]

[Total Marks : 70

Instructions to the candidates :

1) Answer Q.1 or Q.2, Q.3 or Q.4, Q.5 or Q.6, Q.7 or Q.8.

2) Figures to the right indicate full marks.

3) Neat diagrams must be drawn wherever necessary

4) Assume suitable data if necessary.

- Q.1** a) How many permutations can be made out of the letter of word "COMPUTER"? How many of these : i) begin with C ii) end with R iii) begin with C and end with R iv) C and R occupy the end places. (Refer example 6.3.7) [8]
- b) From 10 programmers in how many ways can 5 be selected when i) A particular programmer is included every time ii) A particular programmer is not included at all time. (Refer example 6.4.5) [6]
- c) In the expression of $(1+x)^6$. What is the coefficient of x^3 . (Refer Q.17 of section 6.8) [4]

OR

- Q.2** a) If a committee has eight members :
i) How many ways can committee members be seated in a row ?
ii) How many ways can the committee select a president, vice-president and secretary ? (Refer Q.15 of section 6.8) [6]
- b) Solve the following :
i) How many different car number plates are possible with 2 letters followed by 3 digits ? ii) How many of these number plates begin with 'MH' ? (Refer Q.11 of section 6.8) [6]
- c) i) Find 8^{th} term in the expression $(x+y)^{13}$. (Refer Q.21 of section 6.8)
ii) Expand $(3x-4)^4$ using binomial theorem. (Refer Q.23 of section 6.8) [6]
- Q.3** a) Determine whether graphs G and H are isomorphic or not. Justify your answer. (Refer example 7.7.6) [6]

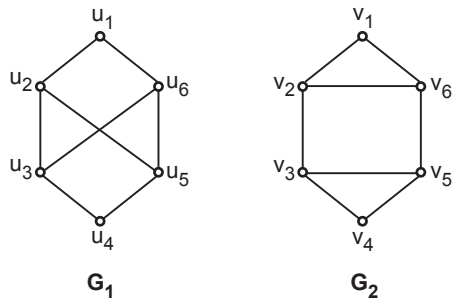


Fig. 1

- b) Use Dijkstra's algorithm to find the shortest path between a and z .
(Refer example 7.13.1)

[6]

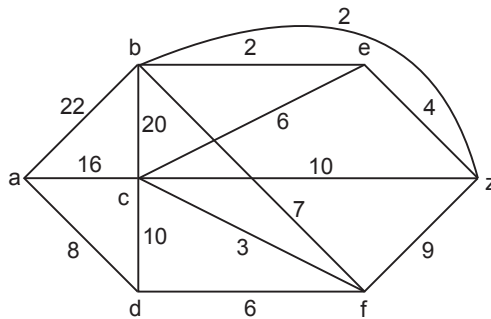


Fig. 2

- c) Determine which of the graphs below represents Eulerian circuit, Eulerian path, Hamiltonian circuit and Hamiltonian path. Justify your answer.
(Refer Q.1 of section 7.22)

[5]

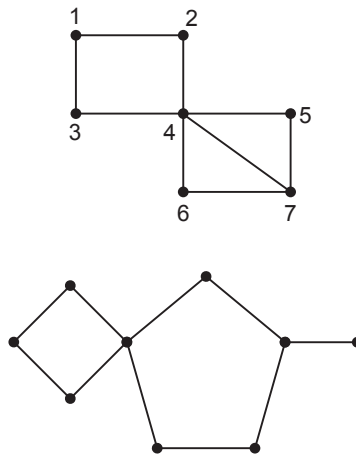


Fig. 3

OR

- Q.4 a)** Use dijkstra's algorithm to find the shortest path between a and z.
(Refer Q.16 of section 7.22)

[6]

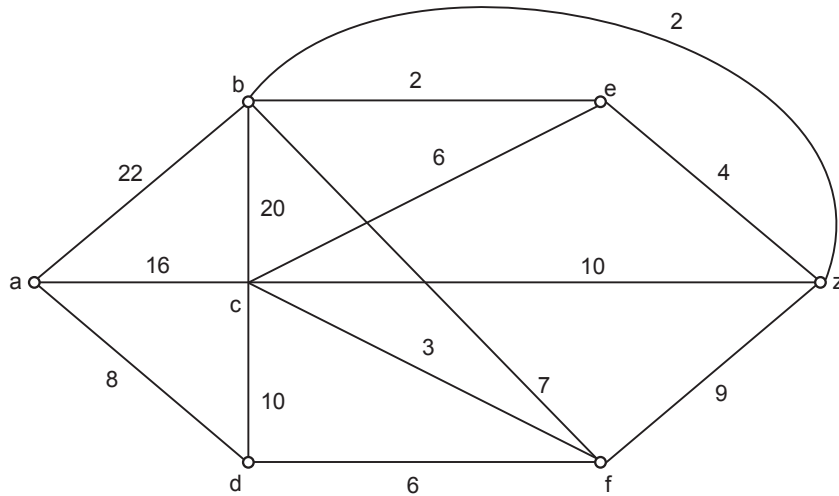
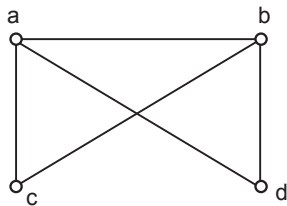


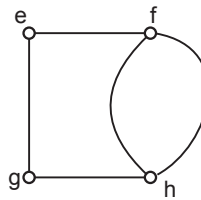
Fig. 4

- b)** Determine whether the following graphs are isomorphic to each other.
(Refer Q.21 of section 7.22)

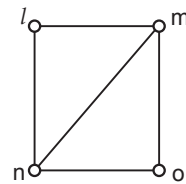
[6]



(a)



(b)



(c)

Fig. 5

- c)** i) Define the graph K_n , K_{mn} . (Refer section 7.5 (ii) and (iv))
ii) How many colors required to color K_{mn} ? (Refer Q.5 of section 7.22)

[5]

- Q.5 a)** Suppose data items A, B, C, D, E, F, G occur in the following frequencies.

Data Items	A	B	C	D	E	F	G
Weight	10	30	5	15	20	15	05

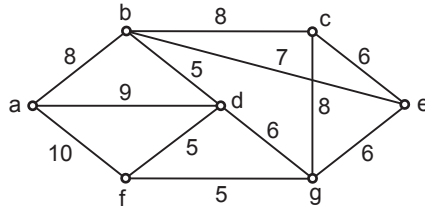
Construct a Huffman code for the data.

What is the minimum weighted path length.

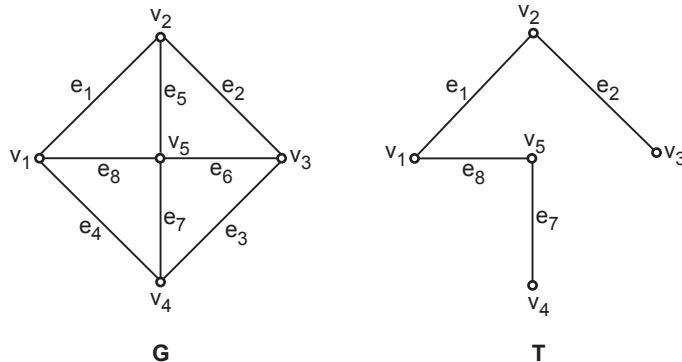
(Refer example 8.6.2)

[6]

- b) Find the minimum spanning tree for the graph given in the following figure using Prim's algorithm. (Refer example 8.7.1) [6]



- c) Find the fundamental cutsets and fundamental circuits of the following graph w.r.t. given spanning tree. (Refer example 8.8.2) [6]



OR

- Q.6 a) Use labelling procedure to find a maximum flow in the transport network shown in the following Fig. 6. Determine the corresponding minimum cut. (Refer example 8.9.3) [6]

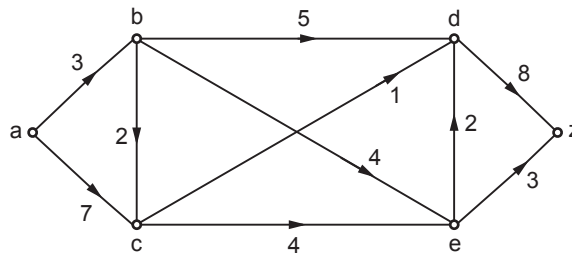
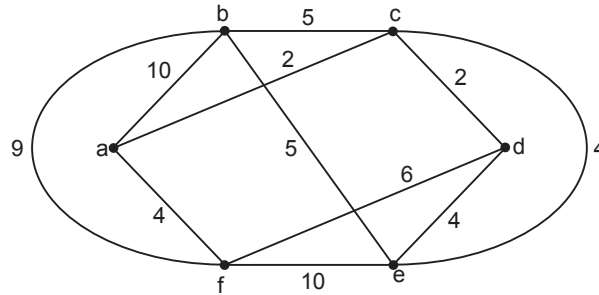


Fig. 6

- b) Use the Kruskal's algorithm to find the minimum spanning tree for the graph shown. (Refer example 8.7.5) [6]



- c) Find the pre, post and inorder traversal of a tree shown in Fig. 7. (Refer Q.14 of section 8.12) [6]

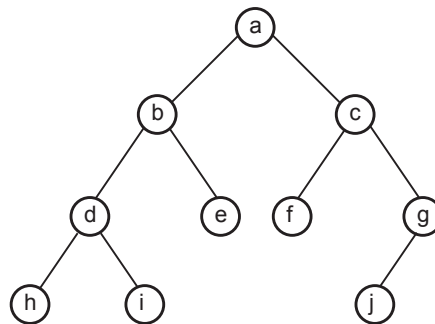


Fig. 7 Tree

- Q.7** a) Let $R = \{0^\circ, 60^\circ, 120^\circ, 180^\circ, 240^\circ, 300^\circ\}$ and $*$ is a binary operation so that a and b in R , $a * b$ is overall angular rotation corresponding to successive rotations by a and then b . Show that $(R, *)$ is a group. (Refer Q.4 of section 9.18) [6]
- b) Prove that $(a + b\sqrt{2}, +, \times)$ where $a, b \in R$ is integral domain. (Refer Q.7 of section 9.18) [5]
- c) Explain the following terms :
 i) Monoids ii) Sub-group iii) Group codes (Refer Q.20 of section 9.12) [6]

OR

- Q.8** a) Show that $(I, \oplus, \square)$ is a commutative ring with identity where $\oplus$ and $\square$ are defined as : $a \oplus b = a + b - 1$ and $a \square b = a + b$. (Refer Q.19 of section 9.18) [6]

- b)** Find the number of code words generated by the parity check matrix H given by

$$H = \begin{bmatrix} 1 & 1 & 0 & 1 & 0 & 0 \\ 0 & 1 & 1 & 0 & 1 & 0 \\ 1 & 0 & 1 & 0 & 0 & 1 \end{bmatrix} \text{ find all the code words generated.}$$

(Refer example 9.16.4)

[6]

- c)** Define :

i) Rings ii) Integral domain iii) Field **(Refer Q.23 of section 9.18)**

[5]

□□□